

The functional element of genome

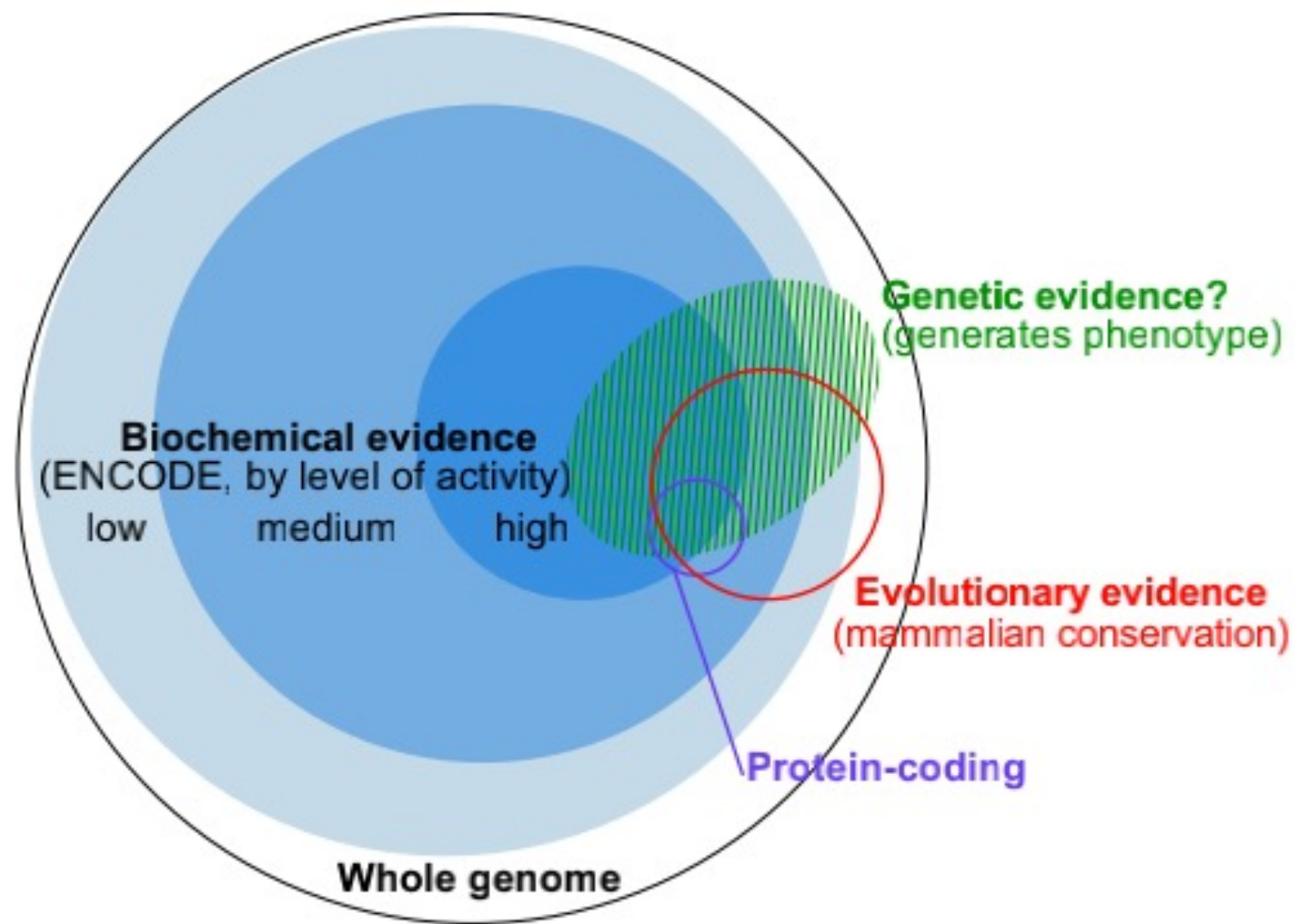
Chapter 6

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Defining functional DNA elements in the human genome



Approach for defining functional DNA elements

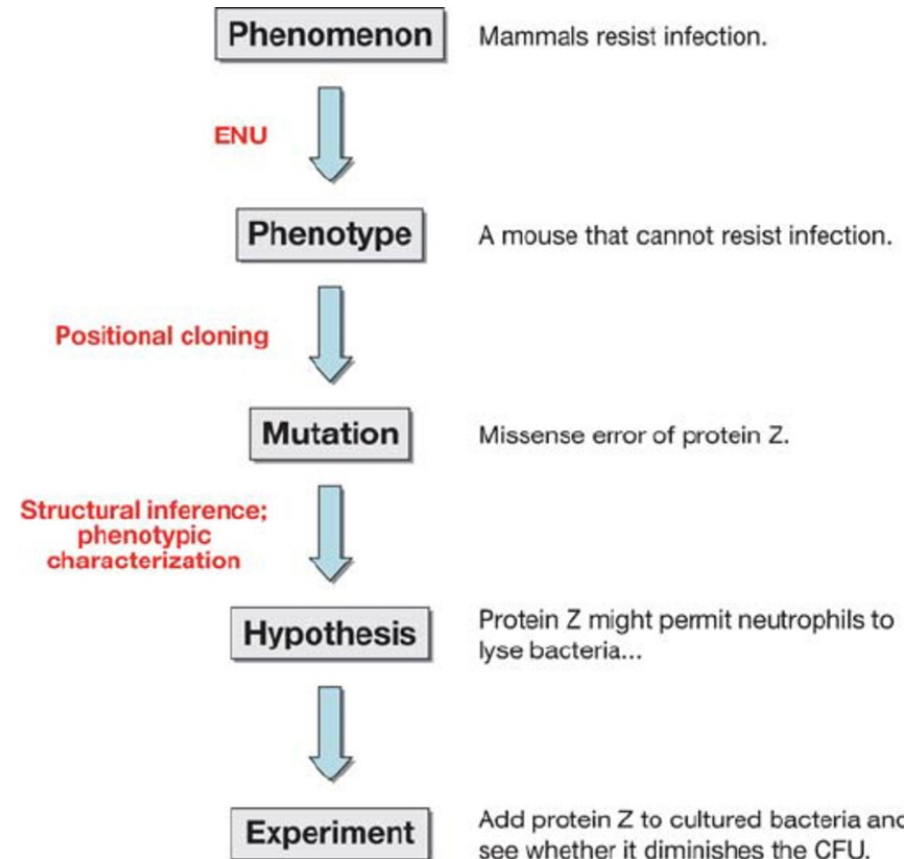
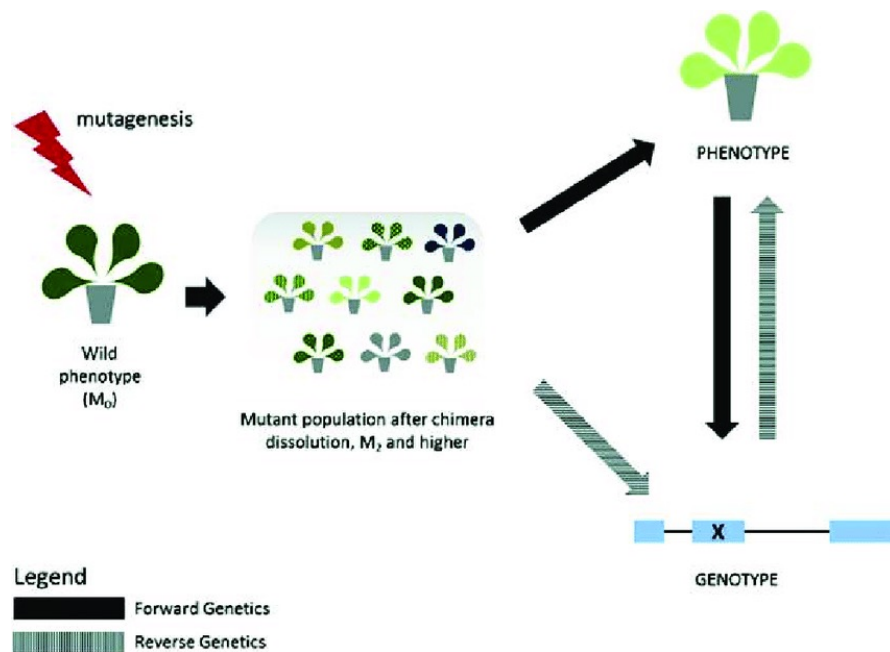
The genetic approach evaluates the phenotypic consequences of perturbations.

The evolutionary approach quantifies selective constraint.

The biochemical approach measures evidence of molecular activity.

Genetic Approach

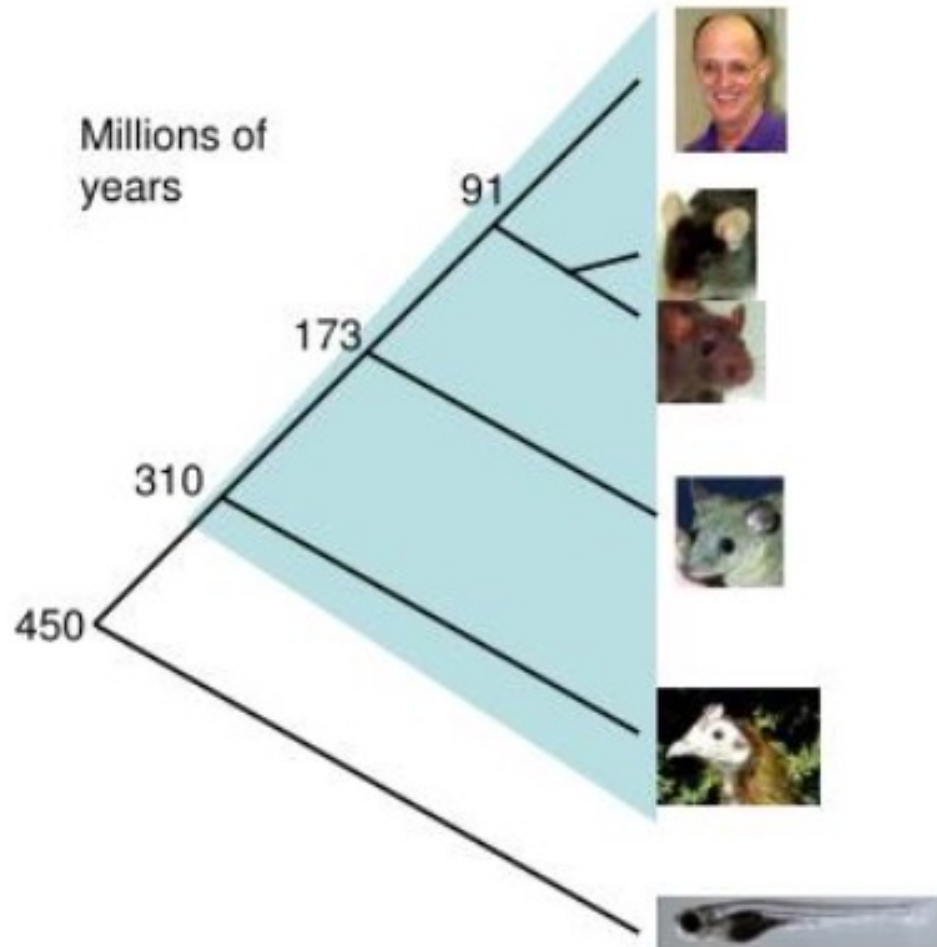
Defining functional DNA elements in the
human genome



Evolutionary Approach

Defining functional DNA elements in the
human genome

A small fraction of *cis*-regulatory modules are conserved from human to chicken

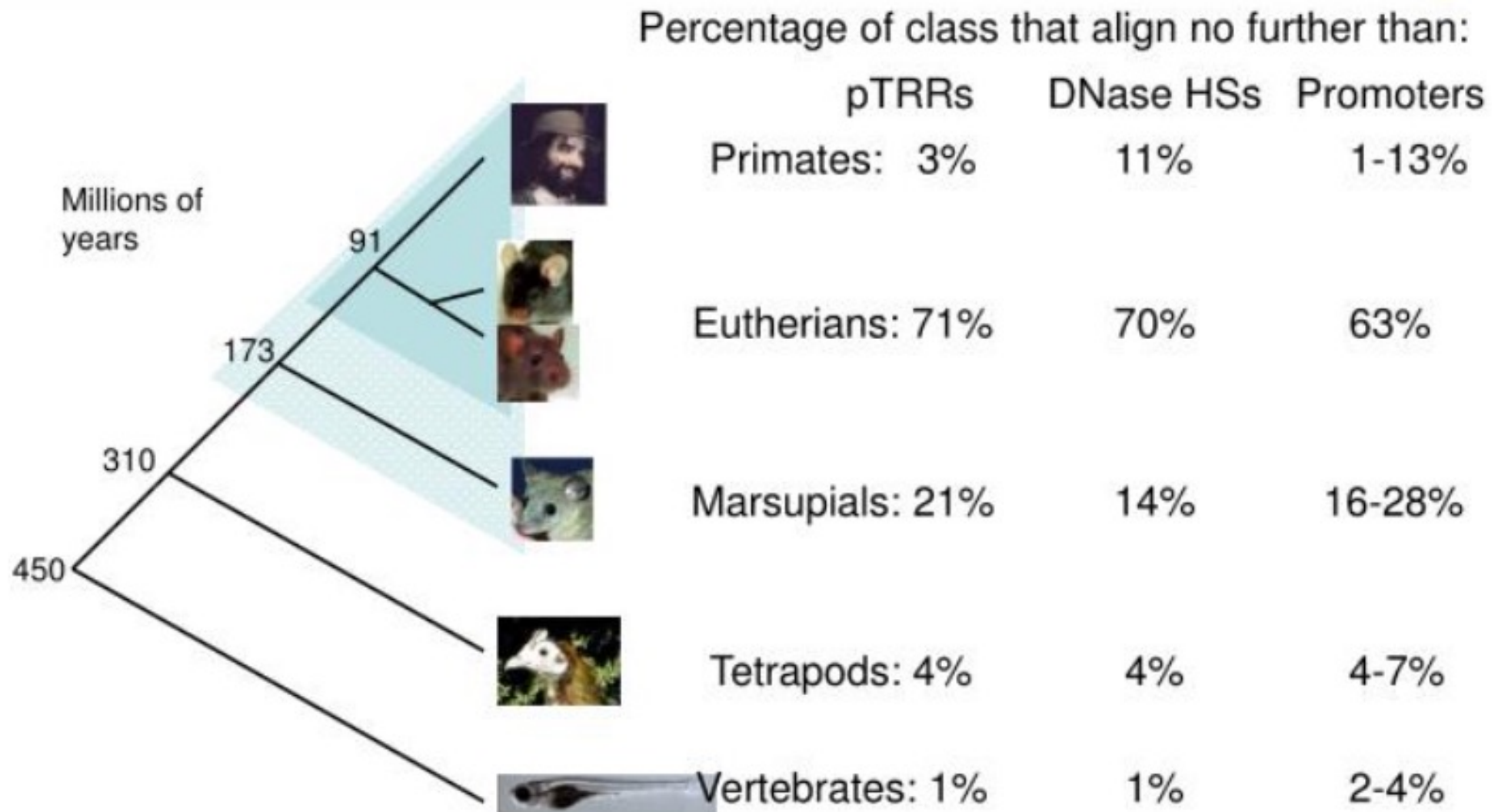


- About 4% of pTRRs, 4% of DNase HSs, 4-7% of promoters active in multiple cell lines
- Tend to regulate genes whose products control transcription and development



David King ¹¹

Most pTRRs are conserved in eutherian mammals



Within aligned noncoding DNA of eutherians, need to distinguish constrained DNA (purifying selection) from neutral DNA.

Regulatory potential (RP) to distinguish functional classes

Regulatory potential (RP) scores exploit a mix of **conservation**, **composition** and **pattern** information.

	seq1	G	T	A	C	C	T	A	C	T	A	C	G	C	A
Alignment	seq2	G	T	G	T	C	G	-	-	A	G	C	C	C	A
	seq3	A	T	G	T	C	A	-	-	A	A	T	G	T	A
Collapsed alphabet		1	2	1	3	4	5	7	7	6	8	3	6	3	9

Train 2 variable order Markov models to estimate the probability that a given alignment column is preceded by other designated alignment columns in

Positive training set: Alignments of known regulatory regions

Negative training set: Alignments of ancestral repeats, a model for neutral DNA

$$RP = \sum_{\substack{a \text{ in} \\ \text{segment}}} \log \left(\frac{p_{\text{reg}}(s_a | s_{a-1} \dots s_{a-t})}{p_{\text{AR}}(s_a | s_{a-1} \dots s_{a-t})} \right)$$

s_a is a symbol from the collapsed alphabet at position a

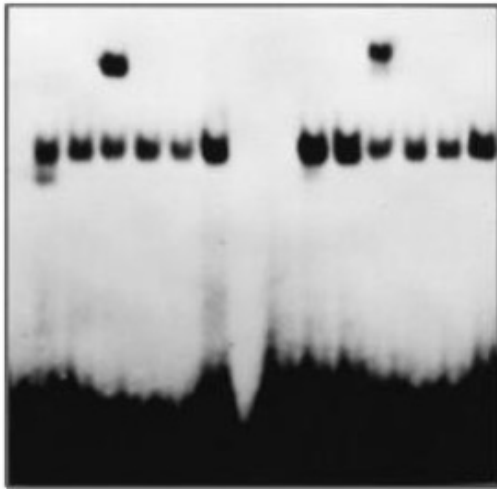
t is the order of the Markov model

reg is regulatory, AR is ancestral repeat

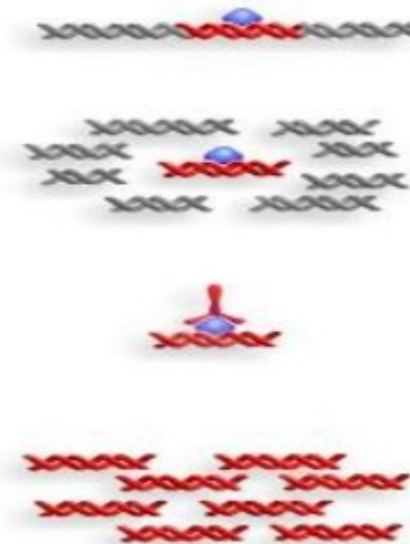
Biochemical Approach

Defining functional DNA elements in the
human genome

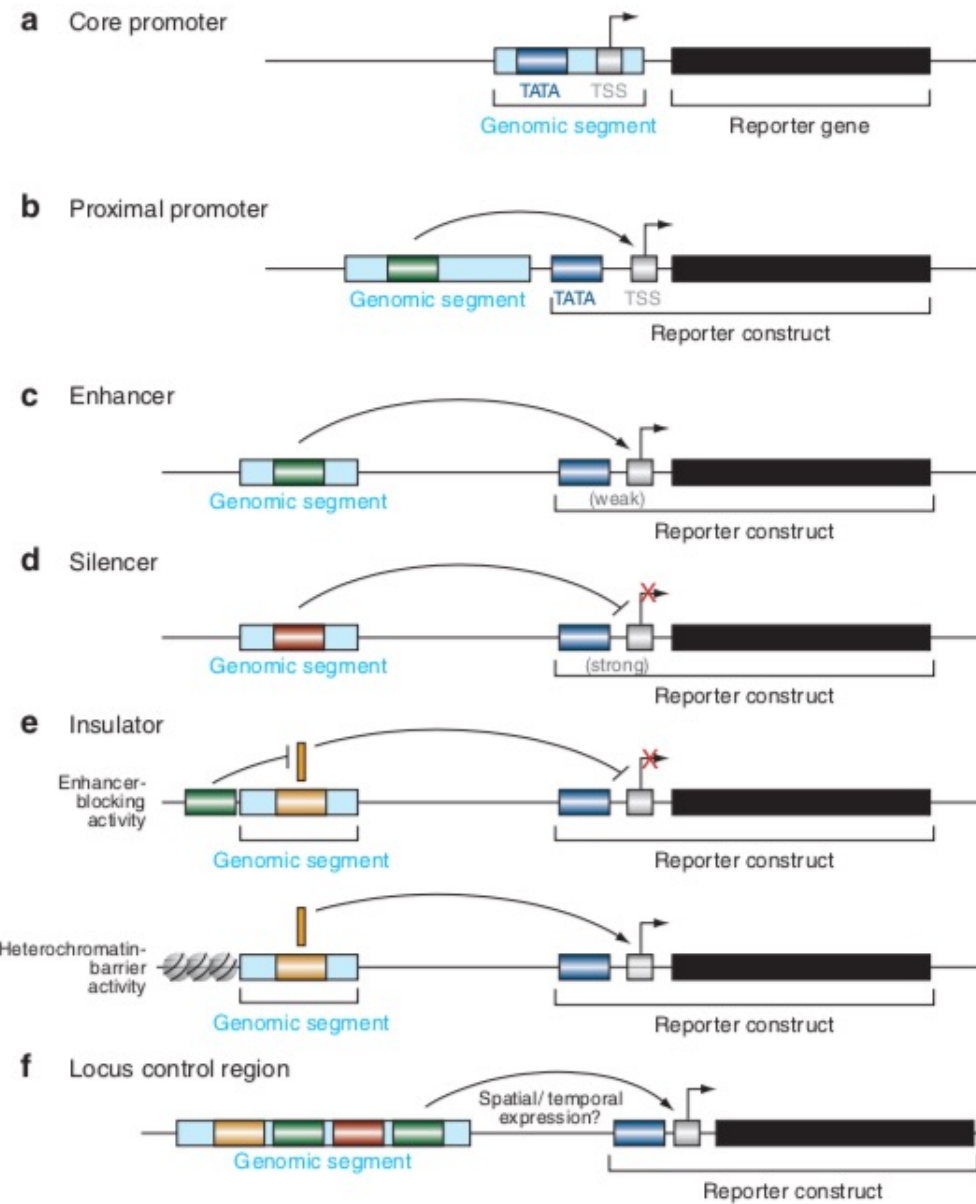
Biochemical assays for protein-binding sites in DNA



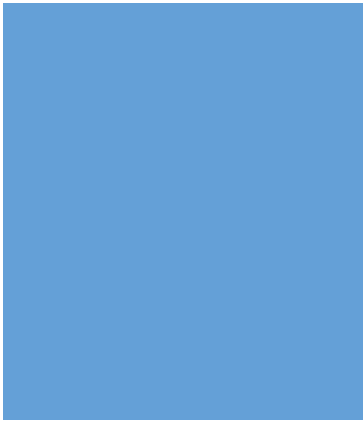
Purified protein
& Naked DNA



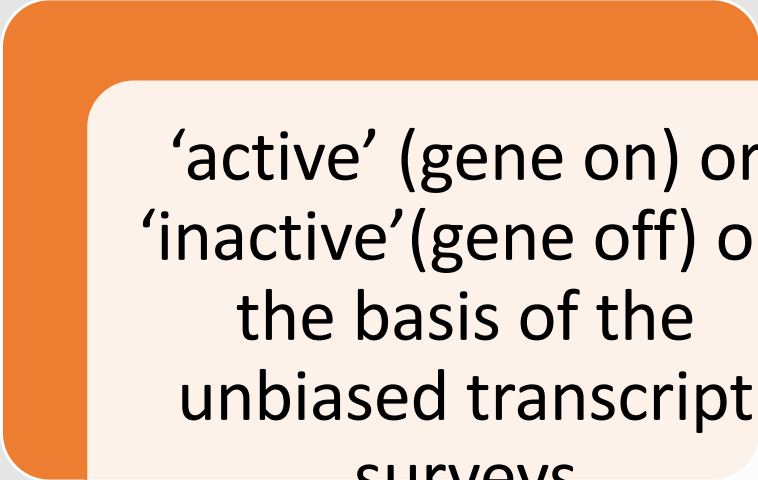
Chromatin Immunoprecipitation:
DNA sites occupied by a protein
inside cells.



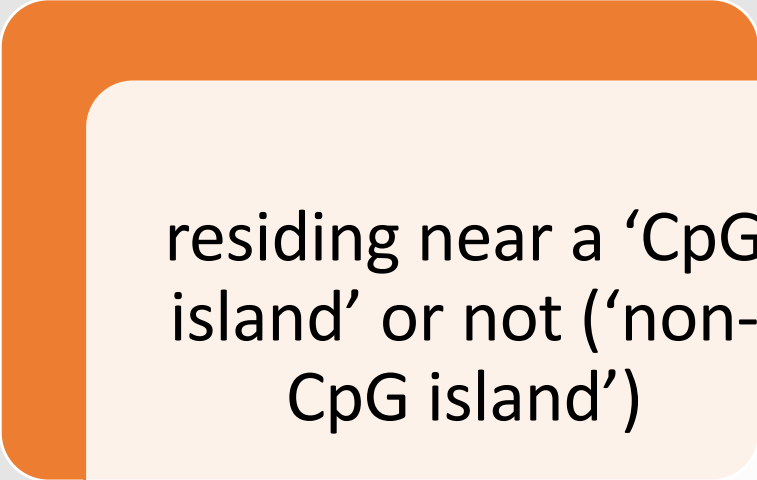
Transcription start site catalogue



Each transcript was
classified as

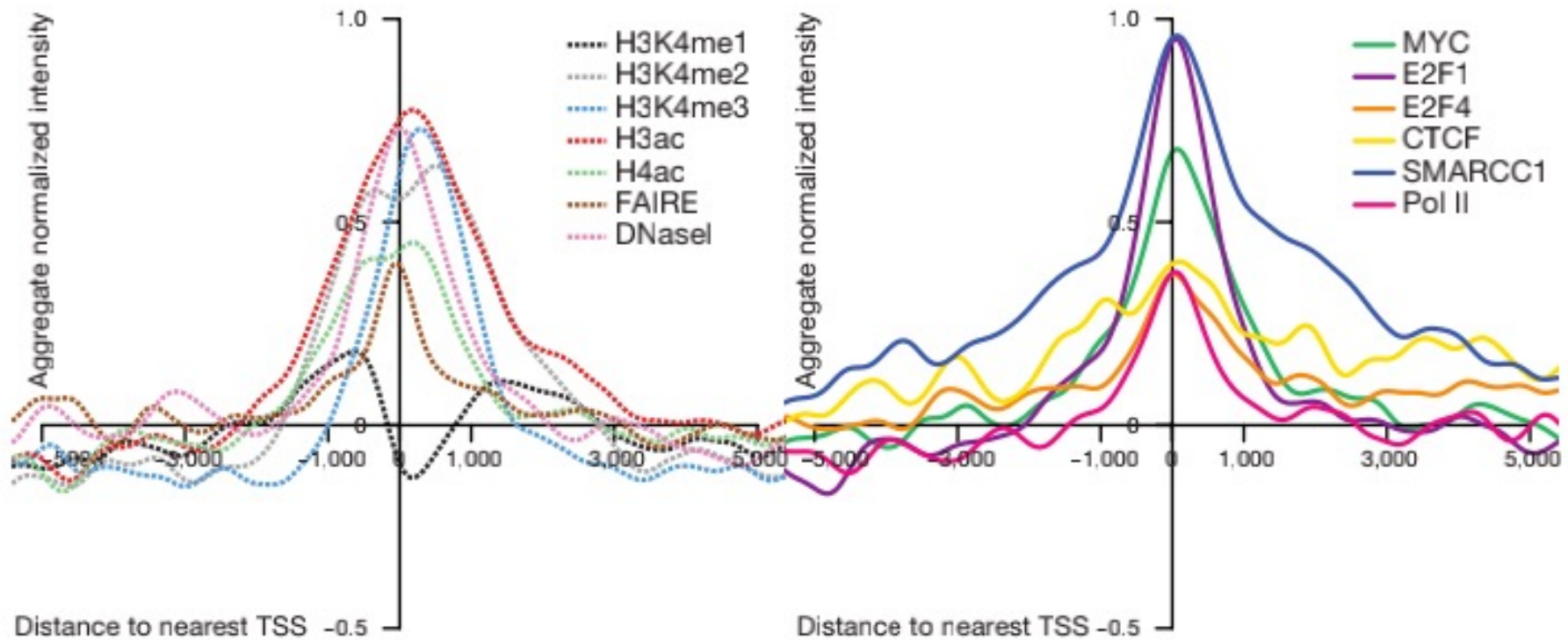


‘active’ (gene on) or
‘inactive’ (gene off) on
the basis of the
unbiased transcript
surveys

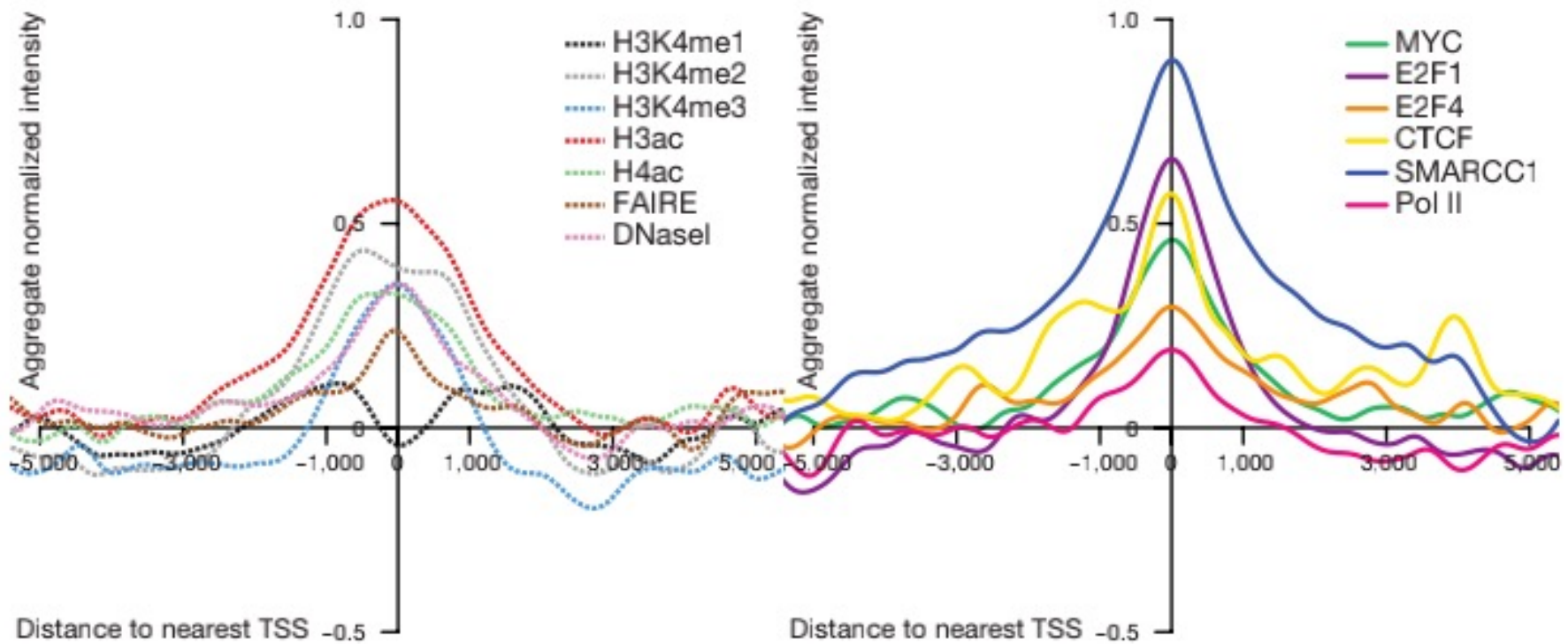


residing near a ‘CpG
island’ or not (‘non-
CpG island’)

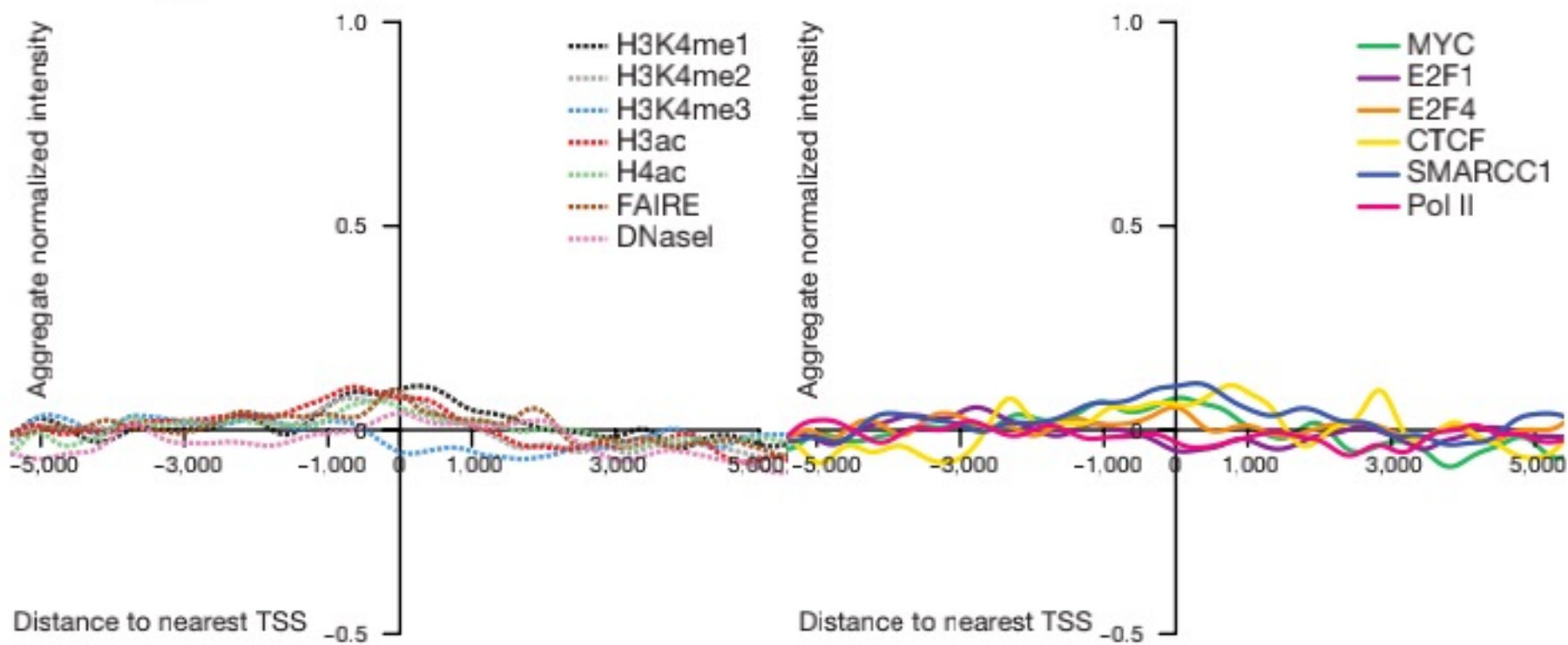
a GENCODE TSS



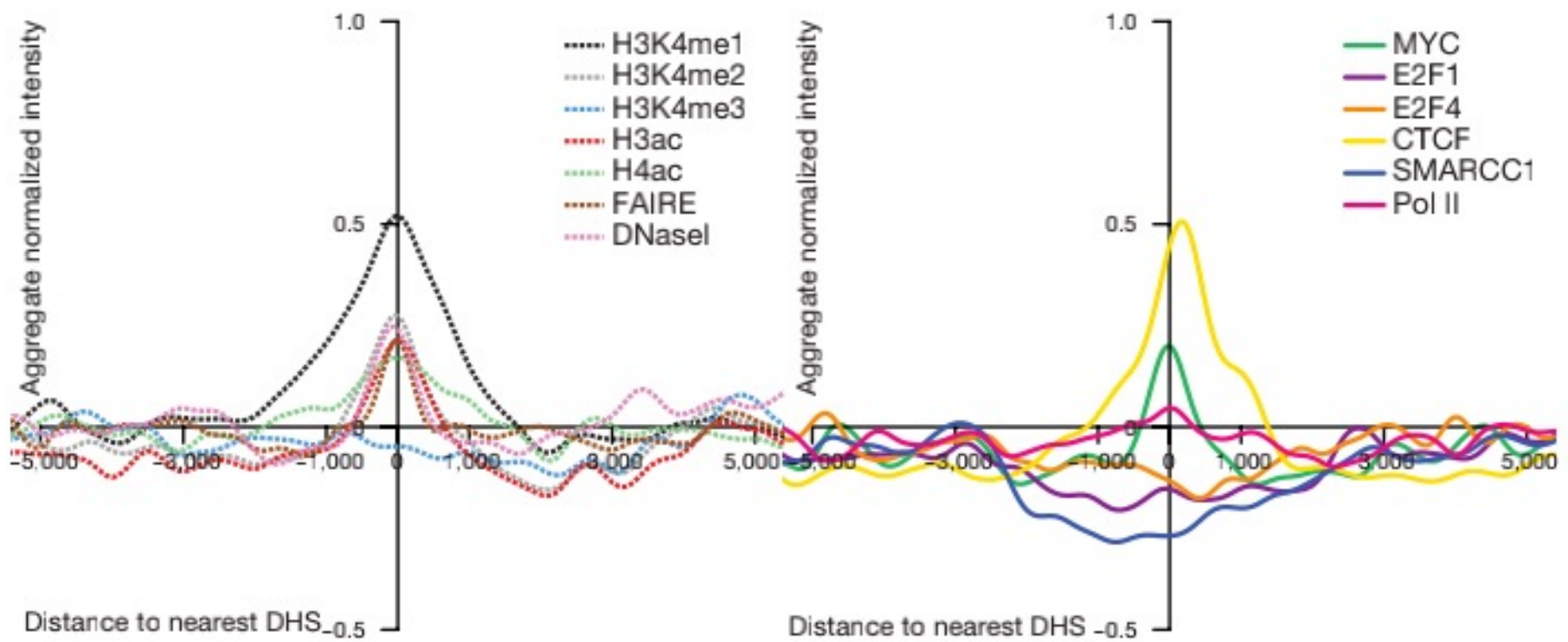
b Novel TSS



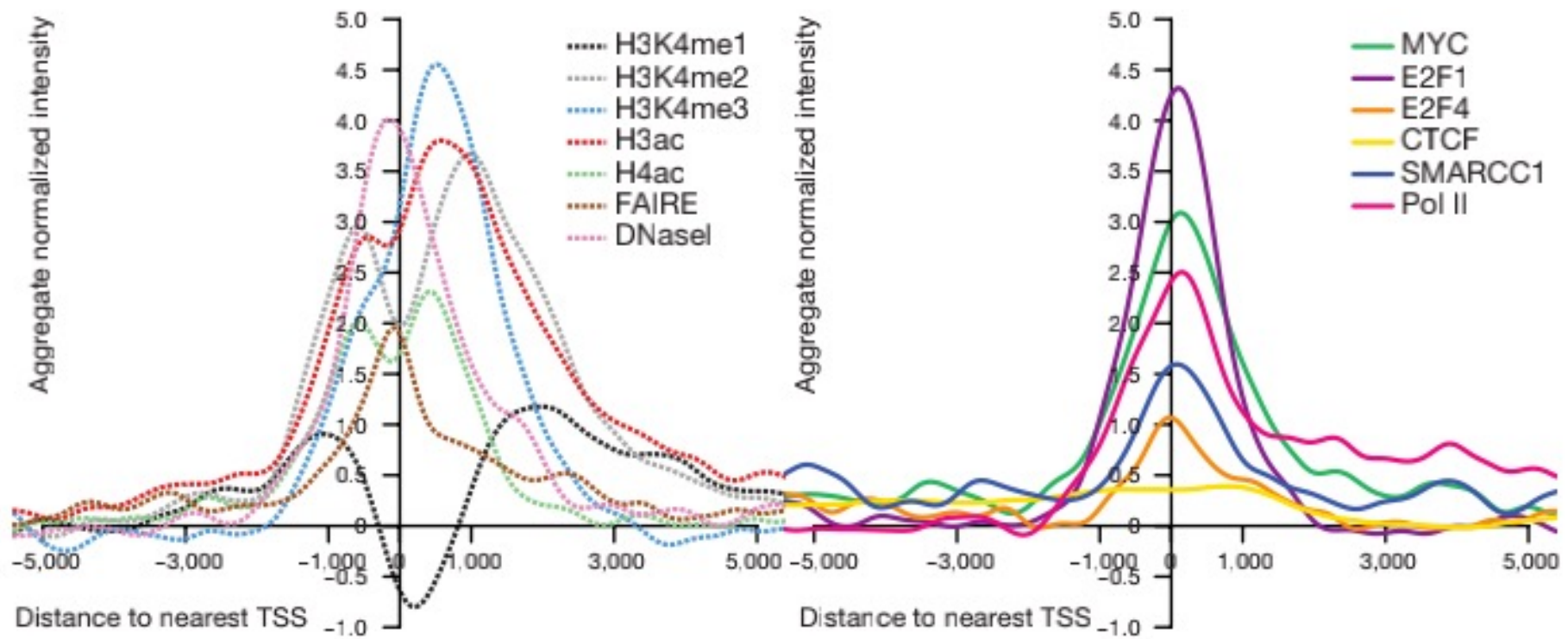
c Unsupported tags



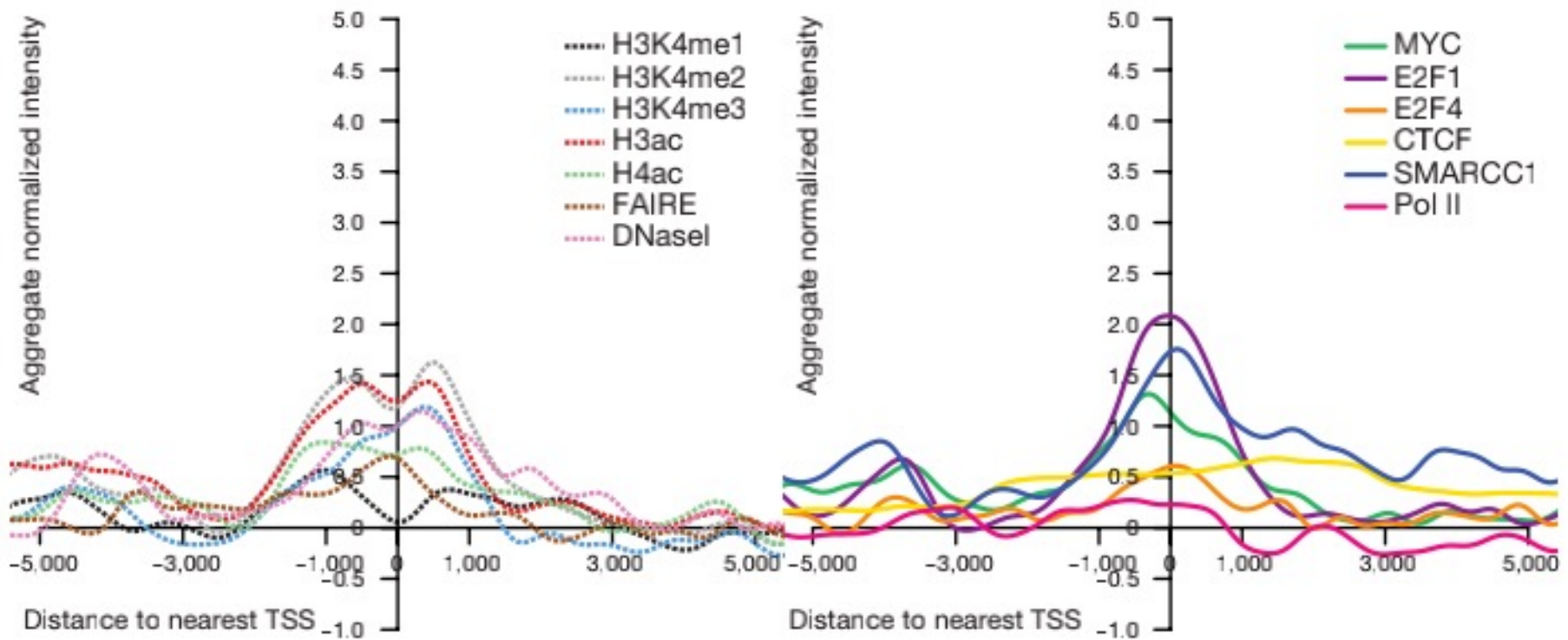
d Distal DHS



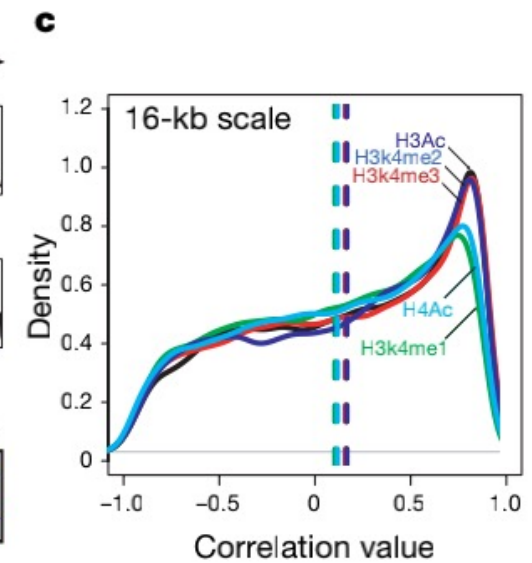
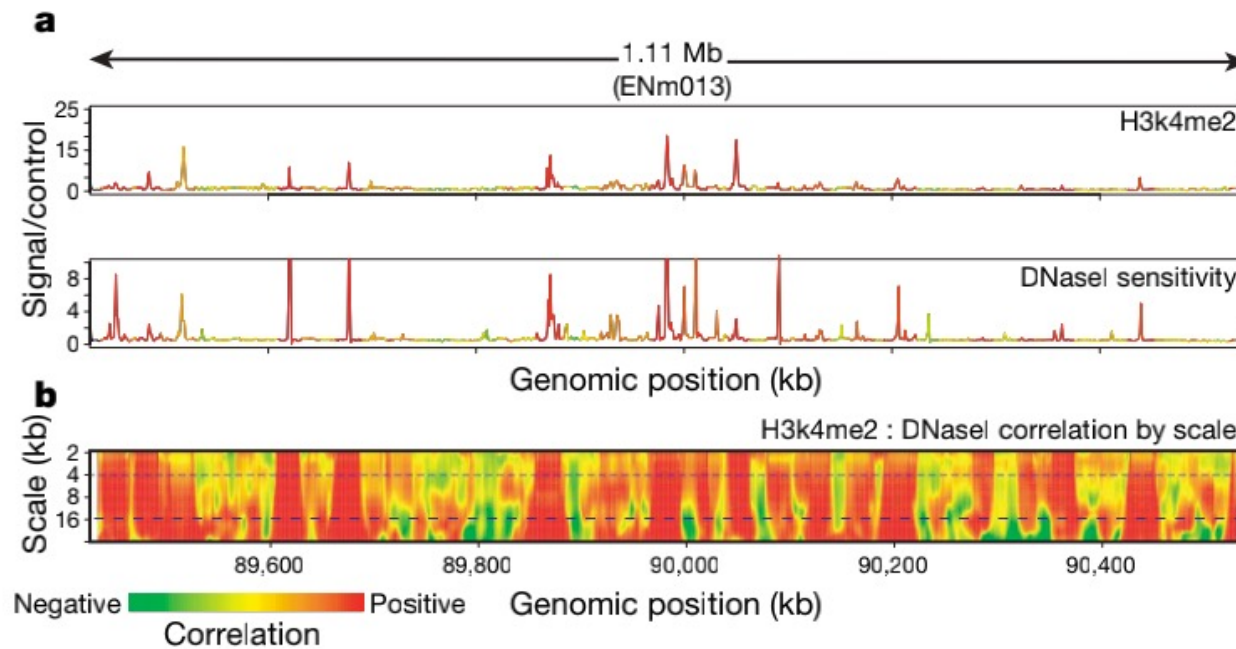
e Gene on CpG

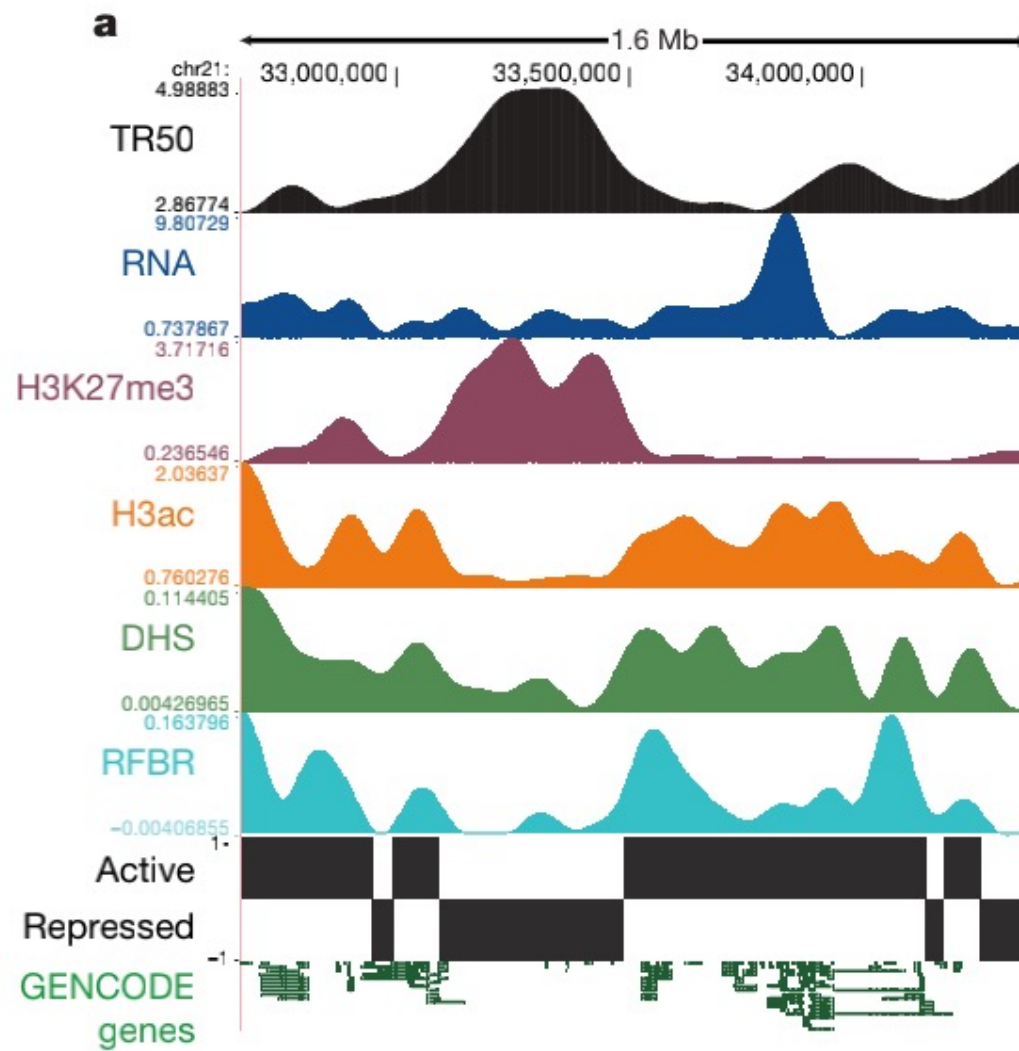


f Gene off CpG



Chromatin architecture and genomic domains





Coding
sequence

Repetitive
element

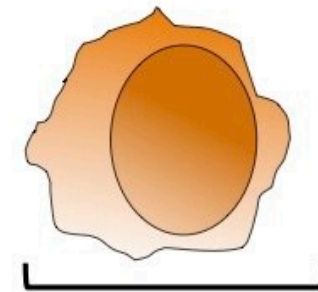
Retrovirus

Regulatory
sequence

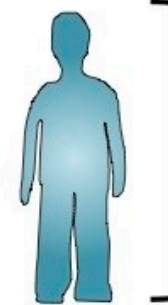
Average gene length:
 5×10^{-7} meters

Condensed
(10,000x) human
chromosome:
 5×10^{-6} meters

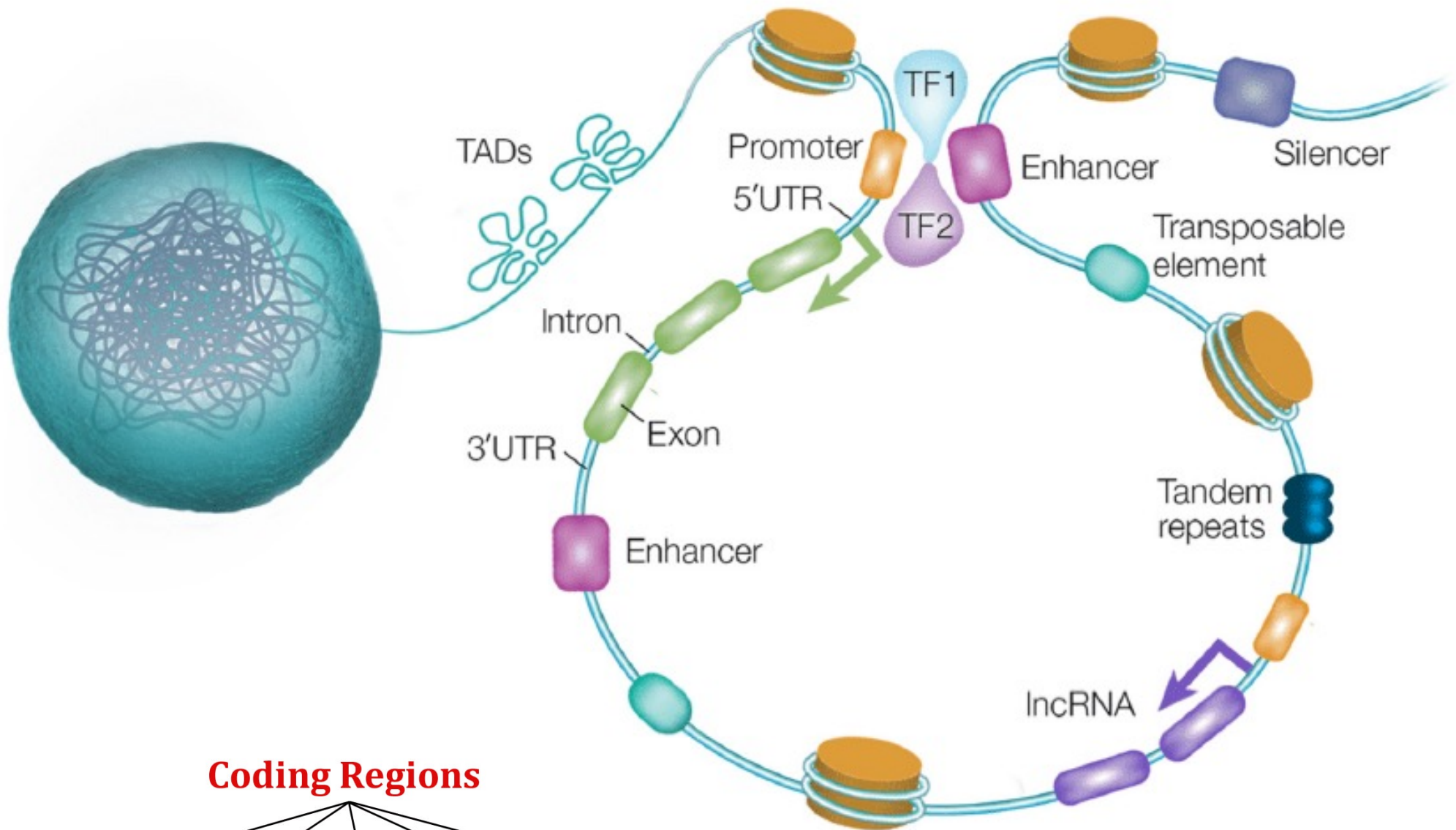
Distance
between bases:
 3×10^{-10} meters



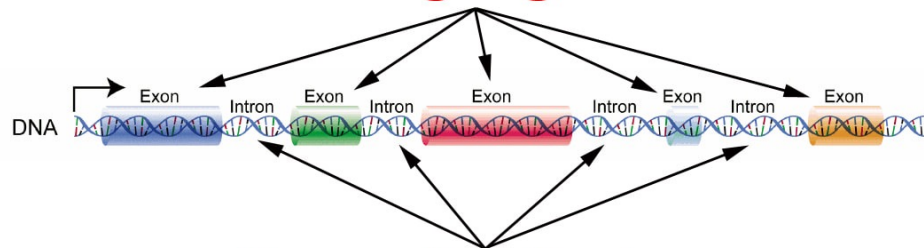
Mammalian cell
diameter:
 2×10^{-5} meters



Human
height:
1.5-2 meters



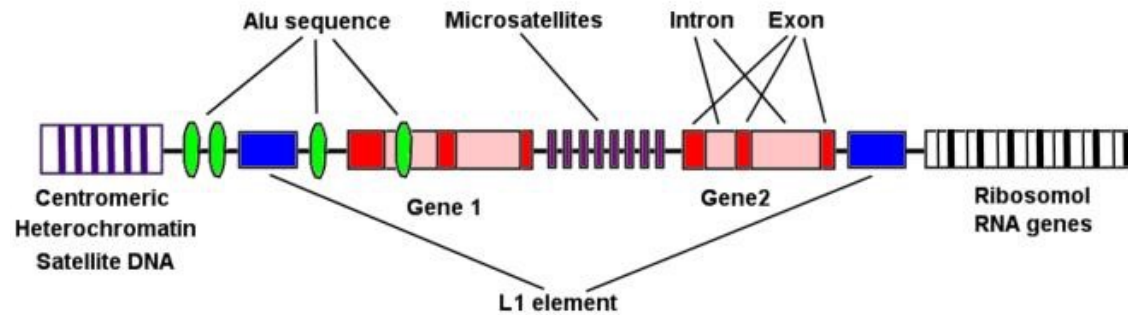
Coding Regions



Non-coding Regions

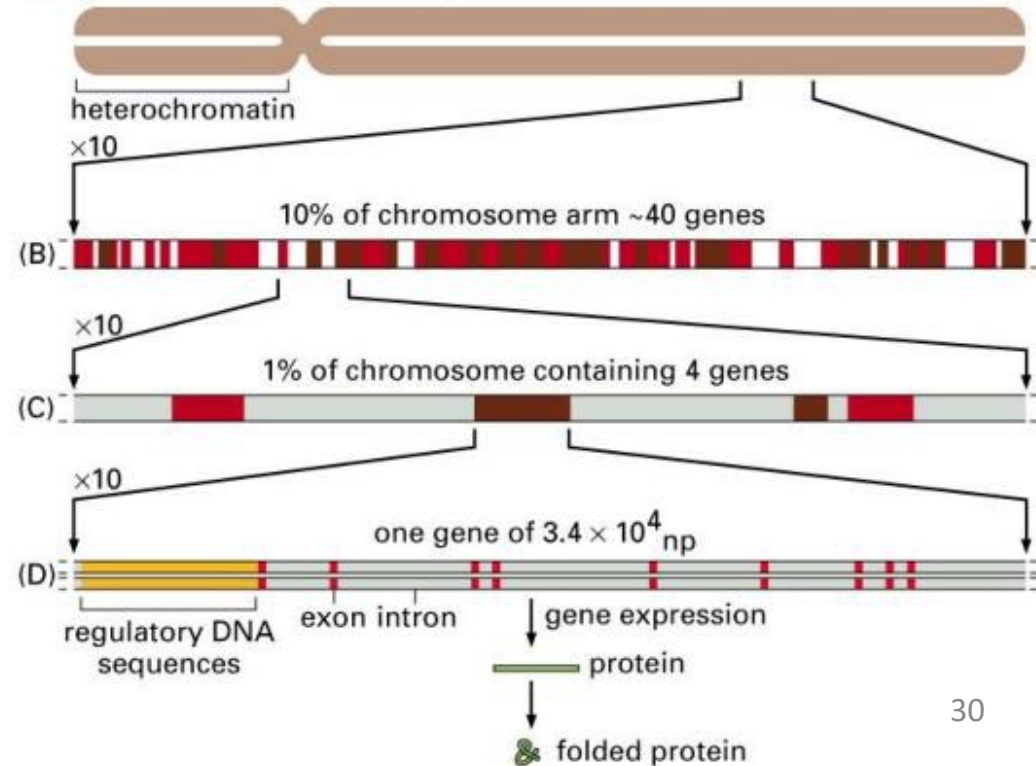
(Containing large TE content)

Schematic diagram of the human genome



Chromosome 22

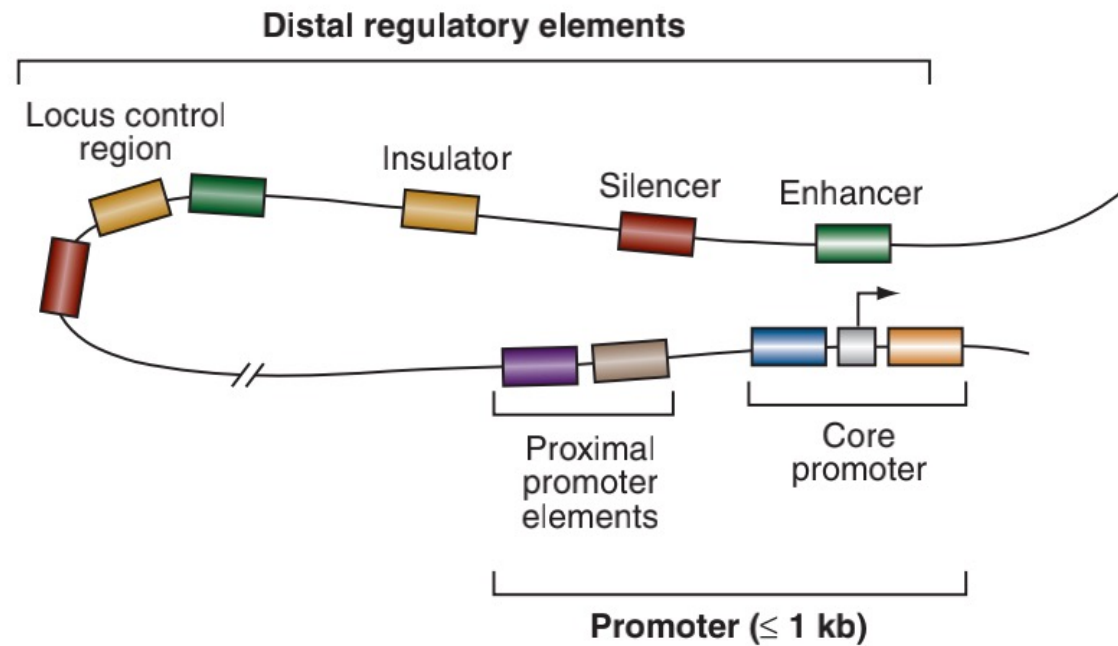
(A) human chromosome 22— 48×10^6 nucleotide pairs of DNA



Regulation of transcription

—promoters, enhancers, silencers, insulators and locus control regions

Transcriptional Regulatory Elements in the Human Genome



Noncoding regulatory sequences

nucleotide
changes

Differences in gene expression

Human variation

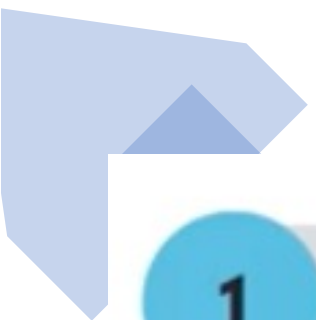


Pharmacogenomics



Human disease





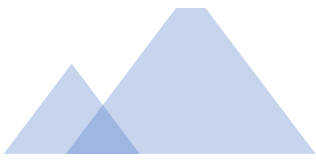
1 Coding and Non-coding DNA

2 Structural DNA

- Telomeres
- Centromeres

3 Functional RNA

4 Introns

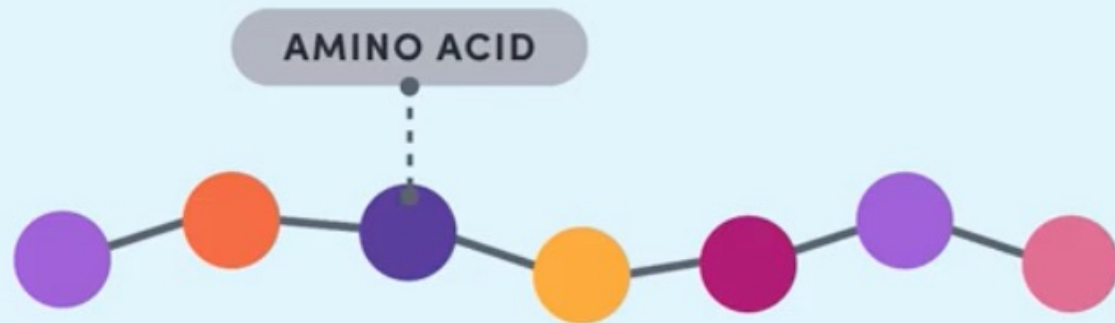


Coding and Non-coding DNA

GENE

A **section of DNA** which contains the **instructions** for making an **RNA molecule or a protein**

The **order of amino acids** in each protein follows a **sequence** which is coded for by its **gene**



Coding and Non-coding DNA

The **order of amino acids** in each protein follows a **sequence** which is coded for by its **gene**

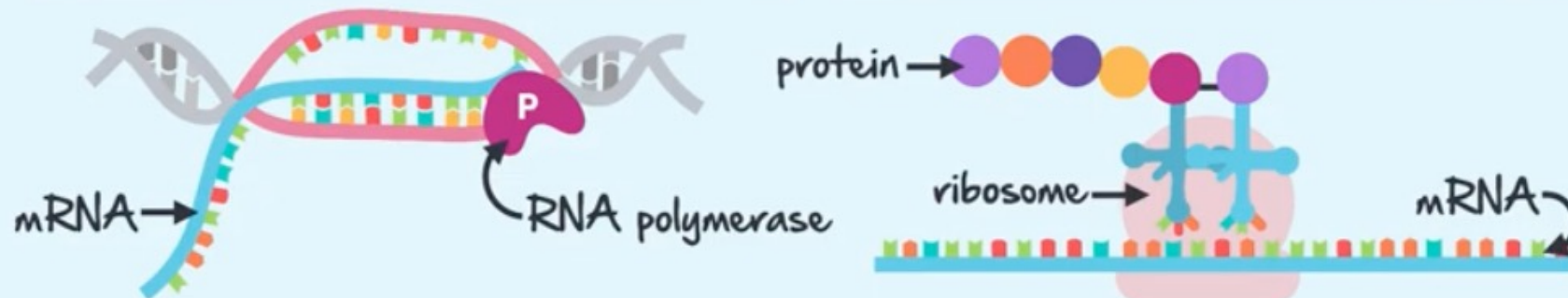


TRANSCRIPTION

Involves the production of **RNA** from a **gene**

TRANSLATION

Involves the production of **proteins** from **amino acids** using **mRNA**



Coding and Non-coding DNA

TRANSCRIPTION

Involves the production of **RNA** from a **gene**



TRANSLATION

Involves the production of **proteins** from **amino acids** using **mRNA**



CODING DNA

DNA that is **transcribed** and **translated**

NON-CODING DNA

- 1 **Structural DNA**
↳ not transcribed
- 2 **Functional RNA**
↳ transcribed, but not translated
- 3 **Introns**
↳ transcribed, but RNA is removed before translation

Coding and Non-coding DNA

CODING DNA

DNA that is **transcribed and translated**

NON-CODING DNA

DNA which **does not** get translated into mRNA, and so **does not code for proteins**

→ the role of lots of this non-coding DNA is unknown!

TELOMERES

CENTROMERES

1

2

3

4

Structural DNA

Structural DNA helps to **maintain** the structure of **chromosomes** in **eukaryotes**



CHROMOSOME

TELOMERES

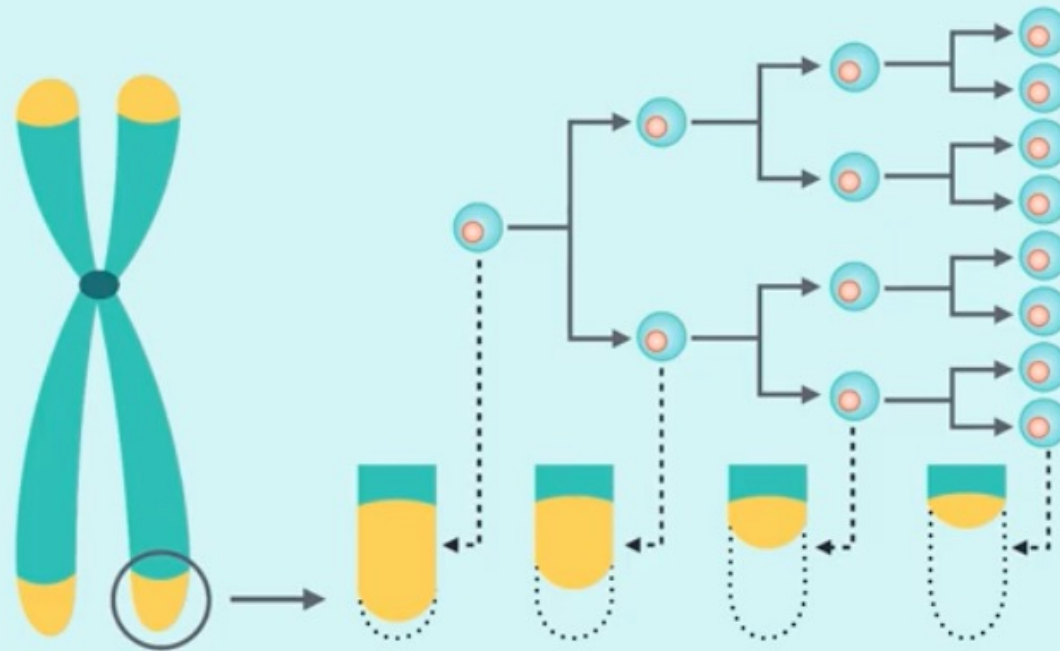
CENTROMERES

1 2 3 4

Structural DNA

Telomeres are the regions of **structural DNA** found at the **ends of chromosomes**

→ act as protective caps for the chromosomes



Each time a **eukaryotic cell divides**, the **DNA** at the ends of each chromosome is **shortened slightly**

Structural DNA

Telomeres are the regions of **structural DNA** found at the **ends of chromosomes**

↳ act as protective caps for the chromosomes



8000 bases long



1500 bases long

TTAGGGTTAGGGTTAGGGTTAGGG
AATCCCAATCCCAATCCCAATCCC

Each time a **eukaryotic cell divides**, the **DNA** at the ends of each chromosome is **shortened slightly**

Non-coding telomeres at the end of chromosomes **prevent** the **coding DNA** from **being lost** in cell division

Over a human's life, their cells undergo more and **more divisions**, and the **telomeres shorten**

TELOMERES

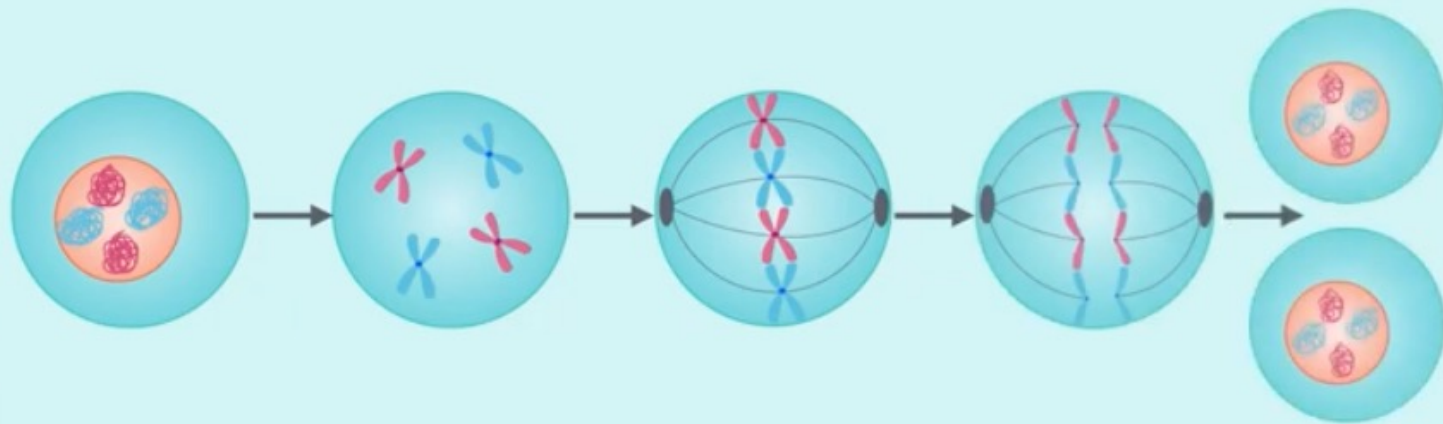
CENTROMERES

1 2 3 4

Structural DNA

The **centromere** is the part of the chromosome which **attaches** to the **spindle fibres** during cell division

↳ allows chromosome to be pulled to opposite ends of cell



TELOMERES

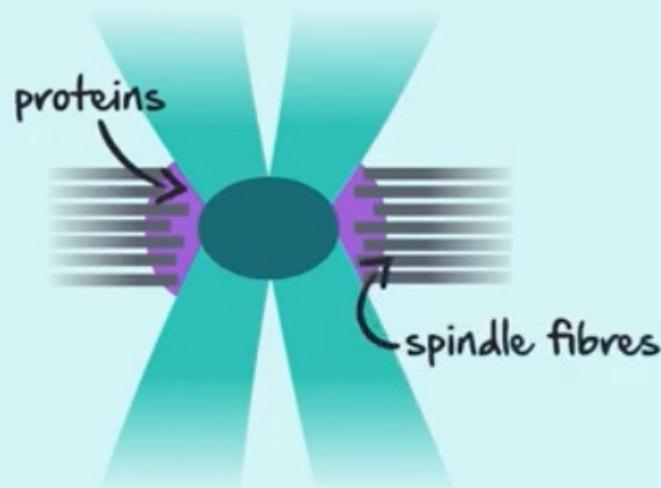
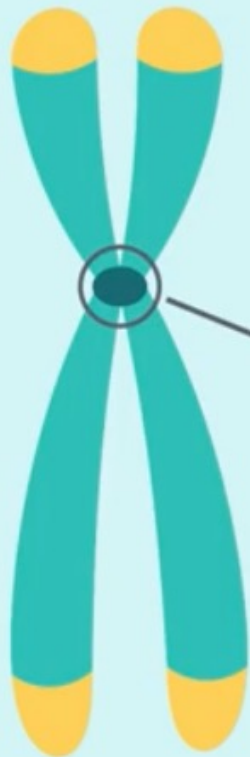
CENTROMERES

1 2 3 4

Structural DNA

The **centromere** is the part of the chromosome which **attaches** to the **spindle fibres** during cell division

→ allows chromosome to be pulled to opposite ends of cell



Centromeres consist of **repeats** of a **simple sequence**

Repeated units allow **proteins** to **bind** to the **DNA**

Proteins bind to the **spindle fibres**

TELOMERES

CENTROMERES

1 2 3 4

Structural DNA

Structural DNA helps to **maintain** the structure of **chromosomes** in **eukaryotes**

Telomeres are the regions of **structural DNA** found at the **ends of chromosomes**

The **centromere** is the part of the chromosome which **attaches** to the **spindle fibres** during cell division



- Both made from **repetitive** base sequences

Functional RNA

Some non-coding DNA is **transcribed**, and made into **RNA**

Functional RNA includes any type of RNA which **isn't messenger RNA**

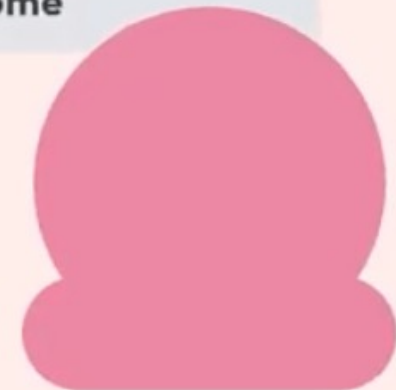


tRNA

Used to carry **amino acids** to **ribosomes** during **protein synthesis**

rRNA

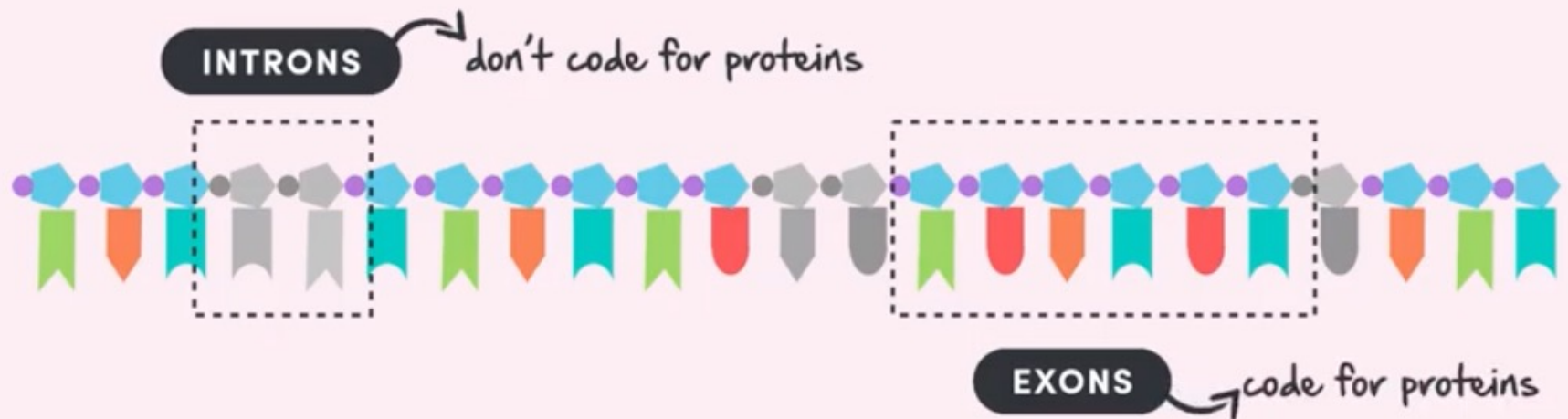
Plays an important role in the **ribosome**



Introns

Genes can contain **regions** of **non-coding DNA**

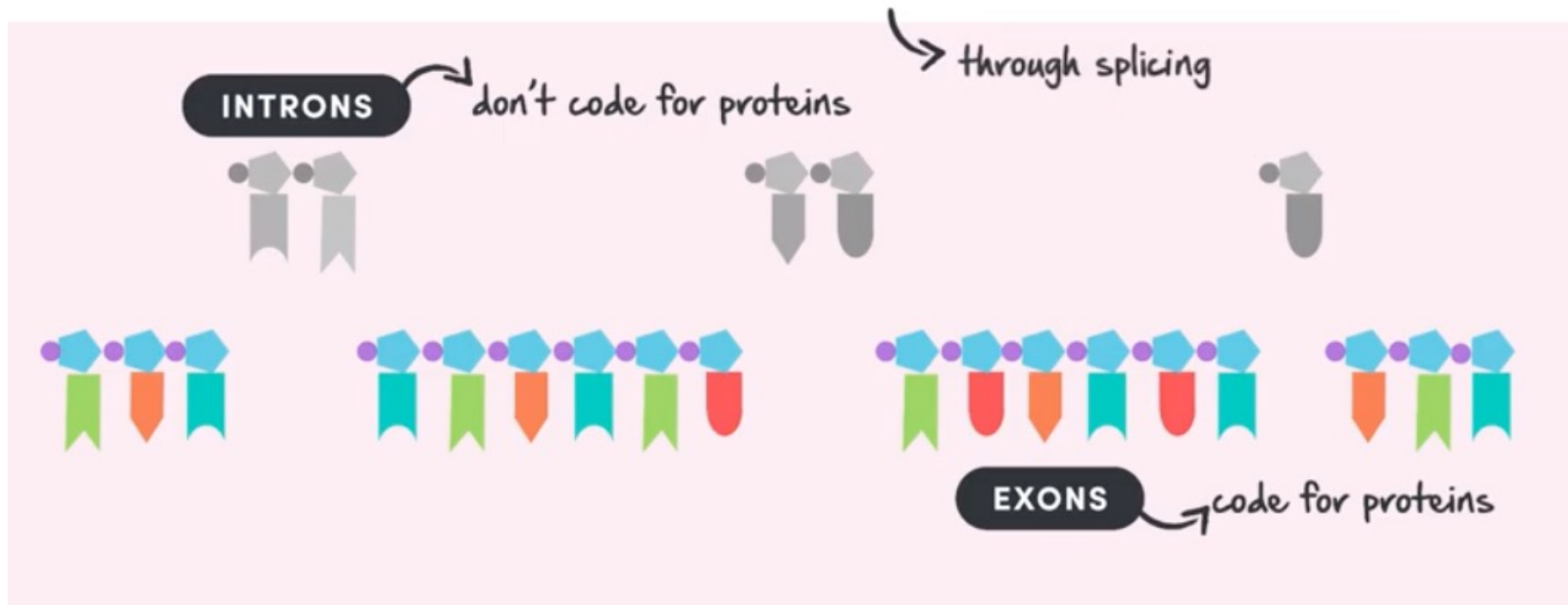
These sections of DNA are **transcribed** but **removed** from mRNA **before translation**



Introns

Genes can contain **regions** of **non-coding DNA**

These sections of DNA are **transcribed** but **removed** from mRNA **before translation**

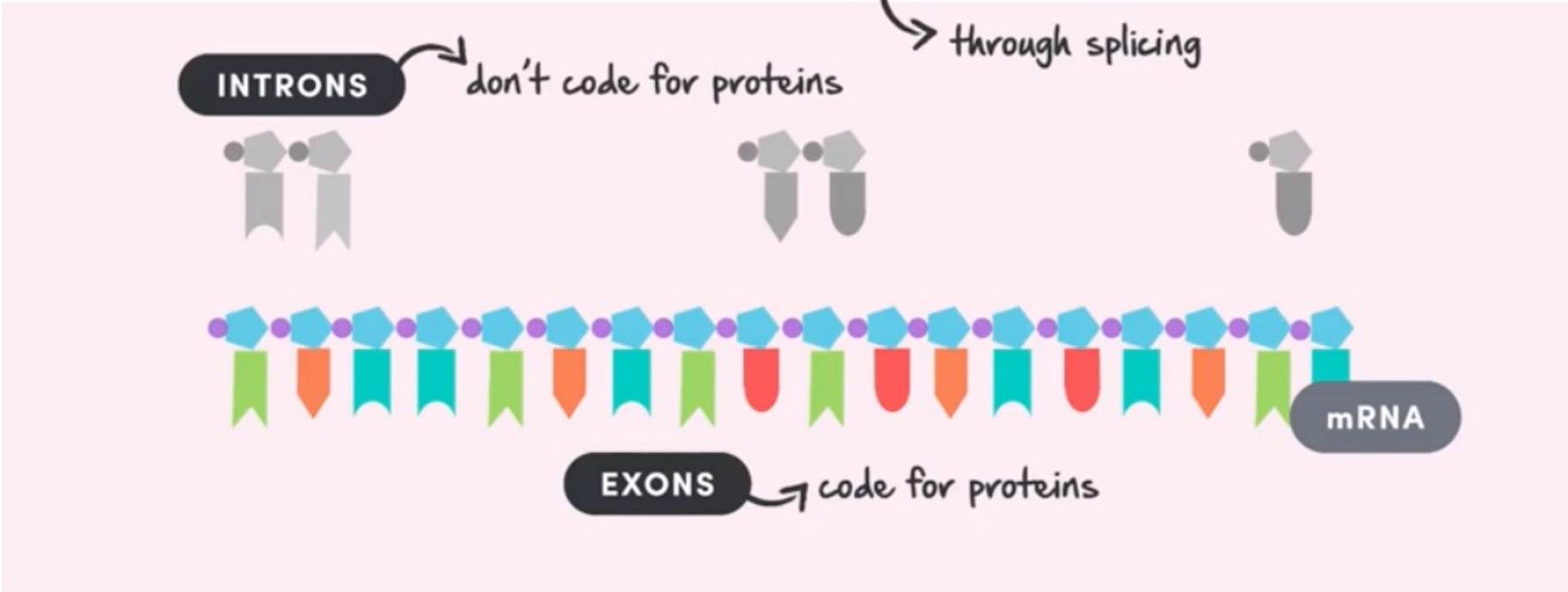


1 2 3 4

Introns

Genes can contain **regions** of **non-coding DNA**

These sections of DNA are **transcribed** but **removed** from mRNA **before translation**



Coding and Non-coding DNA

CODING DNA

Coding DNA is DNA that is **transcribed** and **translated** into **proteins**

NON-CODING DNA

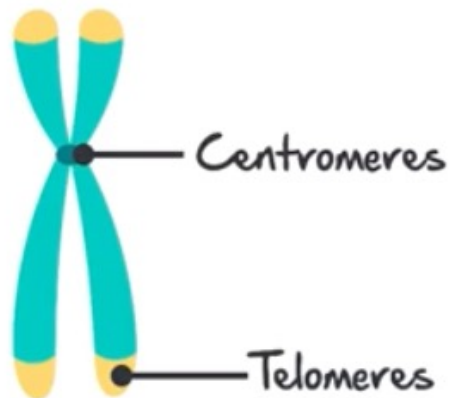
Non-coding DNA is DNA that **does not** **code** for **proteins**

Coding and Non-coding DNA

Non-coding DNA is DNA that **does not code** for proteins

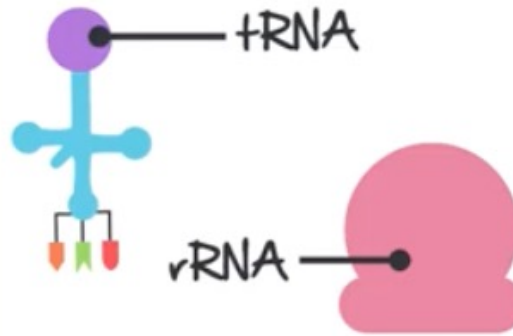
STRUCTURAL DNA

Structural DNA helps to **maintain** the structure of **chromosomes** in **eukaryotes**



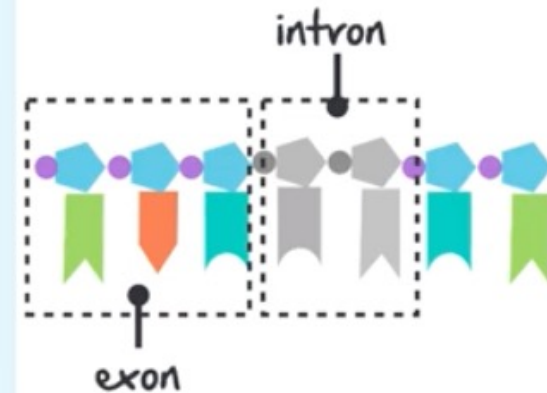
FUNCTIONAL RNA

Functional RNA includes any type of RNA which **isn't** messenger RNA



INTRONS

Introns are regions of RNA **spliced out** before RNA is translated

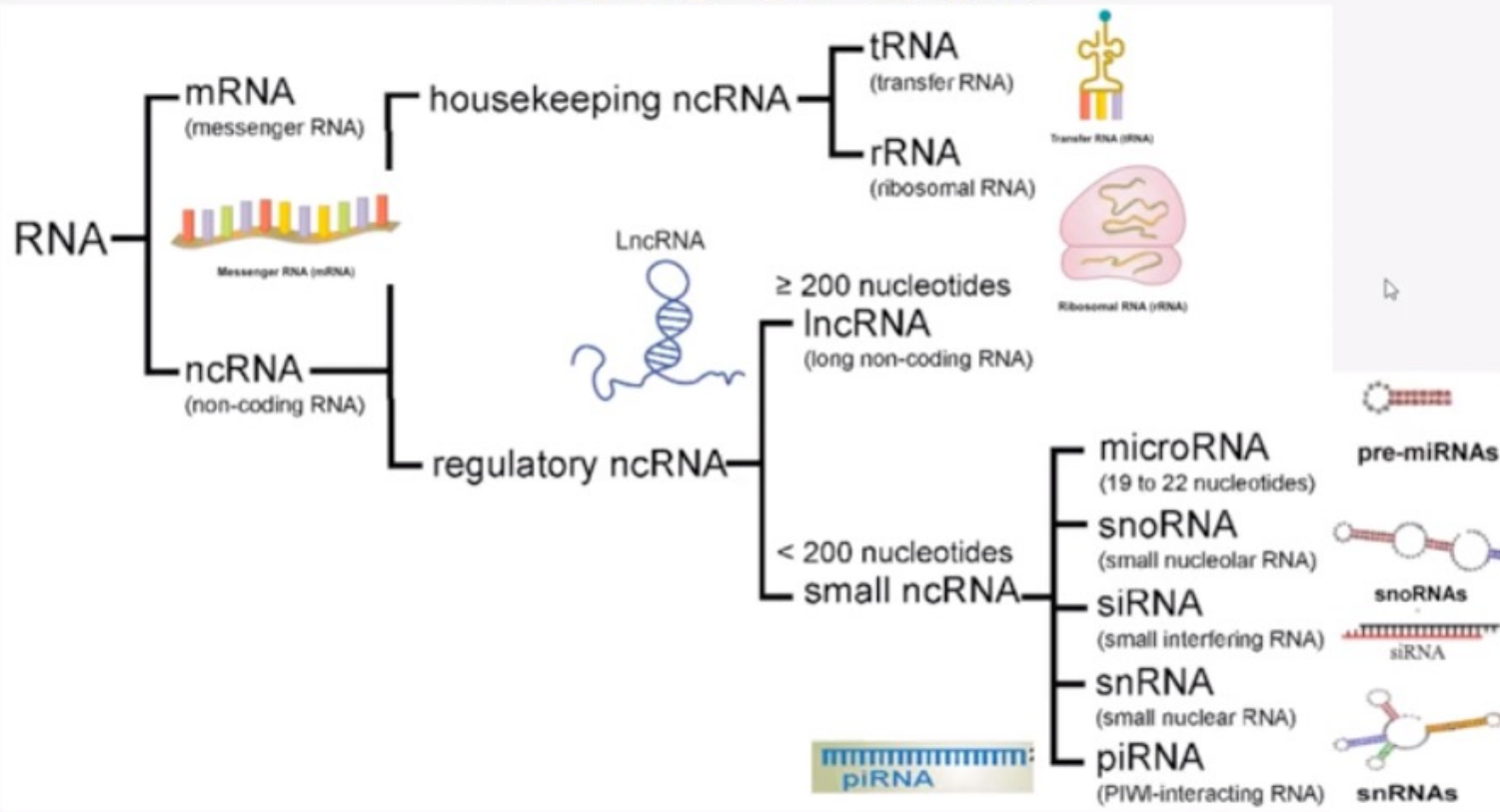


Noncoding region

noncoding RNA genes

microRNAs, piRNAs, structural RNAs, and
regulatory RNAs

TYPES OF RNA

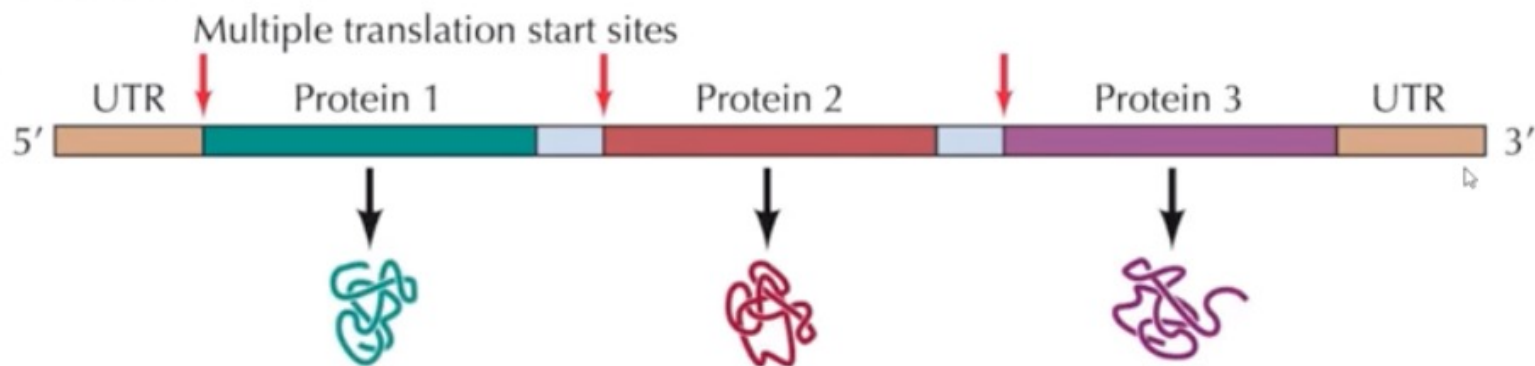


MESSANGER RNA

5% of the
total RNA in
the cell



Prokaryotic mRNA



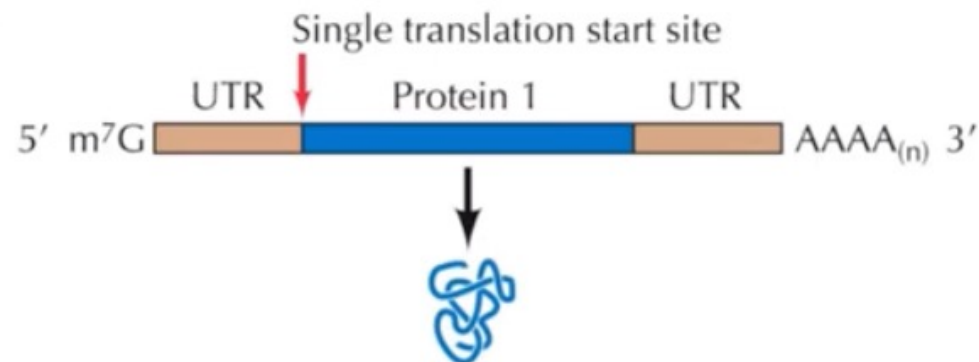
Polycistronic mRNA

MESSANGER RNA

5% of the
total RNA in
the cell



Eukaryotic mRNA



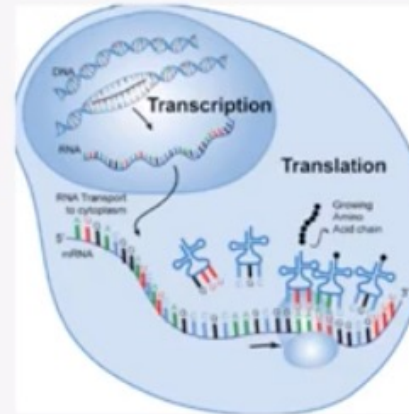
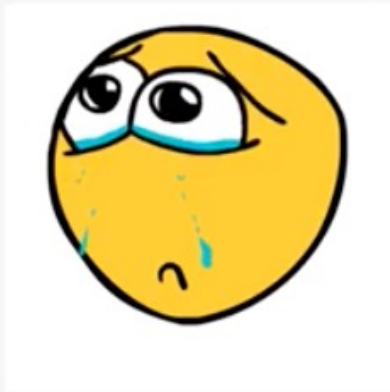
Monocistronic mRNA

MESSANGER RNA

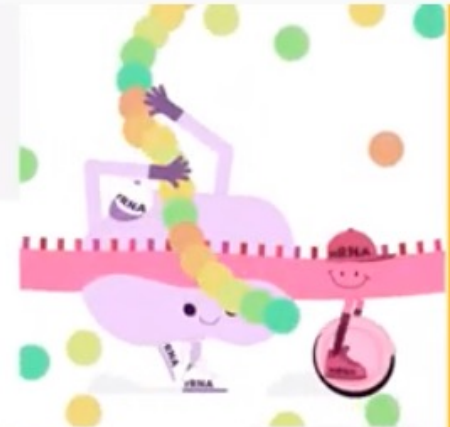
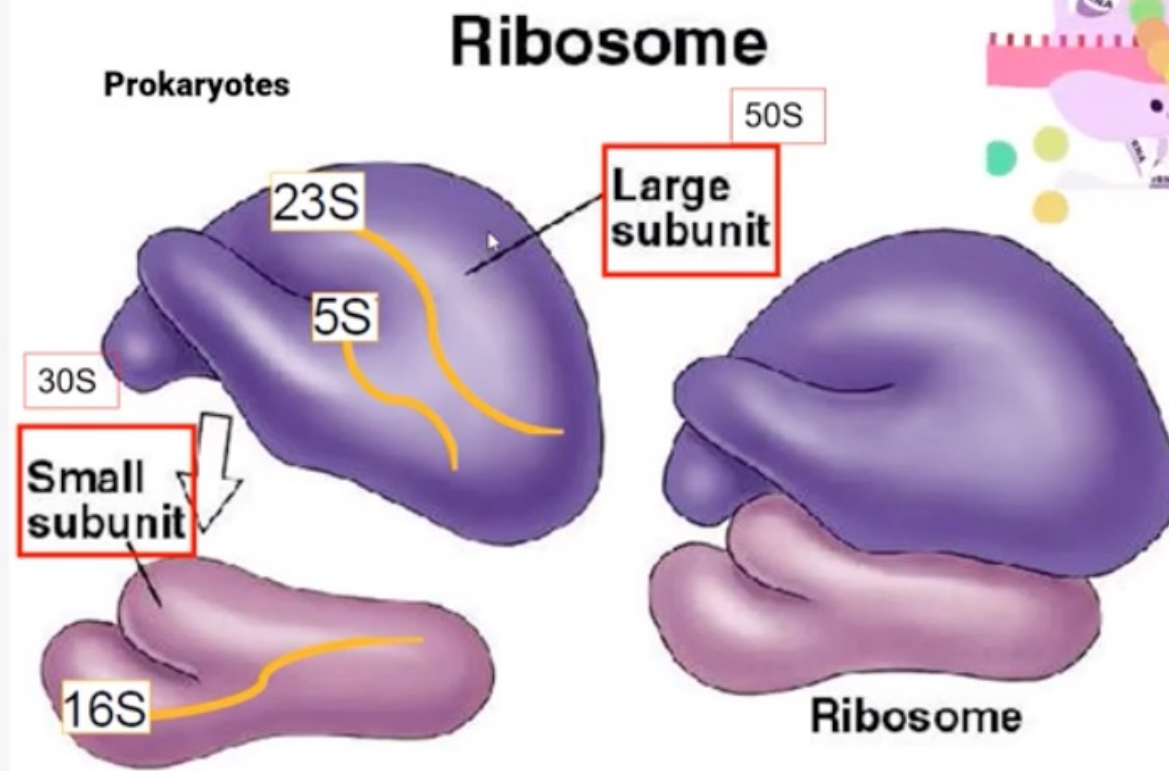
5% of the
total RNA in
the cell



- Recognition of mRNA in protein biosynthesis
- Prevent from 5'-exonucleases activities
- Prevent from 3'-exonucleases activities
- Helps in exporting mRNA from nucleus to cytosol



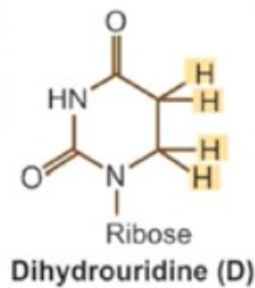
RIBOSOMAL RNA



TRANSFER RNA

4 Main Arms

15% of the total RNA in the cell

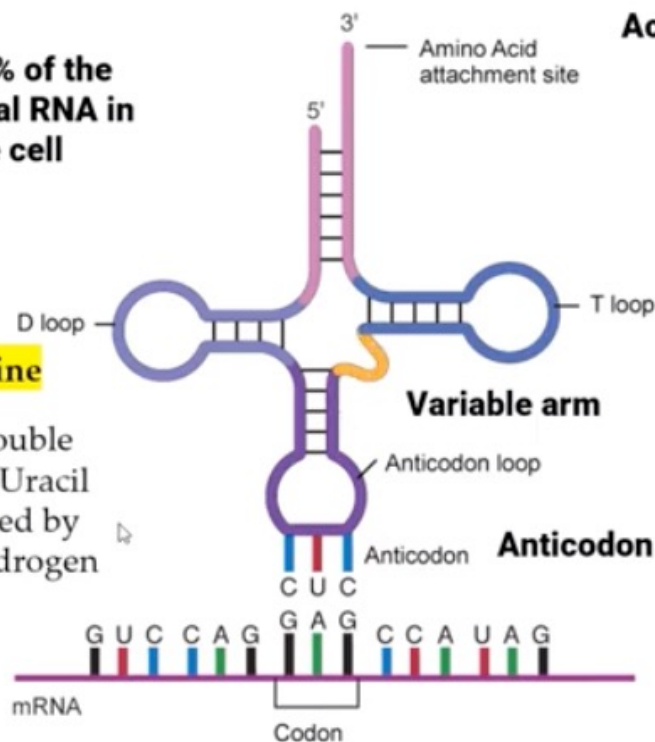


D Arm

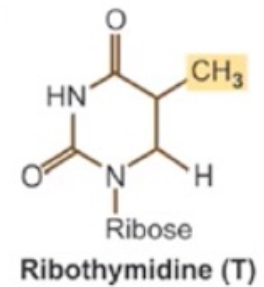
Dihydrouridine

One of the double bonds of the Uracil base is reduced by adding 2 Hydrogen atoms

tRNA



Acceptor Arm



T Arm

Ribothymidine

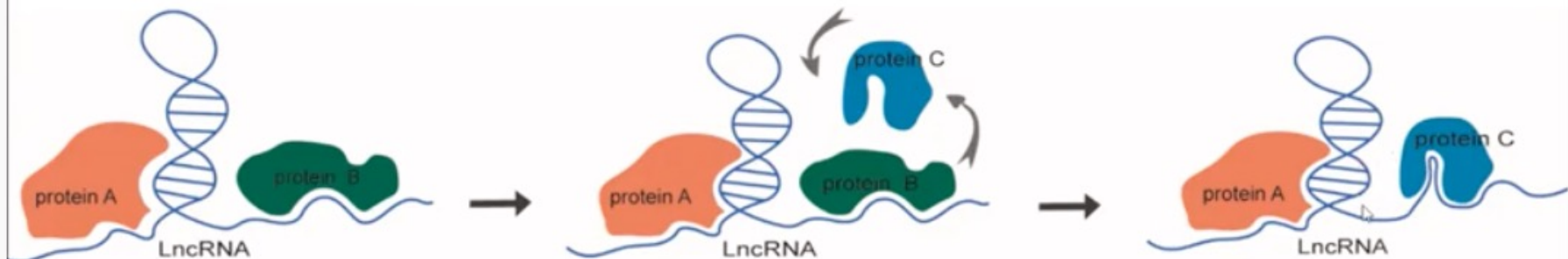
Methyl group is added to uracil to form thymine

Anticodon arm

LONG NON-CODING RNA



- Longer than 200 nucleotides
- Depending on location, sequence, morphology, structure, and function features, lncRNAs can be categorized into different groups.

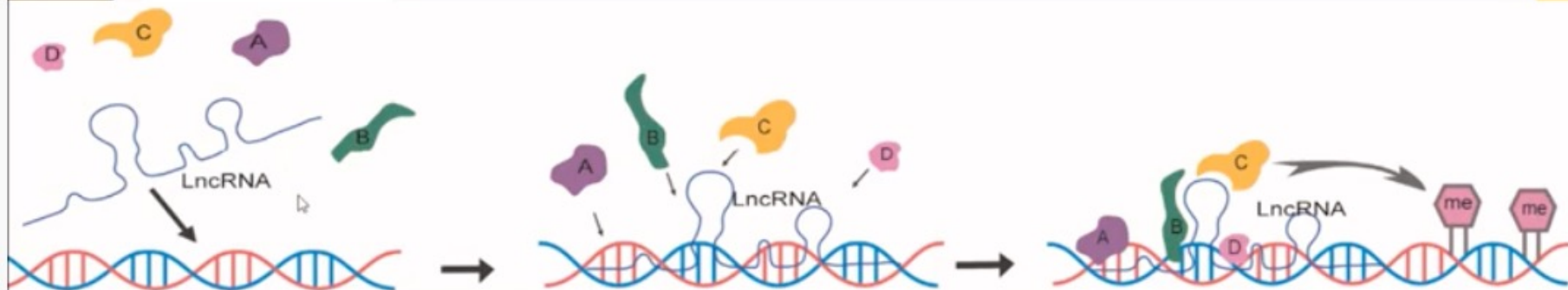


LncRNA allosteric effect for interacting with different ligand proteins.

LONG NON-CODING RNA



- Longer than 200 nucleotides
- Depending on location, sequence, morphology, structure, and function features, lncRNAs can be categorized into different groups.

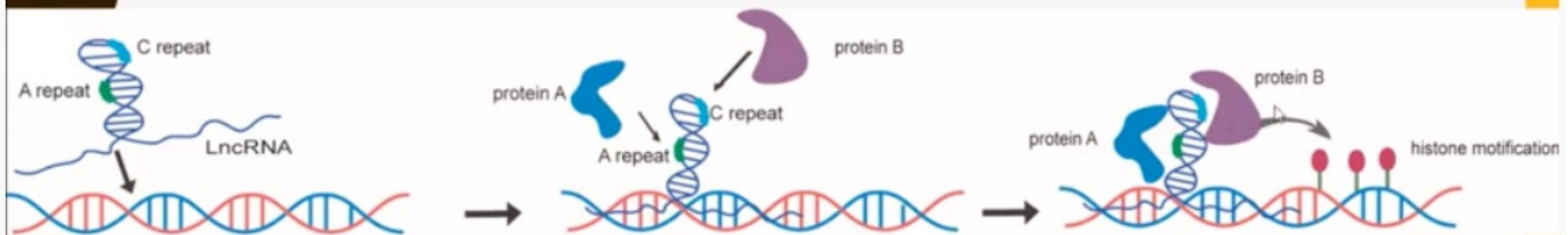


LncRNA acting as molecular scaffold to recruit and combine with multiple regulatory proteins.

LONG NON-CODING RNA



- Longer than 200 nucleotides
- Depending on location, sequence, morphology, structure, and function features, lncRNAs can be categorized into different groups.



LncRNA mediating histone modification by the functional region repeat elements.

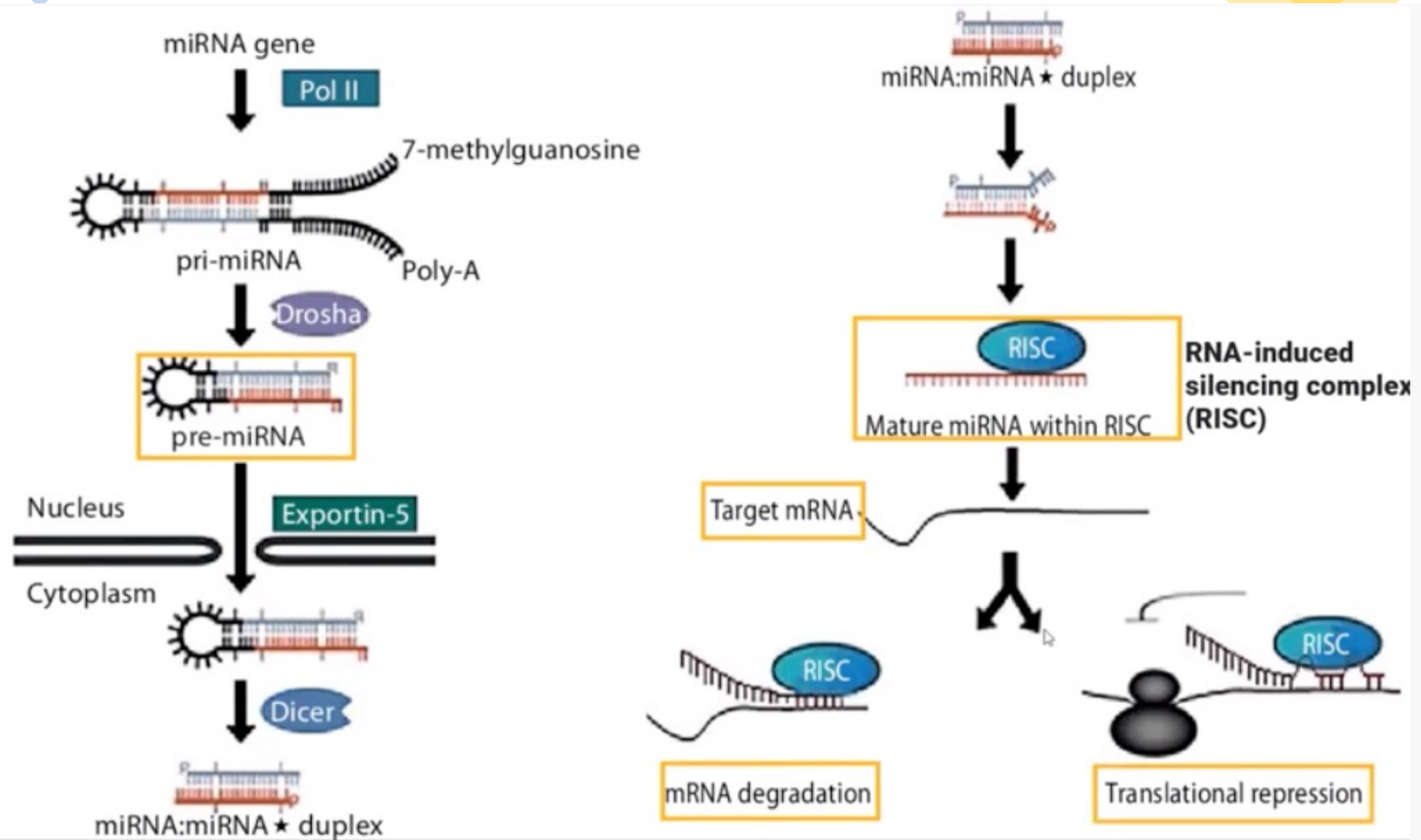
MICRO RNA



- 22 Nucleotides
- More than 2000 miRNAs have been identified in human cells

FUNCTION:

- RNA silencing
- Post-transcriptional regulation of gene expression



SMALL NUCLEOLAR RNA



snoRNAs

60–300 nucleotides

FUNCTION:

Post-transcriptional modification and maturation of ribosomal RNAs (rRNAs), small nuclear RNAs (snRNAs), and other cellular RNAs.

Nucleus

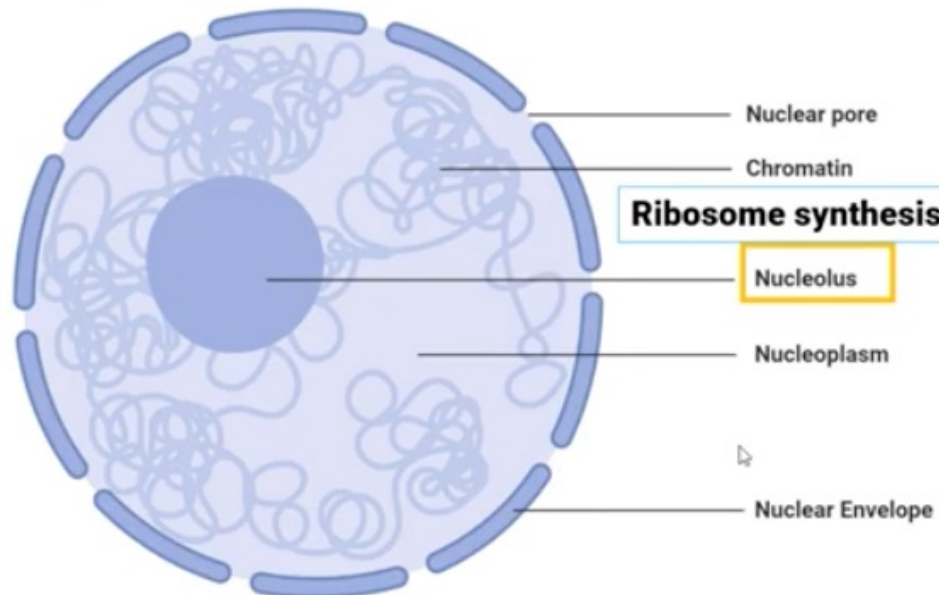
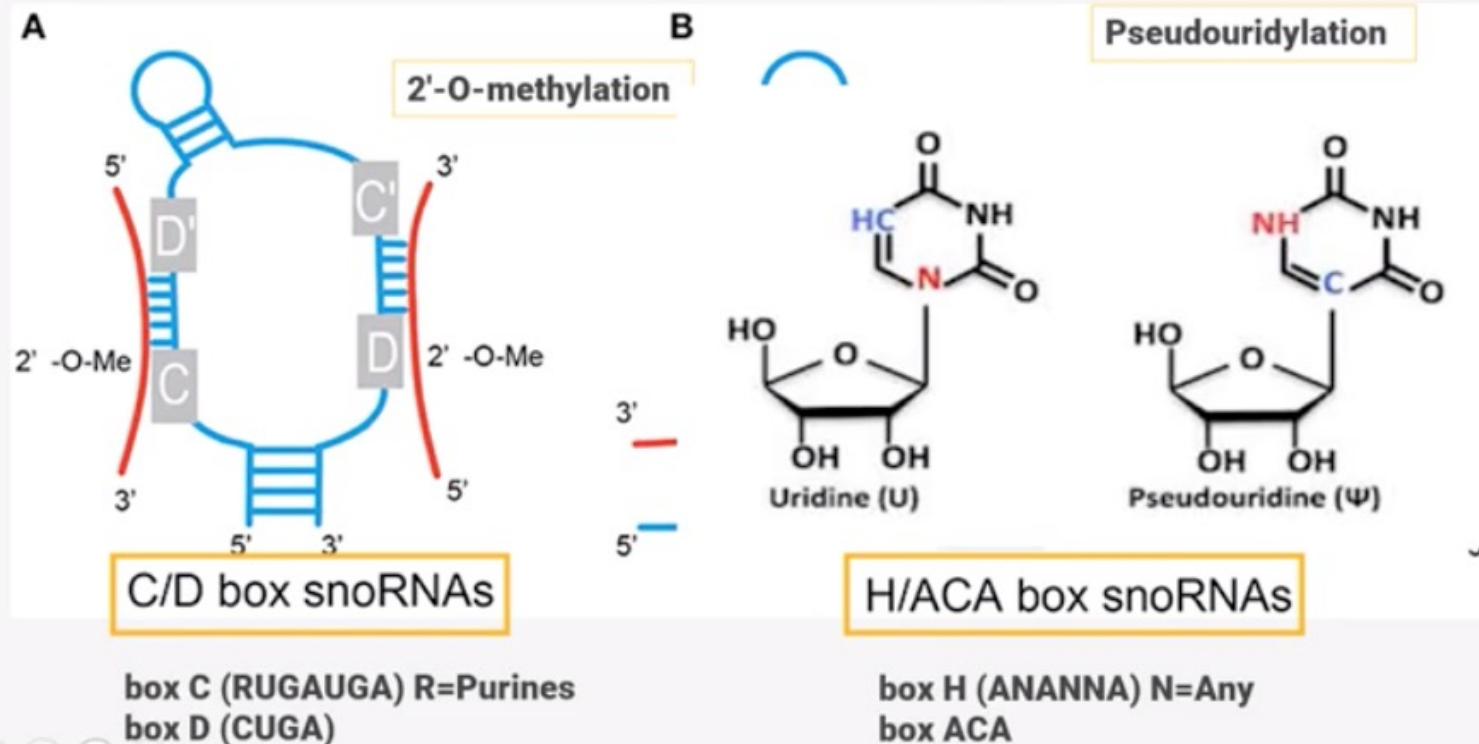
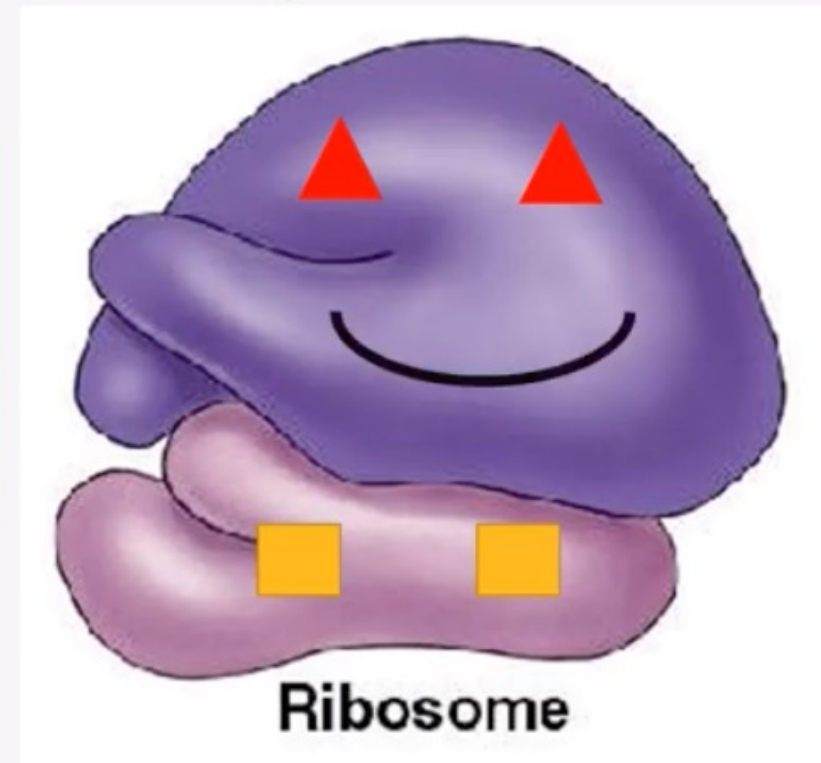
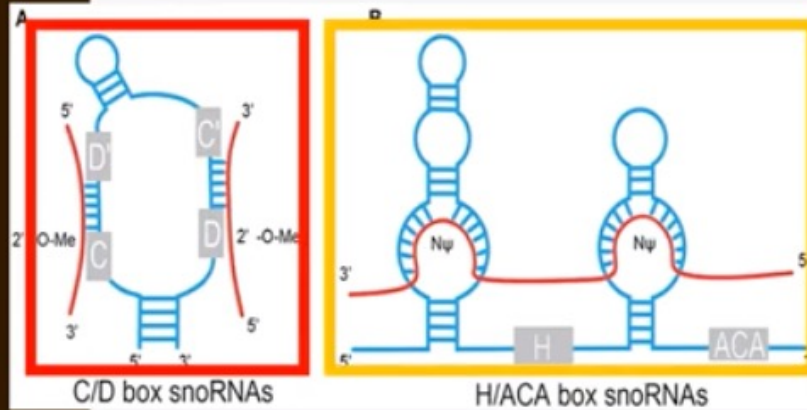


Figure: Nucleus, Image Copyright Sagar Aryal, www.microbenotes.com

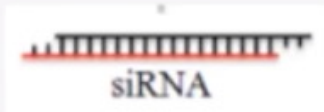
TWO CLASSES OF SNORNA



SMALL NUCLEOLAR RNA



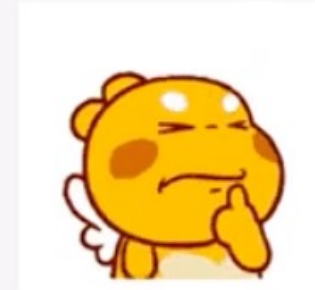
SMALL INTERFERING RNA



- 20-27 Nucleotides

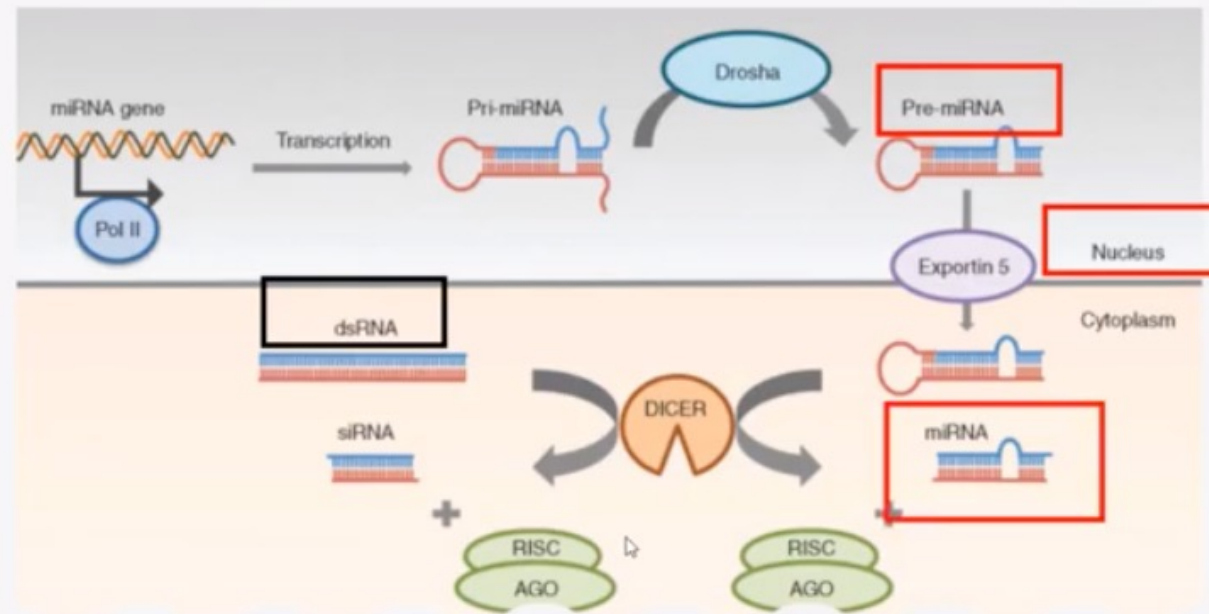
FUNCTION:

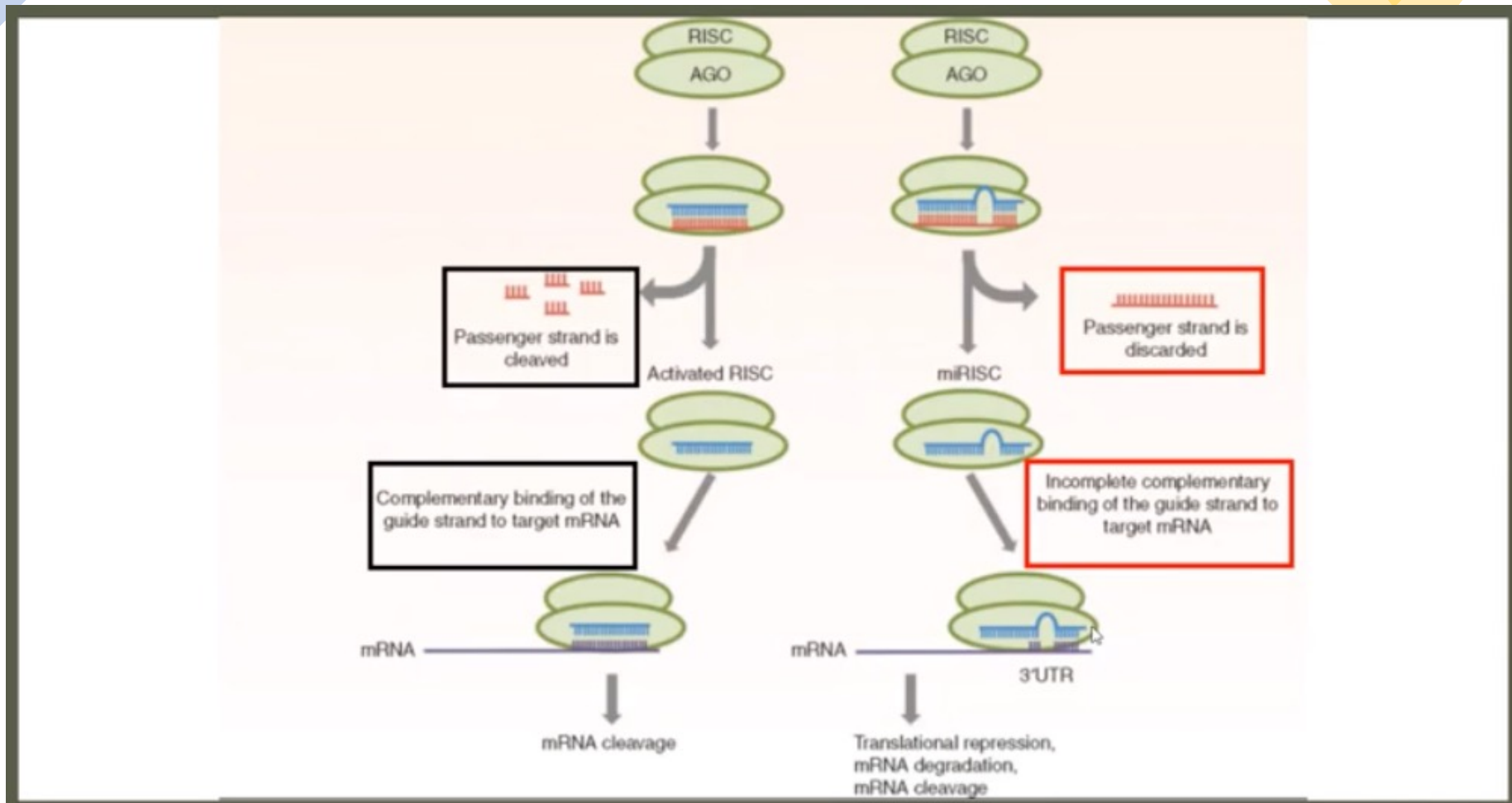
- RNA silencing
- Post-transcriptional regulation of gene expression



WHAT IS THE DIFFERENCE BETWEEN MI-RNA AND SI-RNA?

MIRNA/SIRNA





SMALL NUCLEAR RNA



- 150 Nucleotides

FUNCTION:

- Post transcriptional modifications of mRNA.

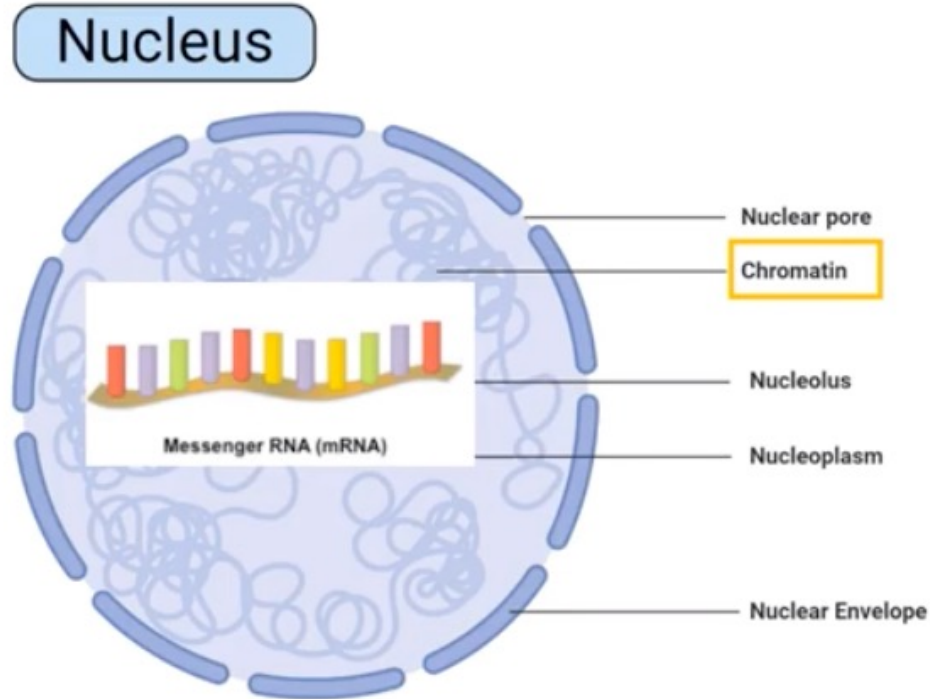
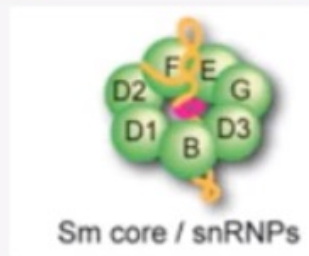


Figure: Nucleus, Image Copyright © Sagar Aryal, www.microbenotes.com

TWO CLASSES OF snRNA

High Uridine

Sm-class snRNA



Lsm-class snRNA



Ribonucleoprotein complexes

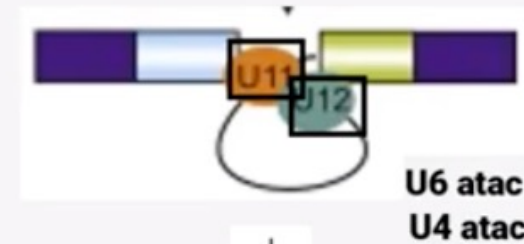
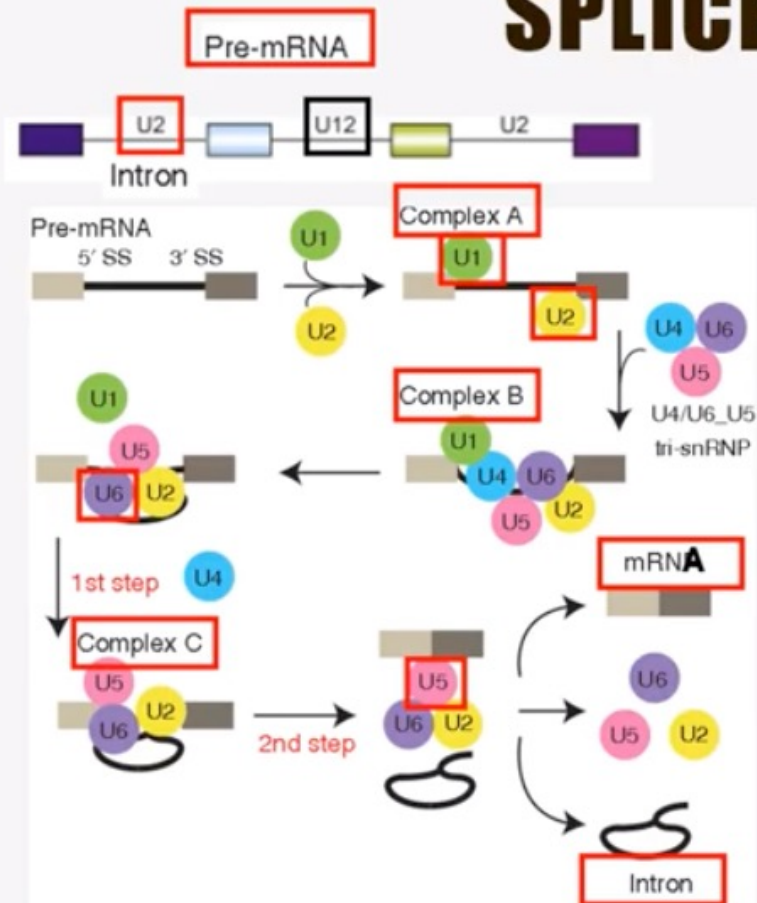
Transcribed by RNA polymerase II

Do leave the nucleus

Transcribed by RNA polymerase III

Never leave the nucleus

SPLICEOSOME



Minor spliceosome

PIWI Interacting RNA

PIRNA



Regulatory proteins responsible for stem cell and germ cell differentiation.

- (piRNAs) 21–35 nucleotides

FUNCTION:

- Silence **transposable elements**
- Regulate gene expression

DNA transposons



Jumping gene
as it can change
its position
within a genome

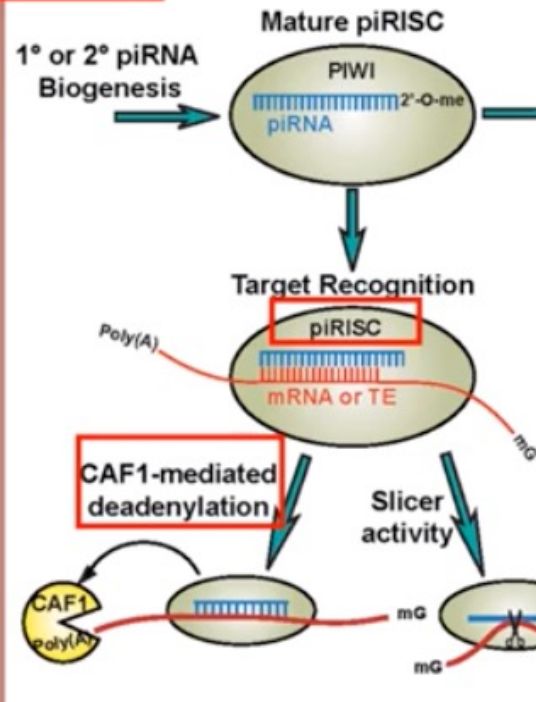
**Reversible
Mutations**

SILENCE TRANSPOSABLE ELEMENTS

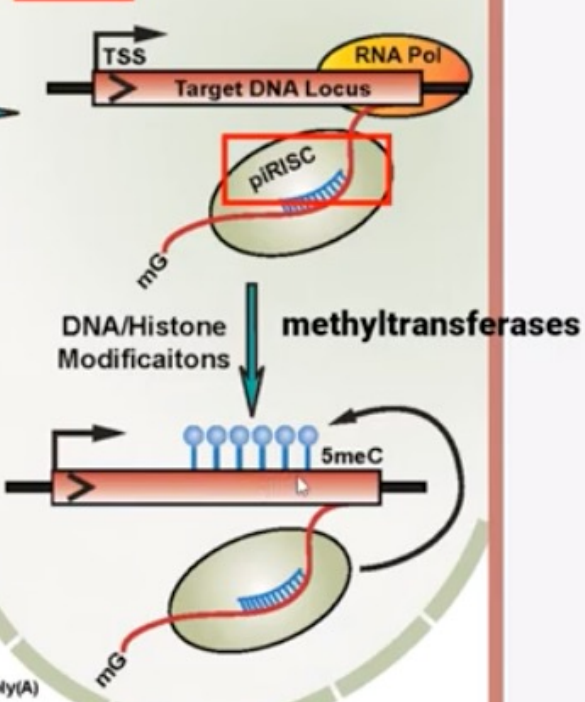
Post-transcriptional gene silencing

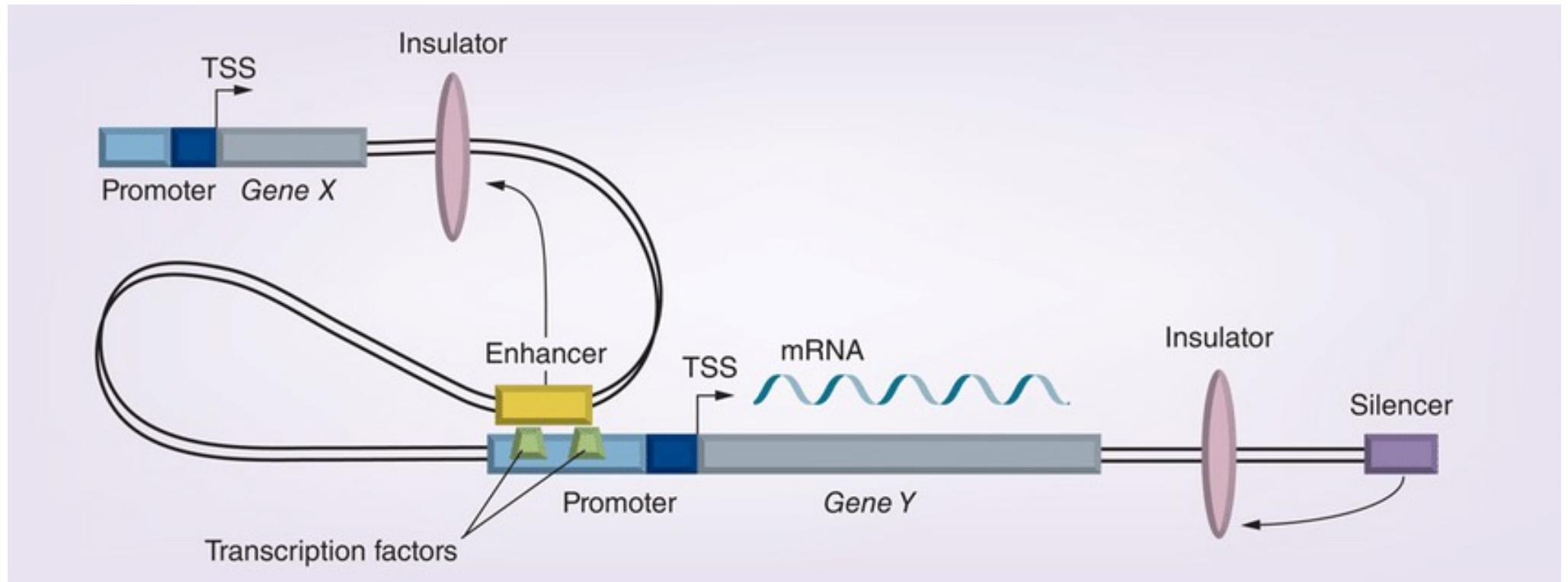
Transcriptional gene silencing

A. PTGS



B. TGS

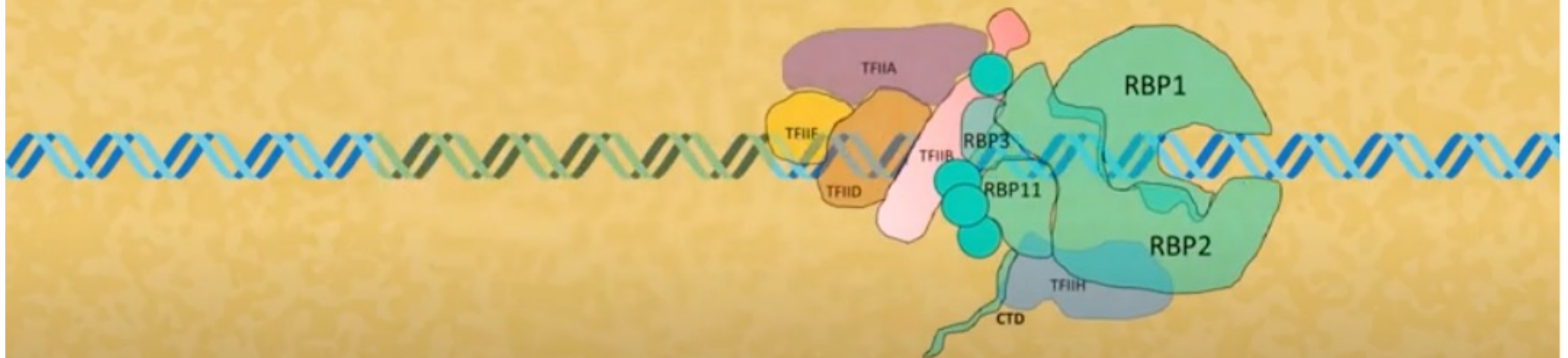
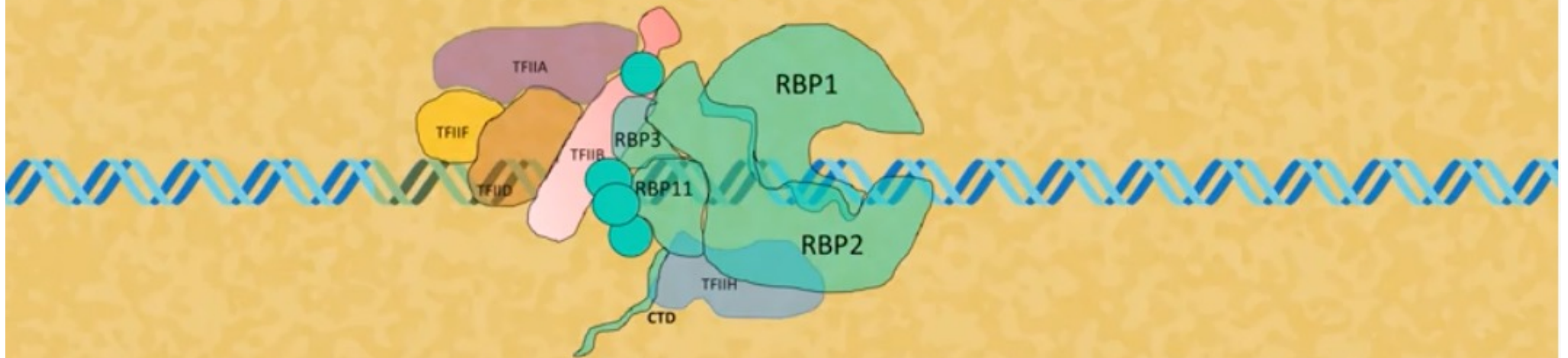






Promoter

RNA polymerase II (Pol II) core promoters are **specialized DNA sequences** near **transcription start sites** of genes that support the assembly of the transcription machinery and transcription initiation.



Promoter element could be upstream

Promoter element could be downstream

BRE element

GGG CGCC
CCA

-28 to -32

-24 to -31

TATA
TATA Box



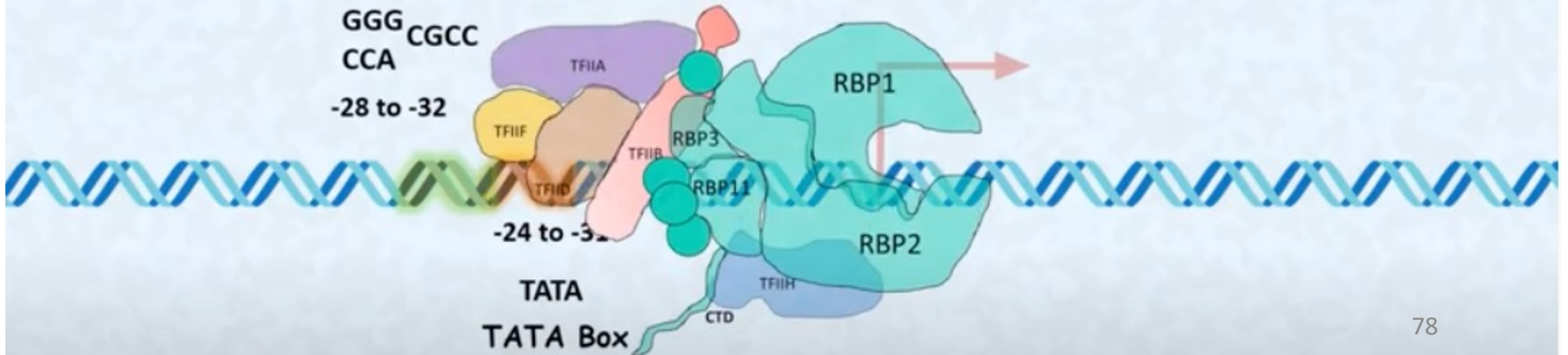
BRE element

GGG CGCC
CCA

-28 to -32

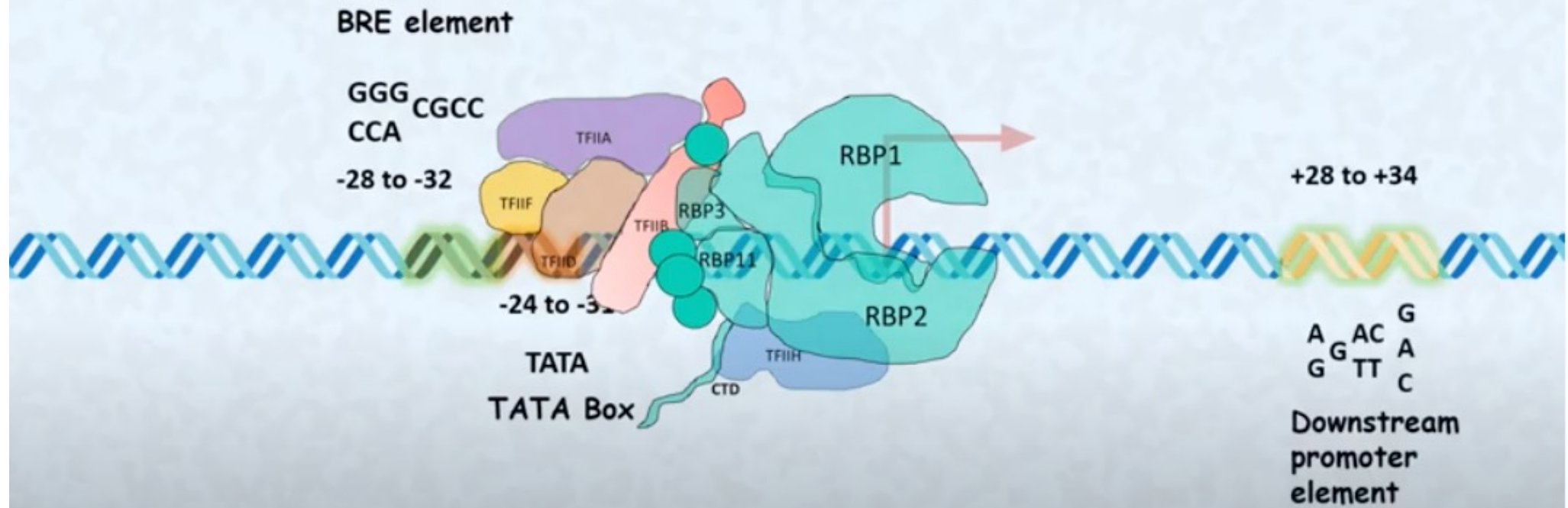
-24 to -31

TATA
TATA Box

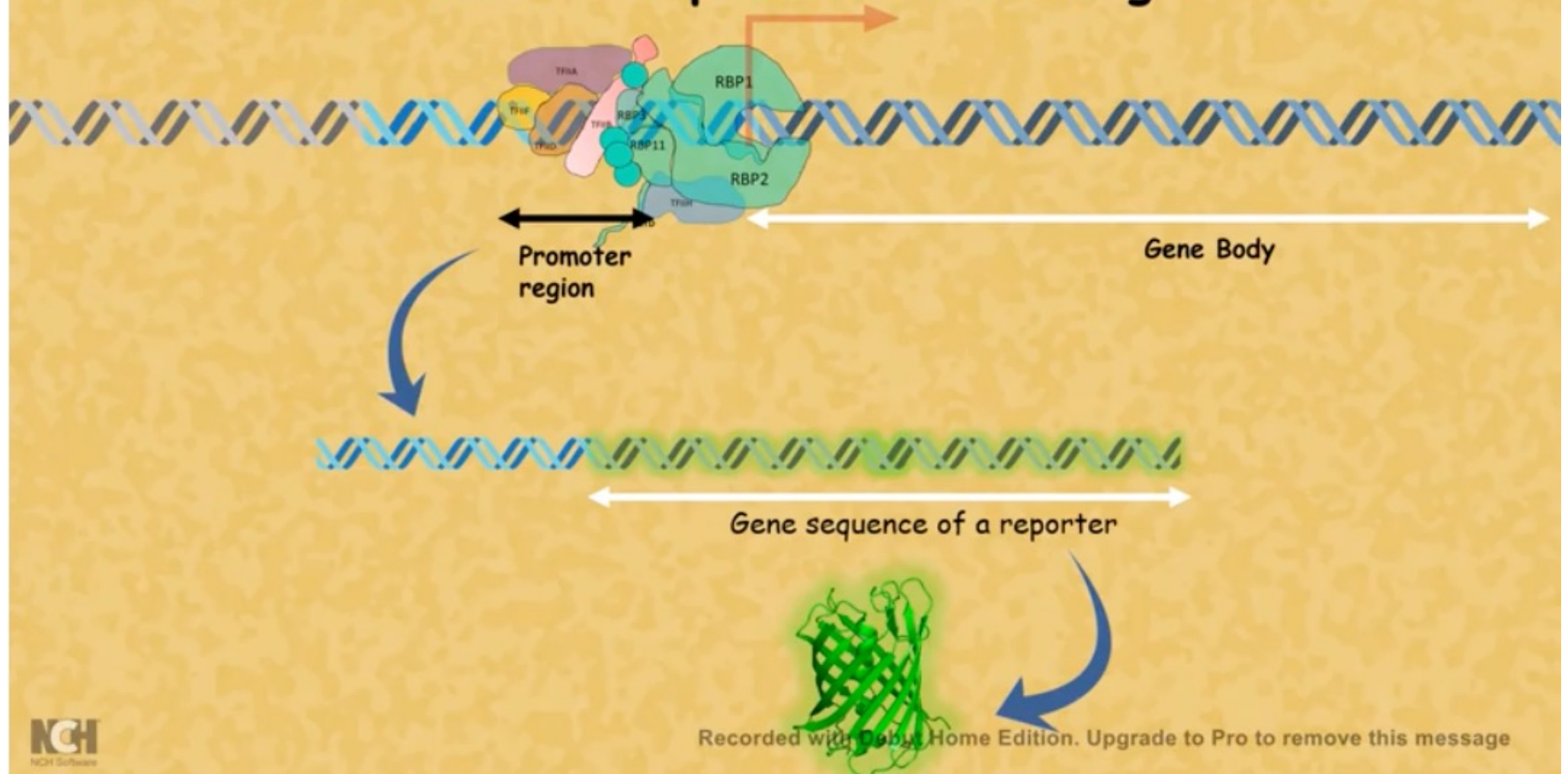


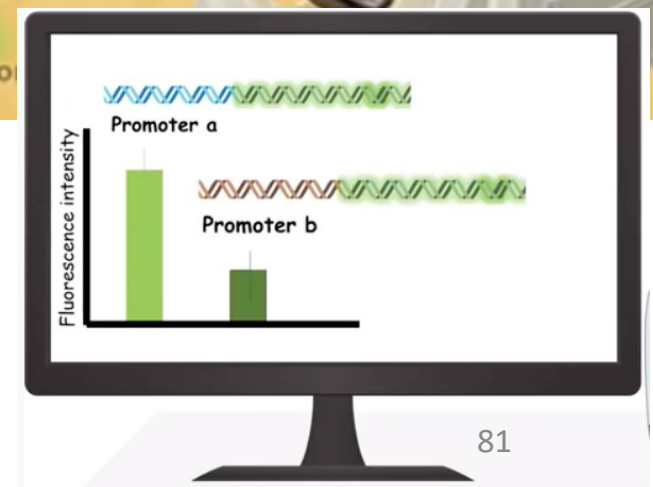
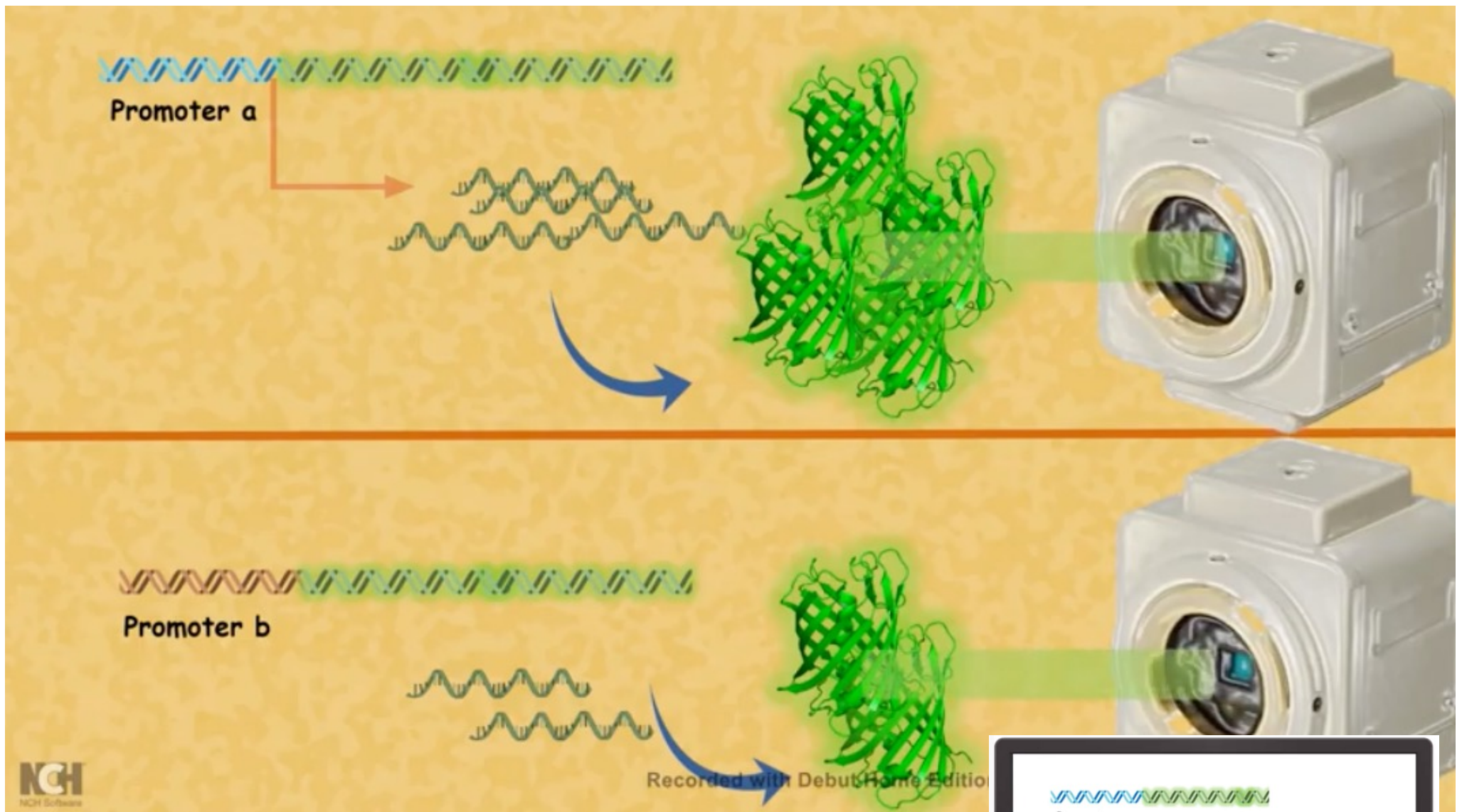
Promoter element could be upstream

Promoter element could be downstream



How to understand the promoter is strong or weak?





initiation pattern,
sequence composition and motifs,
chromatin configuration
and gene function,

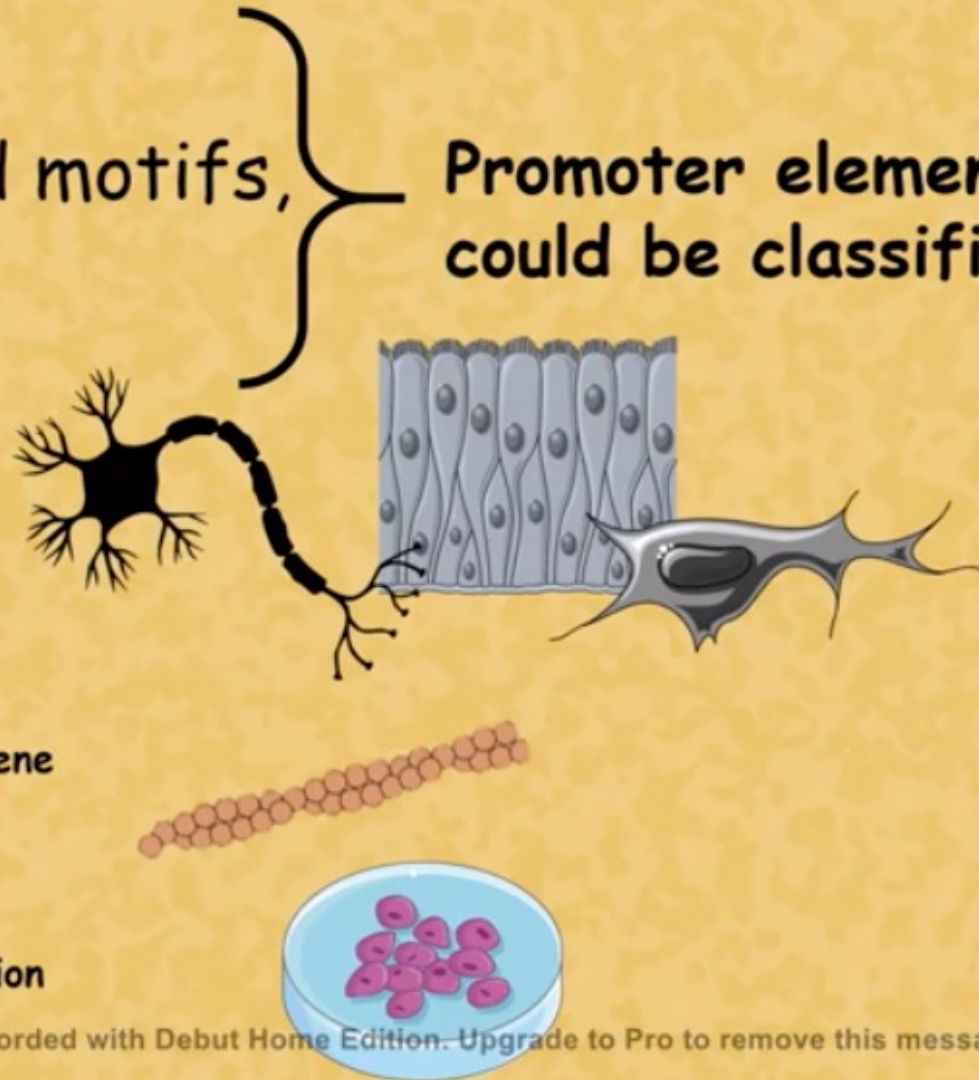
**Promoter elements
could be classified**

Promoters of adult differentiated tissue

Promoters of housekeeping gene

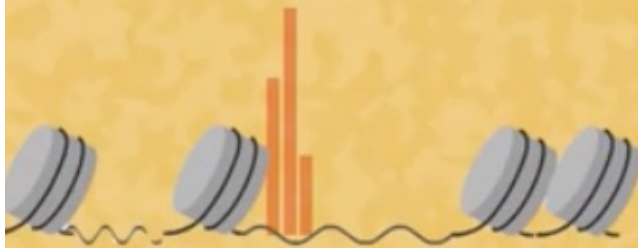
Promoters of development related transcription
factors

Recorded with Debut Home Edition. Upgrade to Pro to remove this message

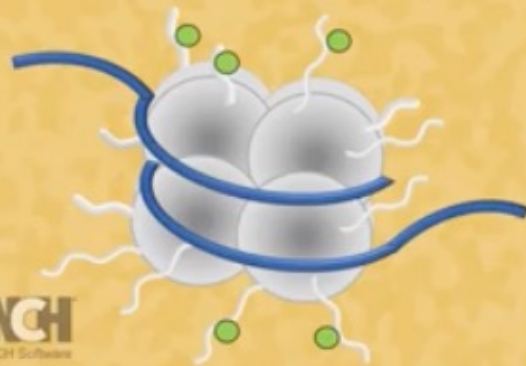


Adult tissue specific terminally differentiated gene promoters

Sharp TSS



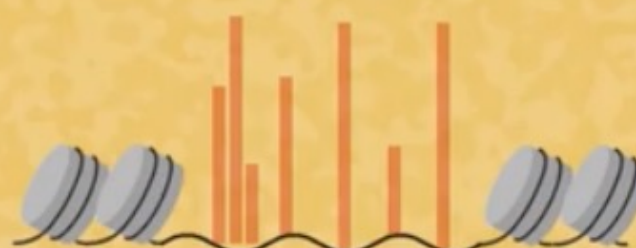
Regularly/scattered placed nucleosomes



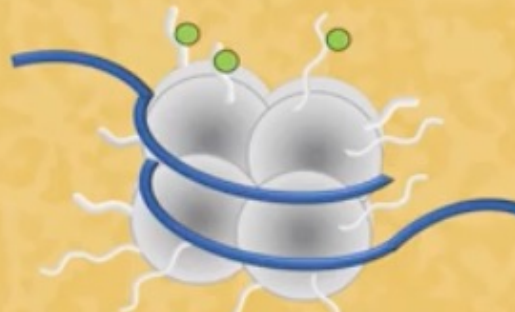
NCH
NCH Software

Housekeeping gene promoters

Broad TSS

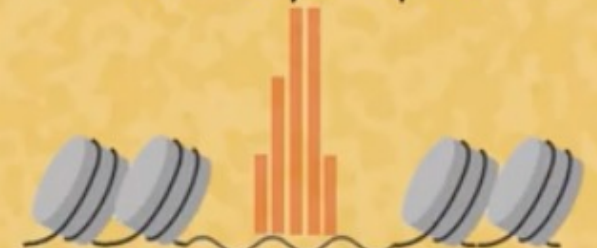


Regularly placed nucleosomes



Development related transcription factor gene promoters

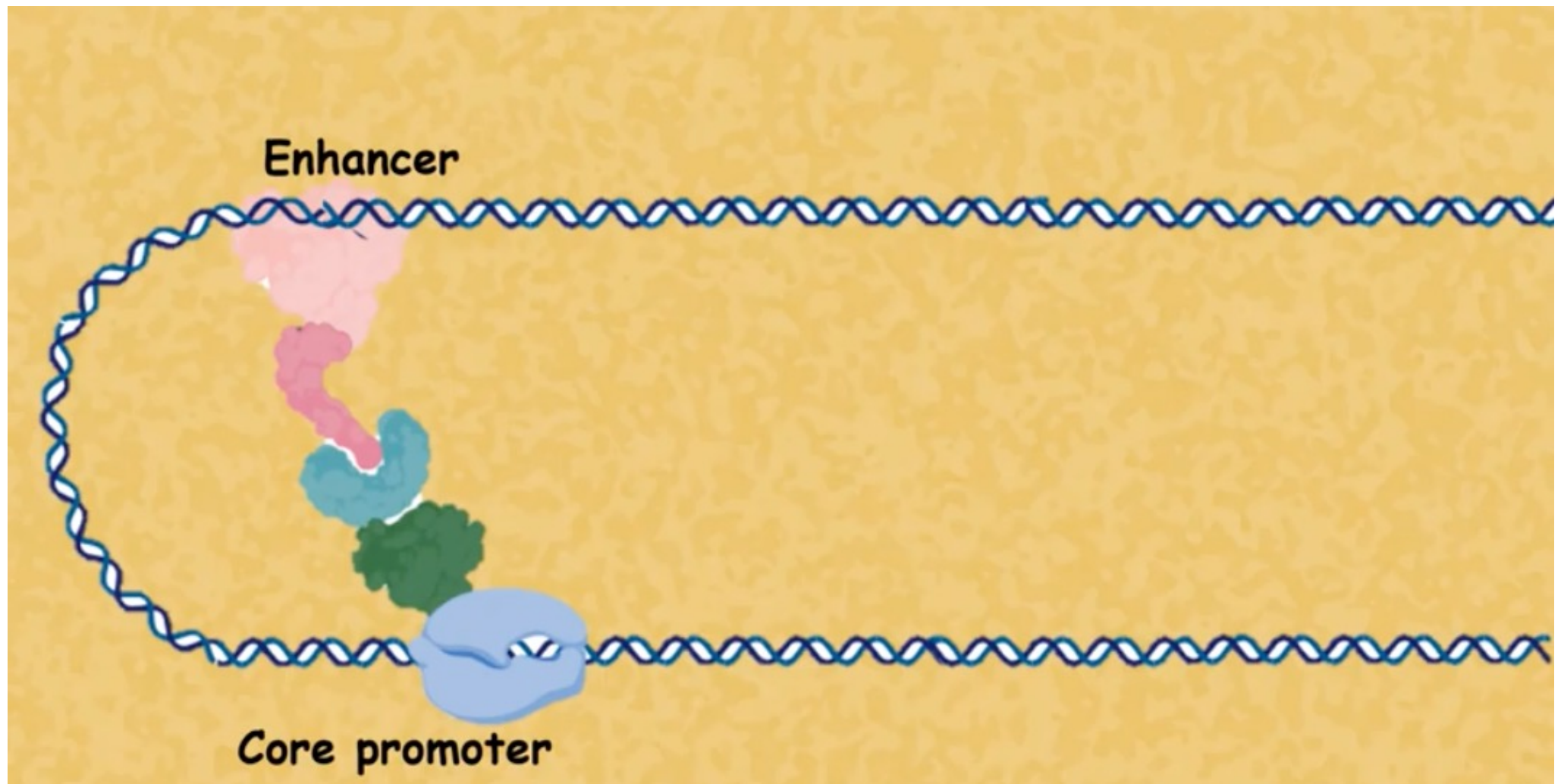
Relatively sharp TSS

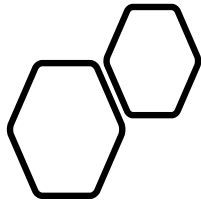


Regularly placed nucleosomes

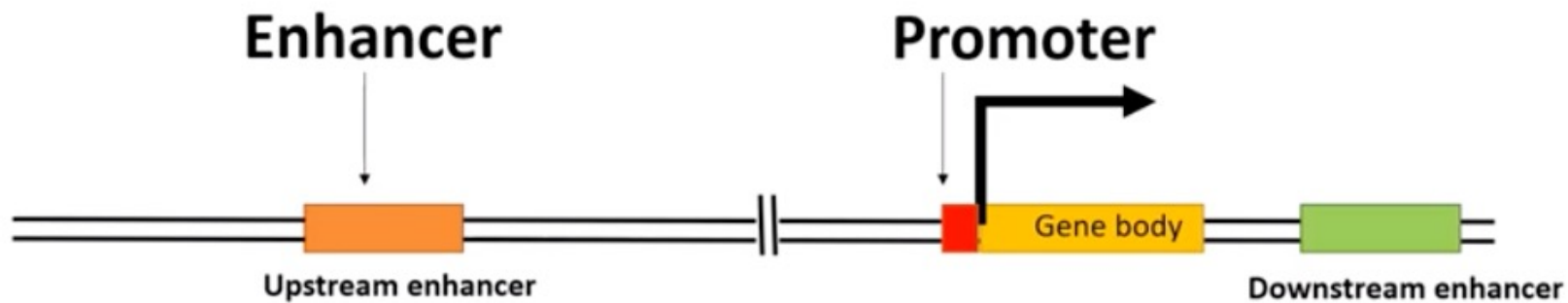


Recorded with Debut Home Edition. Upgrade to Pro to remove this message



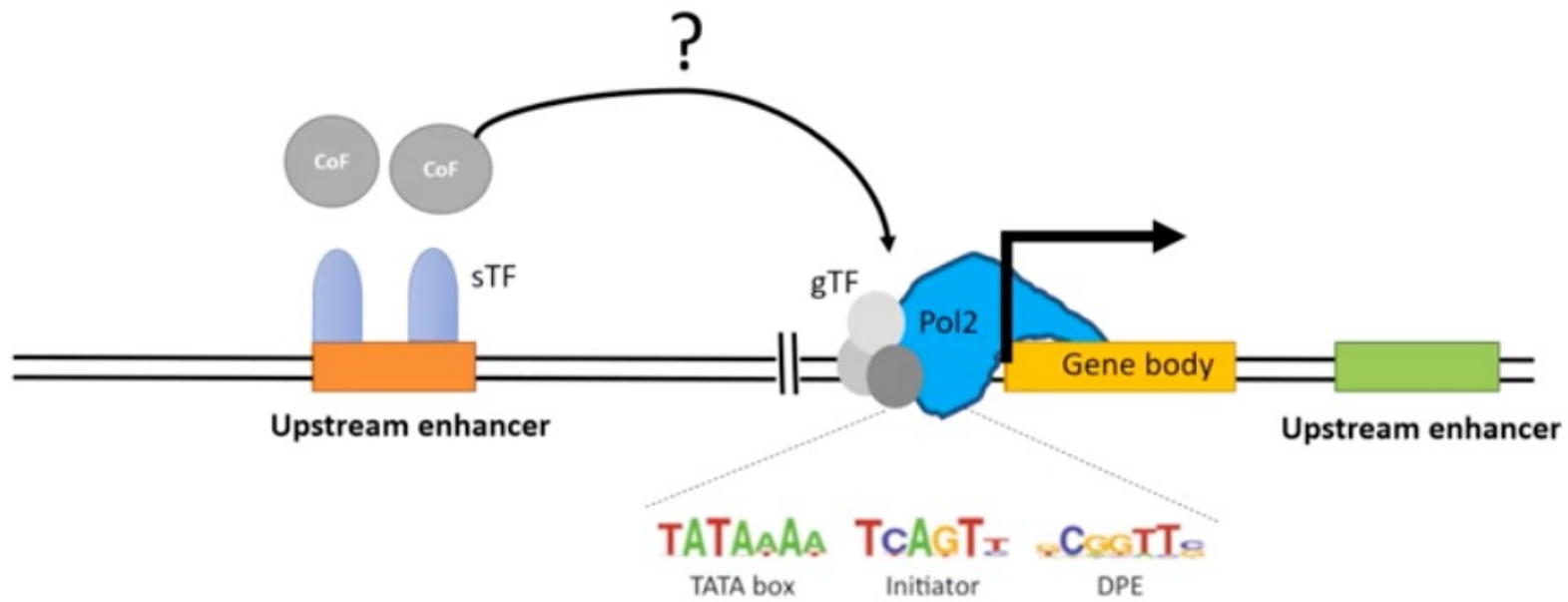


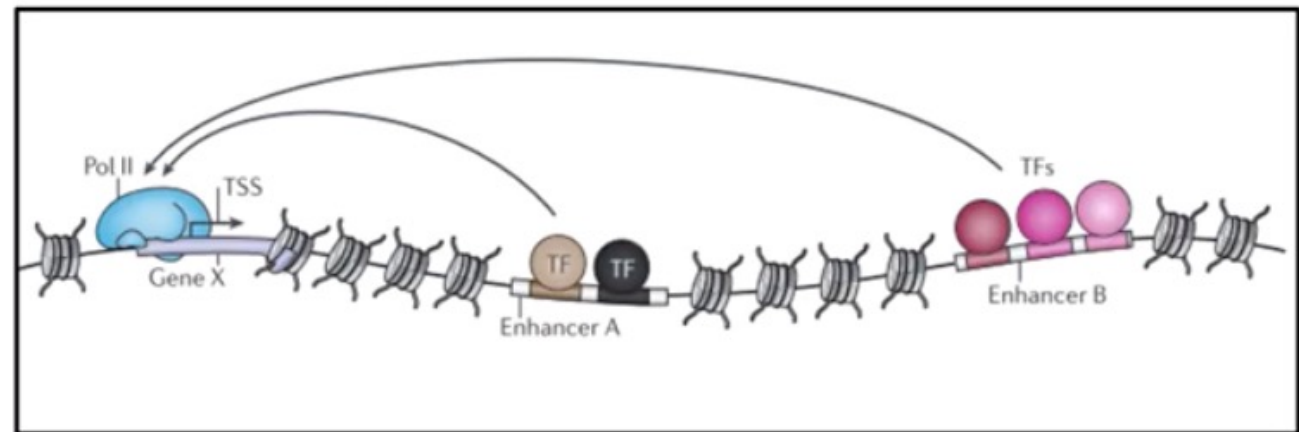
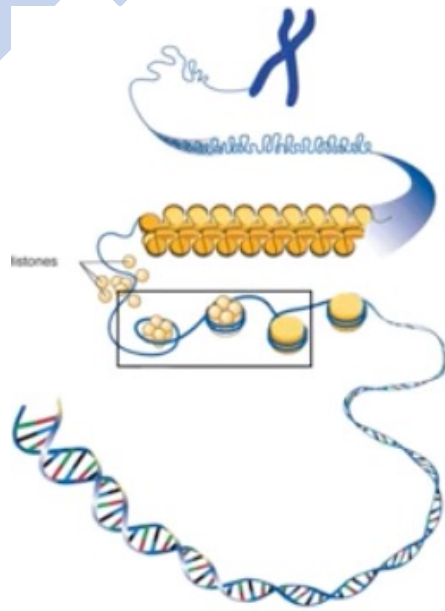
Enhancer



Enhancer: a sequence element that boosts transcription from a target core promoter in a cell type-specific manner and independently of the relative orientation and distance of the enhancer. Known enhancers are several tens to several hundred bp in length.

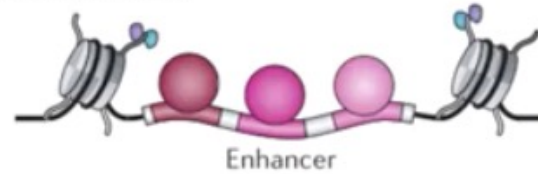
a short sequence around the transcription start-site (TSS) that can direct the recruitment of RNA polymerase II (Pol II) and transcription initiation. Core promoters typically have low basal activities in the absence of enhancers and are also termed minimal promoters



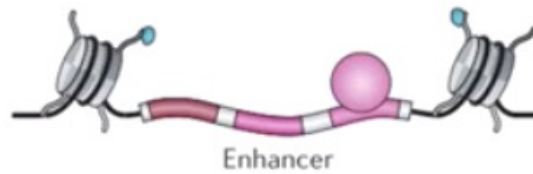


Different Chromatin state of enhancer

Active enhancer



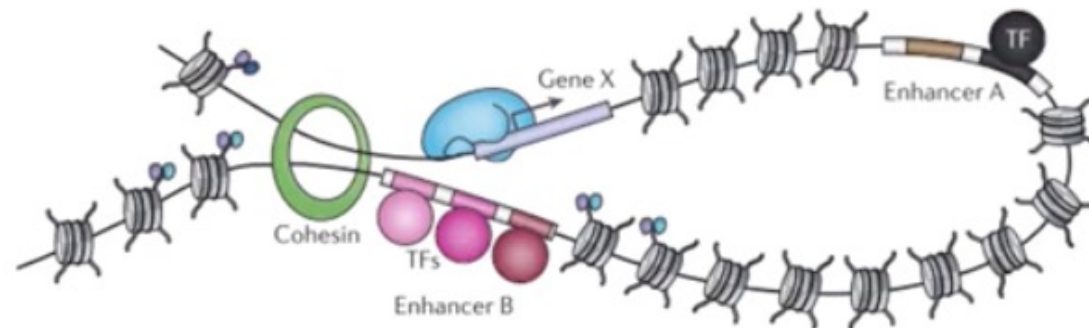
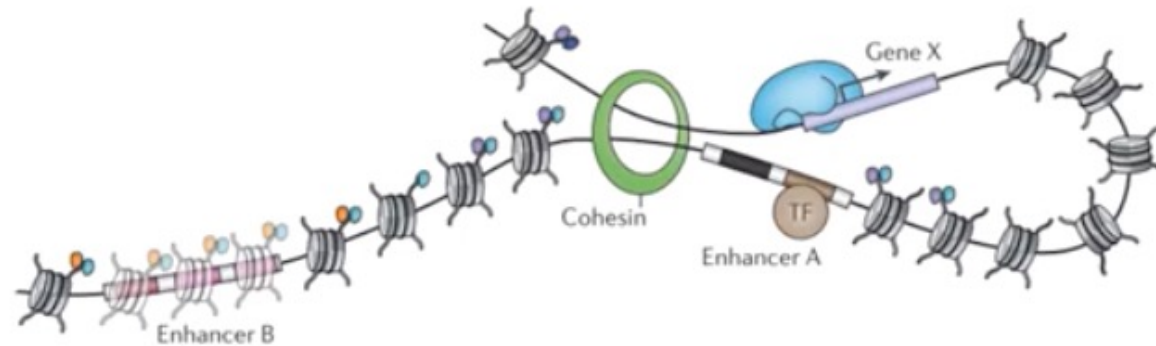
Primed enhancer



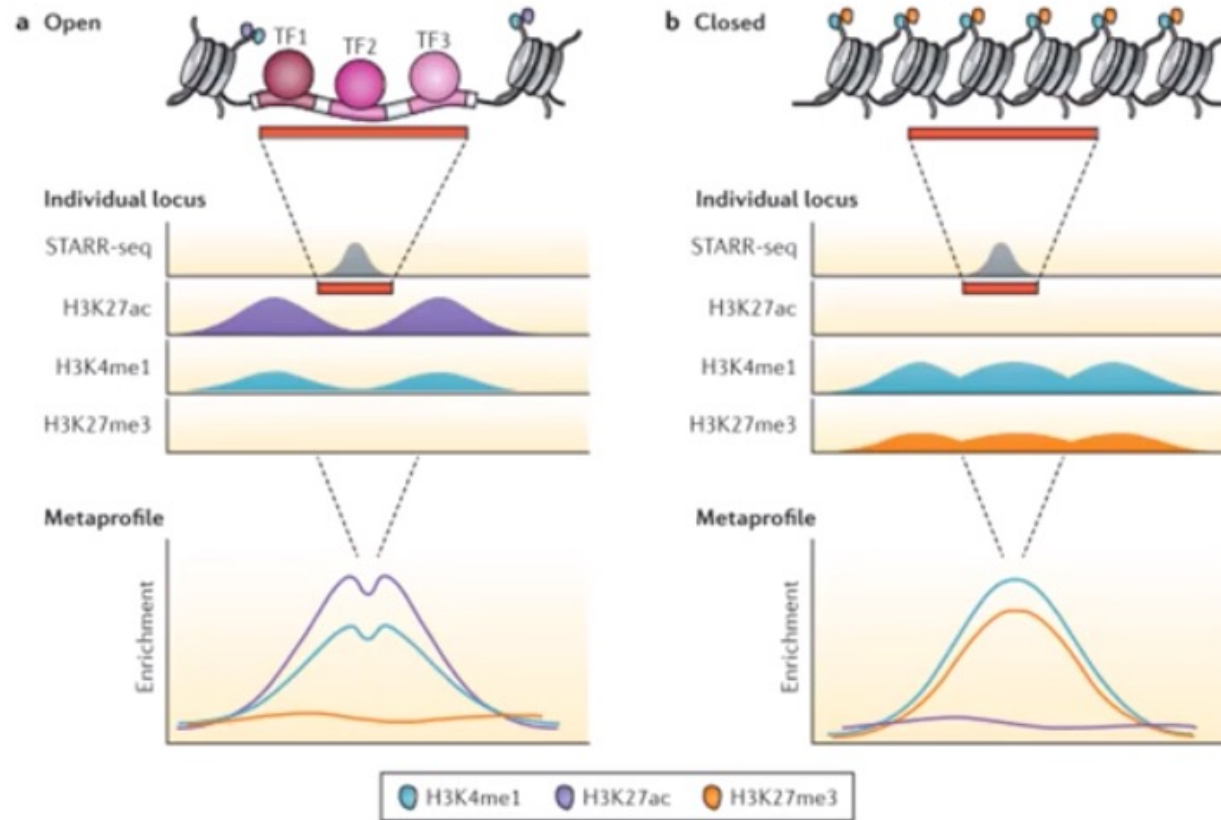
Closed or poised enhancer

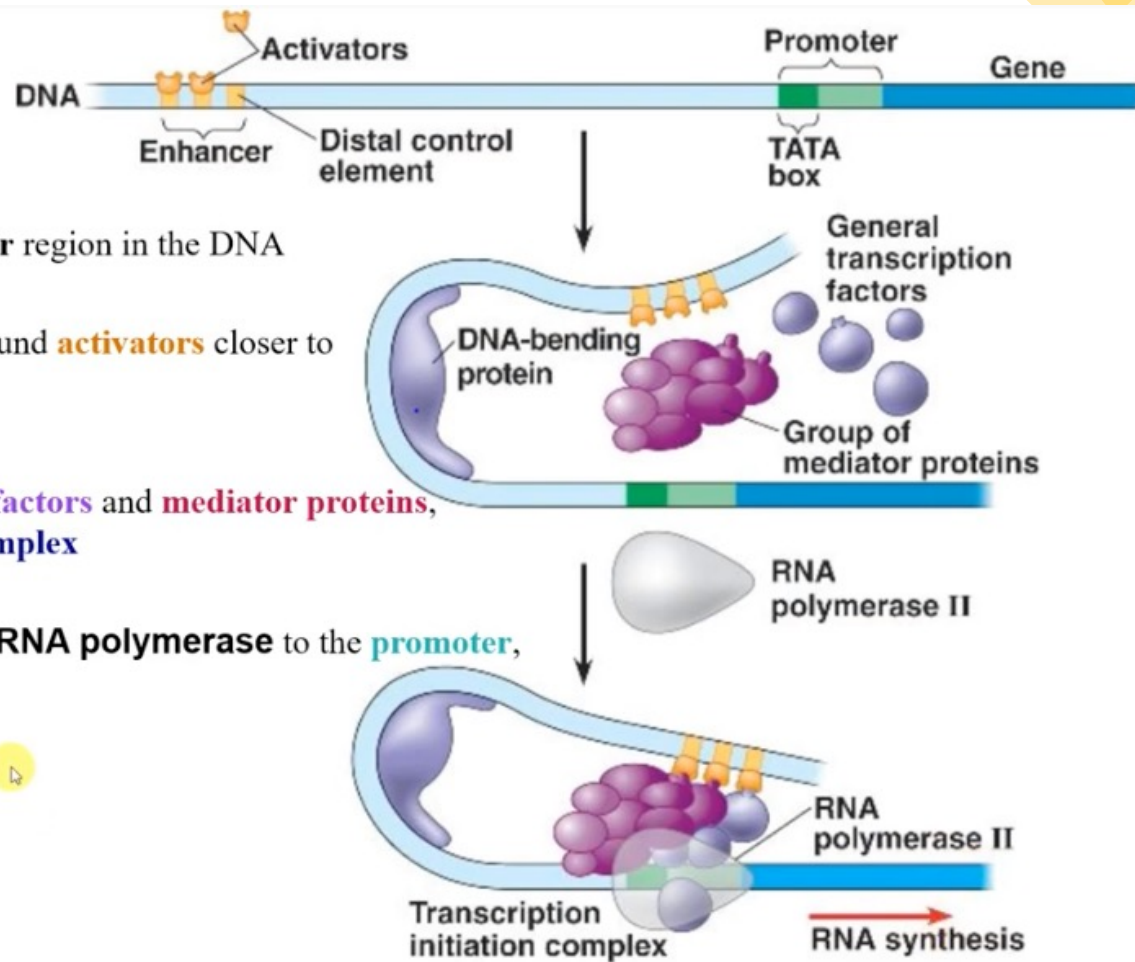


Different enhancer could work upon same gene in different manner depending on which tissue they are present in







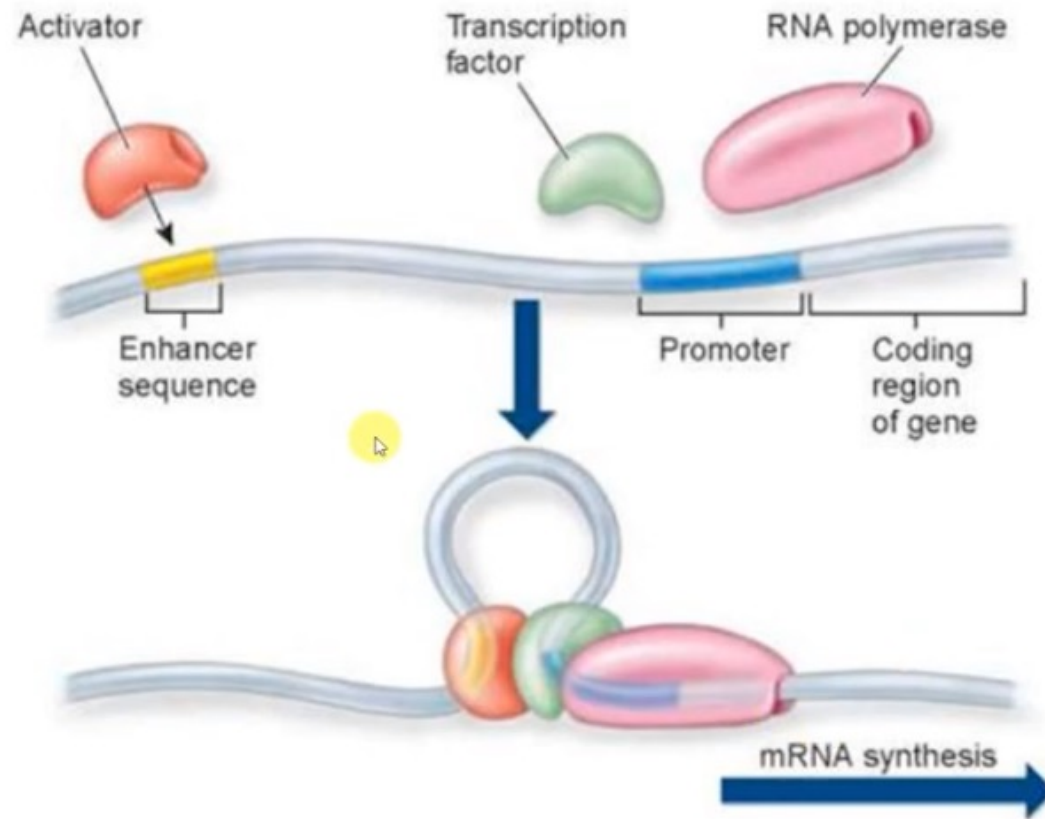


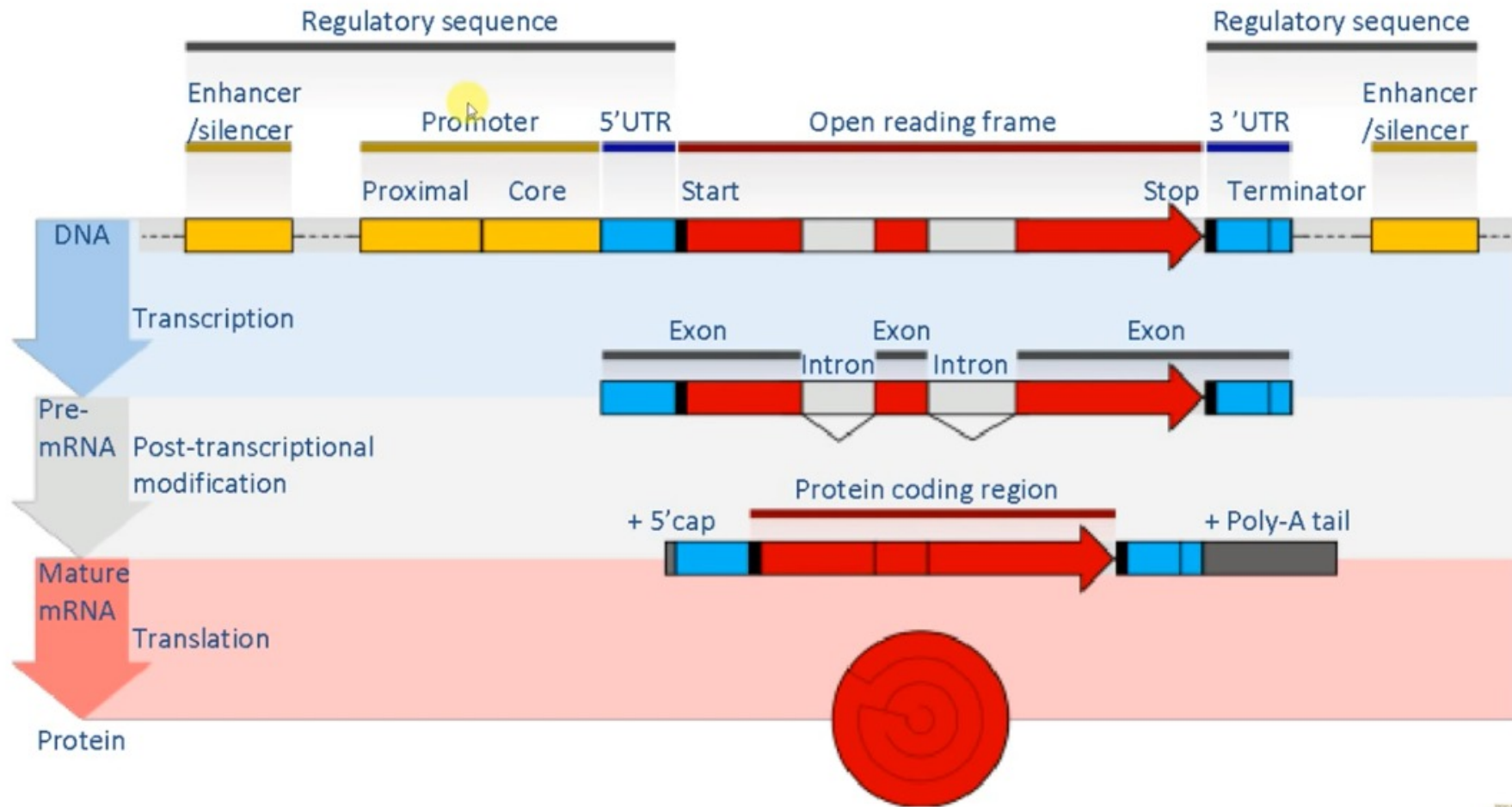
Activator proteins bind to an **enhancer** region in the DNA

A **DNA-bending protein** brings the bound **activators** closer to the **promoter**

The **activators** bind to **transcription factors** and **mediator proteins**, forming a **transcription initiation complex**

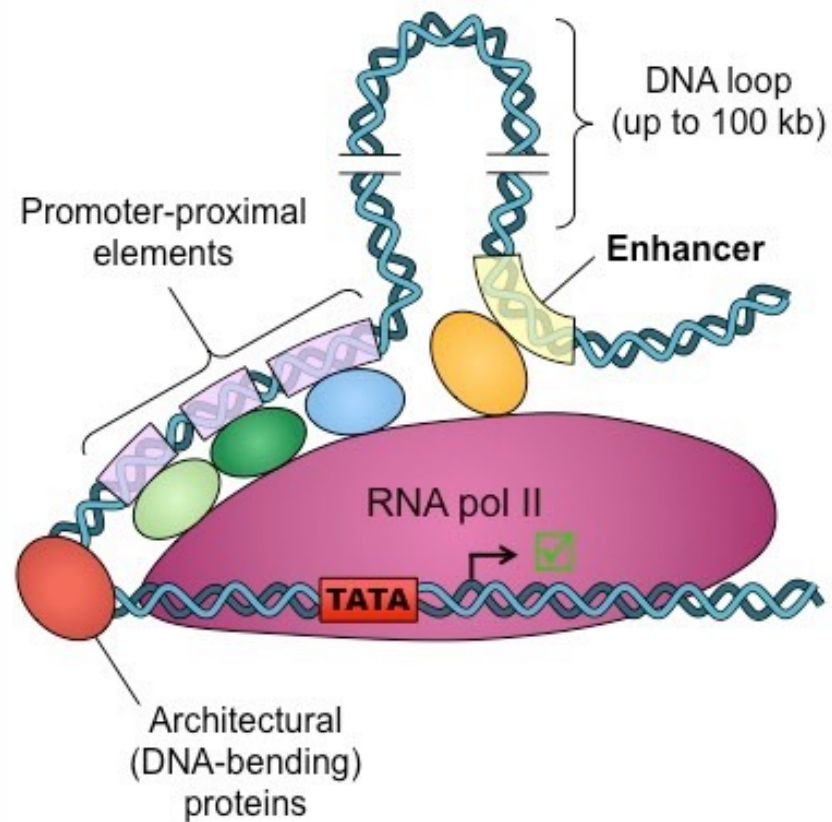
The complex promotes the binding of **RNA polymerase** to the **promoter**, enhancing gene **expression**



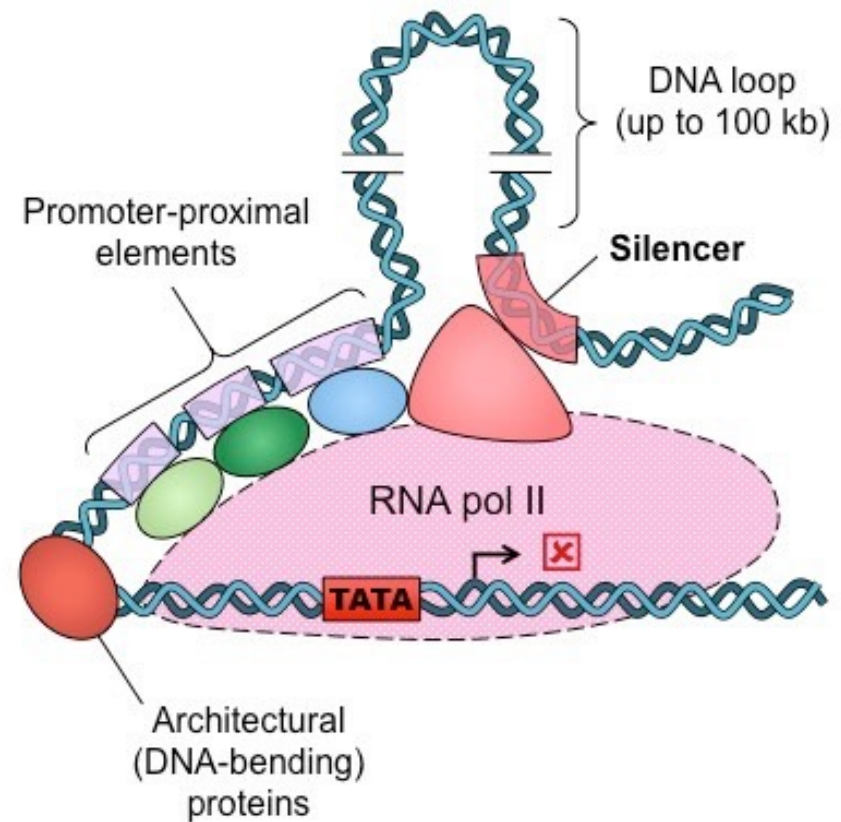


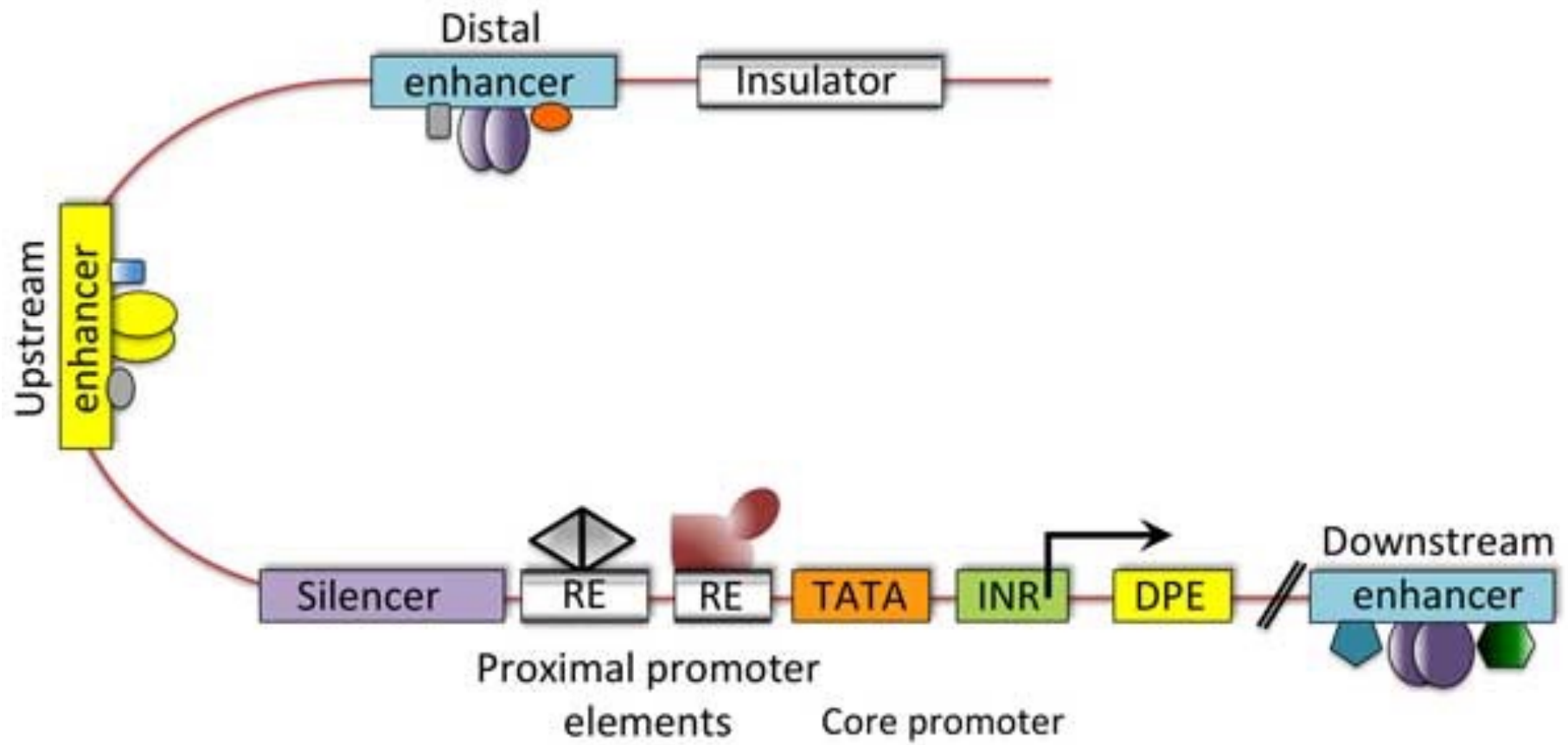
Silencers

Enhancer + Activator Protein

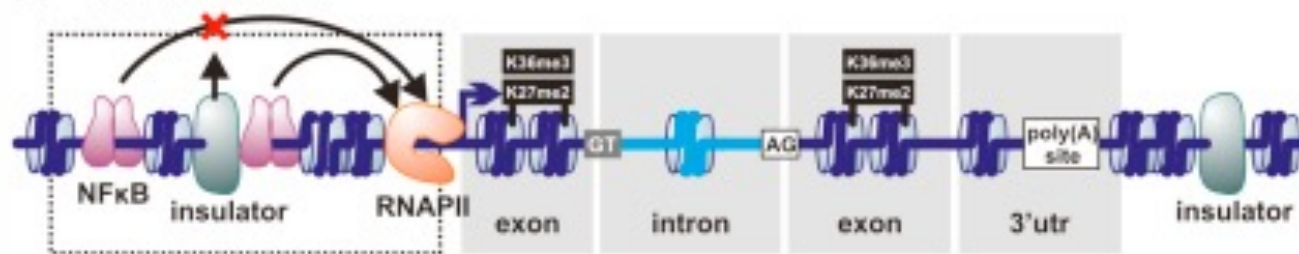


Silencer + Repressor Protein





(a) Linear structure

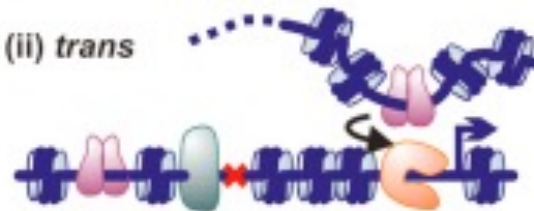


(b) Enhancer-promoter interaction

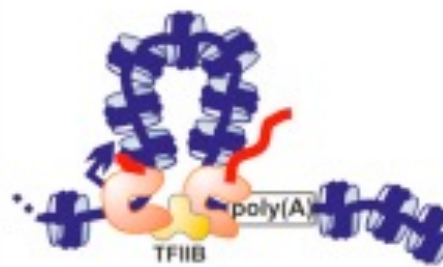
(i) *cis*



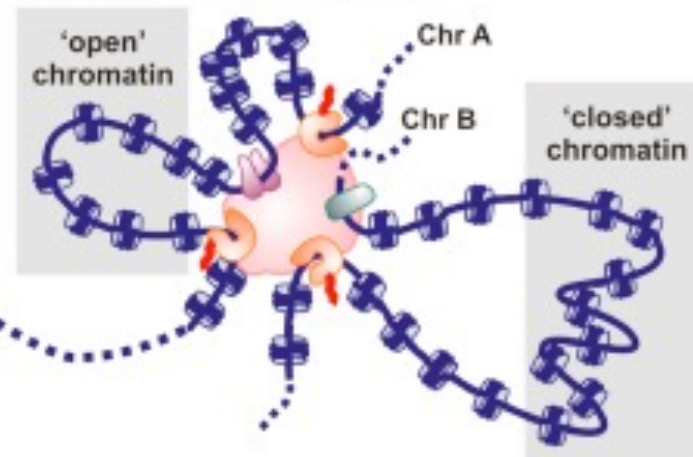
(ii) *trans*



(c) Gene loops



(d) Transcription factories



Current Opinion in Cell Biology

Insulators



Insulators

Insulators function to block genes from being affected by the transcriptional activity of neighboring genes.

They can block such interactions such as block enhancer-promoter

It is thought that an insulator must *reside between* the **enhancer** and **promoter** to *inhibit their subsequent interactions*.

(A) Barrier insulator

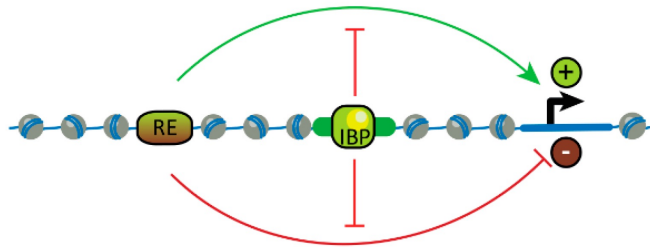


(B) Enhancer blocking insulator

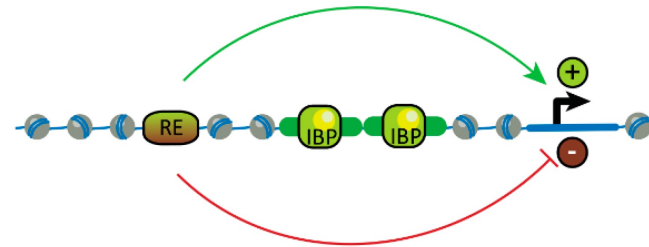


TRENDS in Genetics

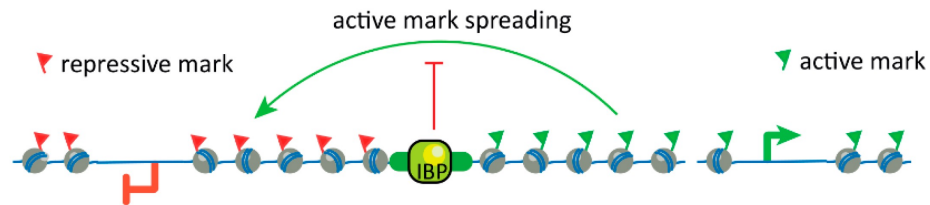
A. Blocking regulatory element-promoter communication



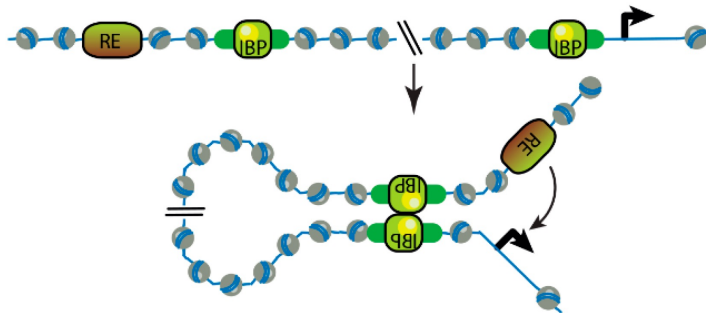
B. Insulator bypass



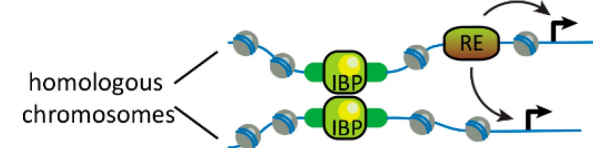
C. Barrier to histone mark spreading



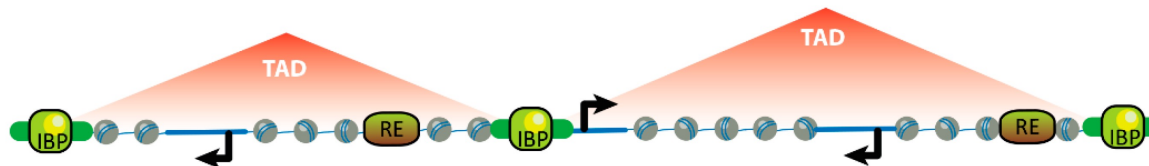
D. Facilitating long-distance regulation



E. *Trans* regulation (Transvection)



F. TAD boundary formation



Insulator Elements (boundary elements) help to coordinate the regulation:

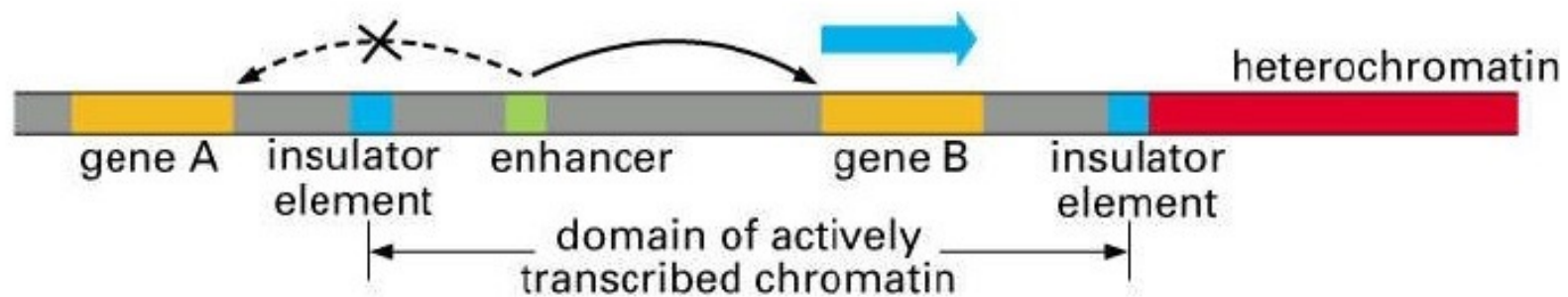
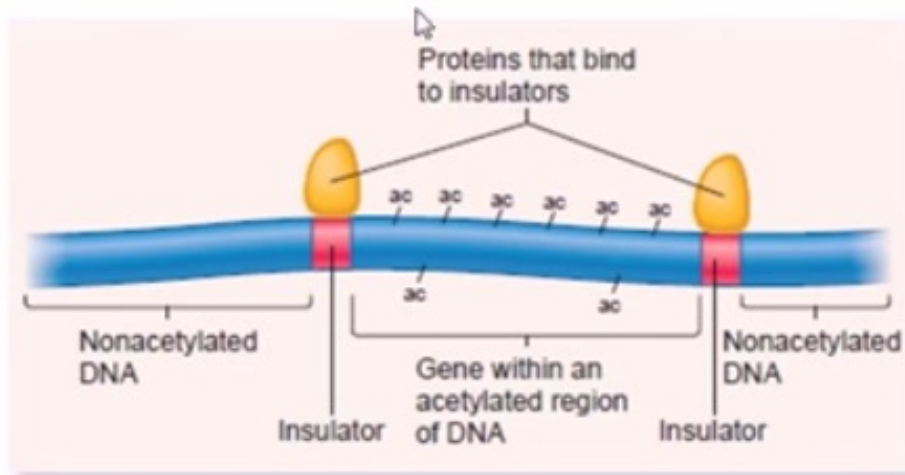
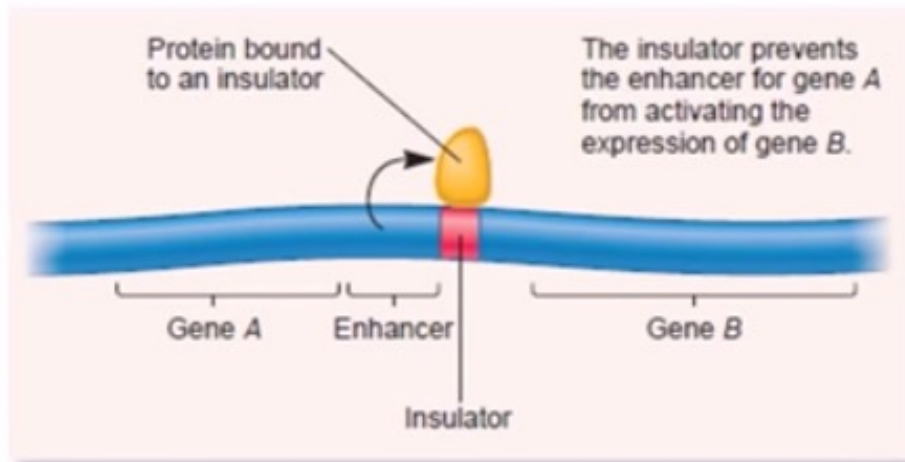


Figure 7-61. Molecular Biology of the Cell, 4th Edition.

Gene insulators as gene silencers



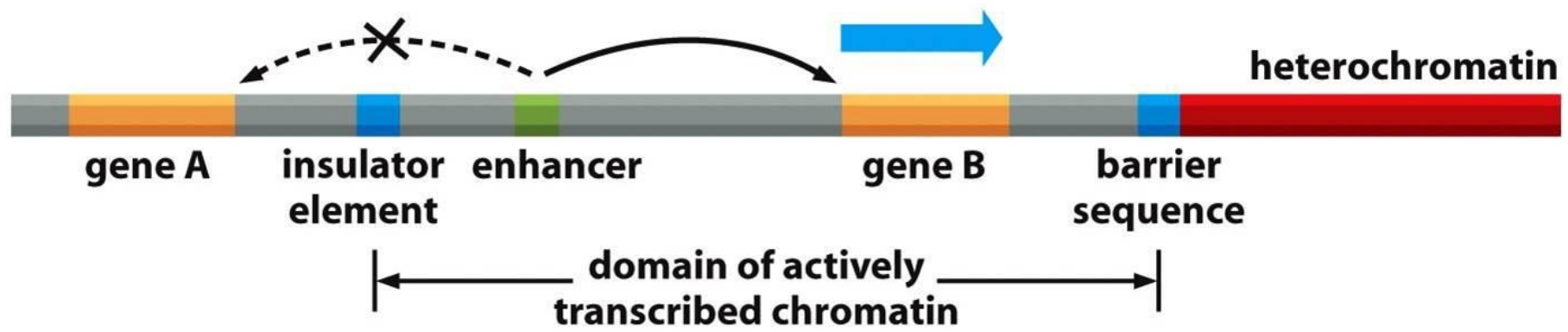
(a) Insulators as a barrier to changes in chromatin structure

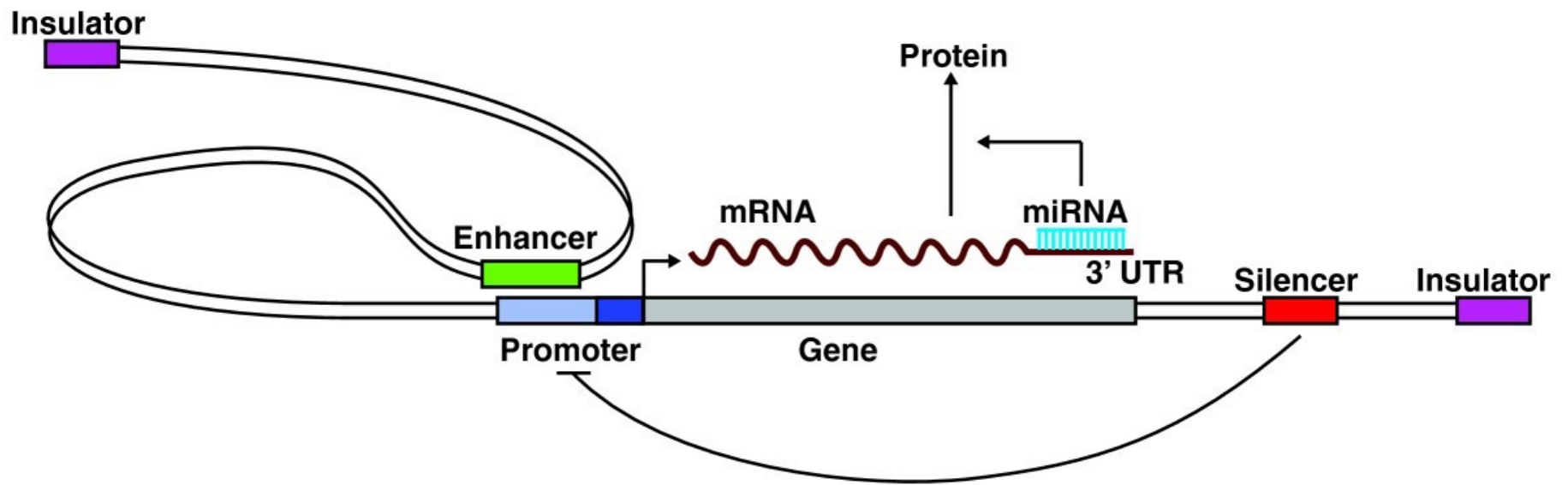


(b) Insulator that blocks the effects of a neighboring enhancer

- An **insulator** is a segment of **DNA** that functions as a boundary between two genes.

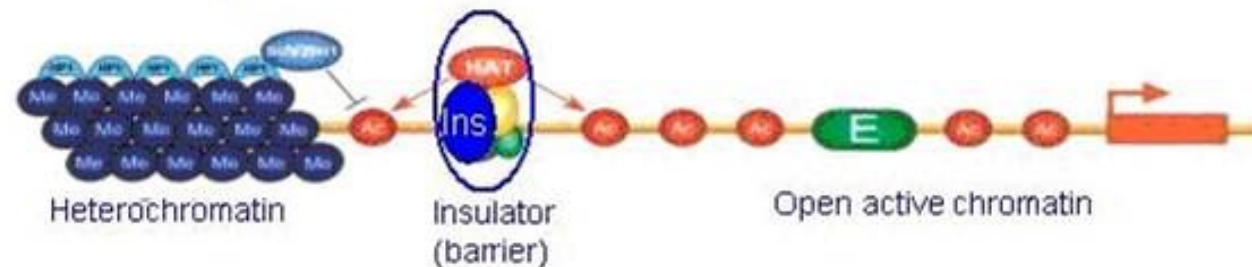
- An insulator is so named because it protects, or “insulates,” a gene from the regulatory effects of a neighboring gene.



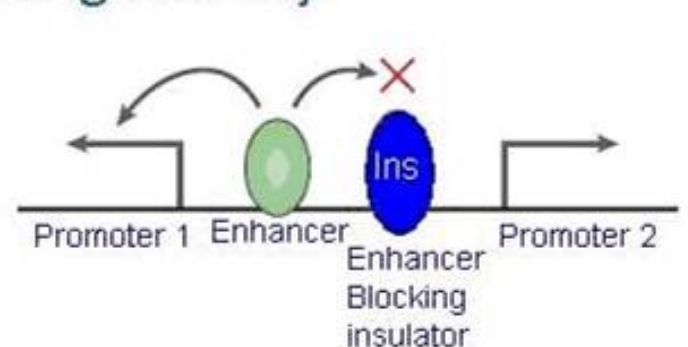


Insulators are a specialized class of DNA elements near chromatin boundaries with two distinct properties:

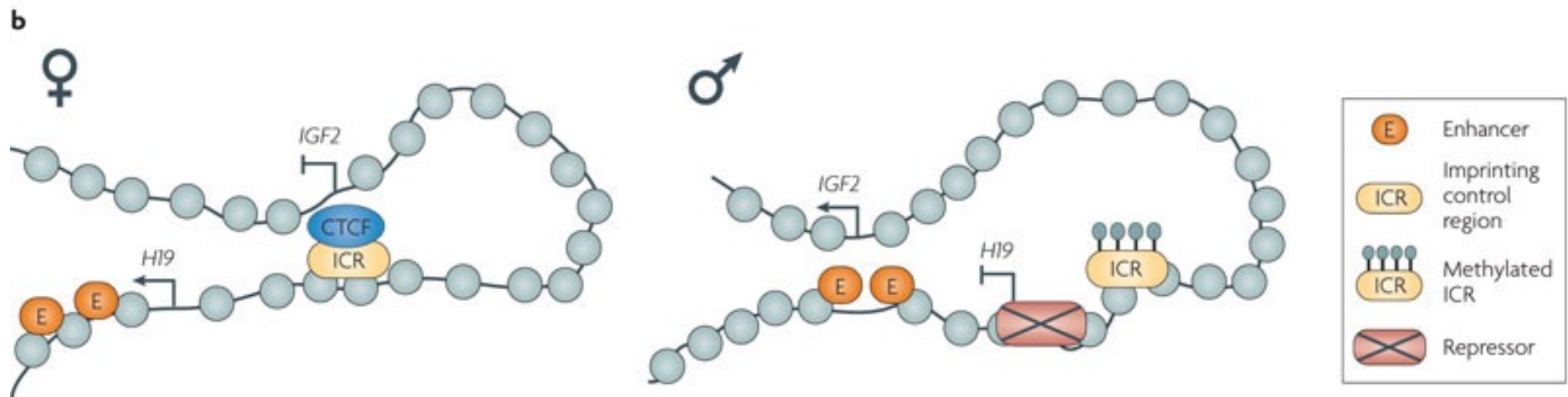
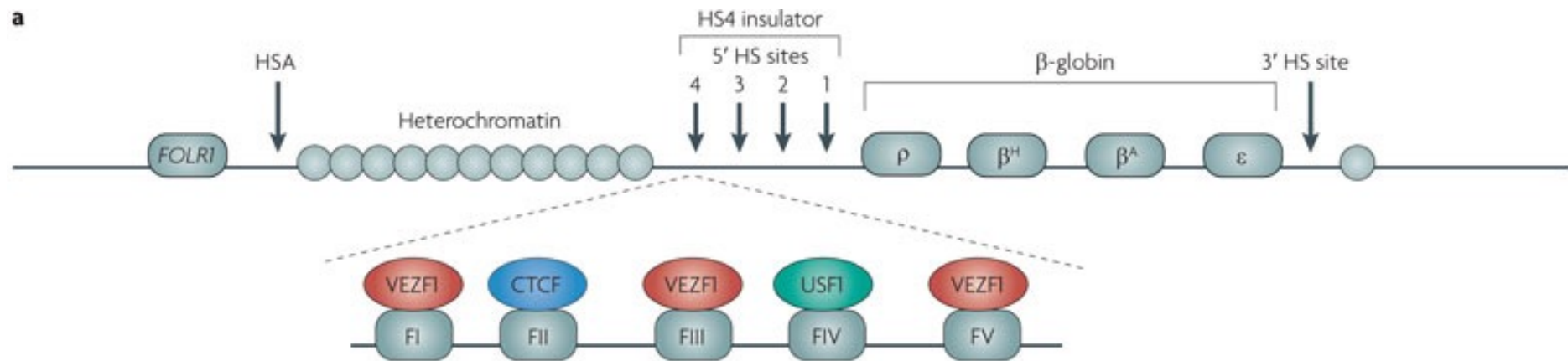
Barrier activity



Enhancer-blocking activity



West et al. 2002, Genes Dev., Liang CC, 2005, Cell Res, Gazner M, 2006 Nat Genetics



Nature Reviews | **Genetics**

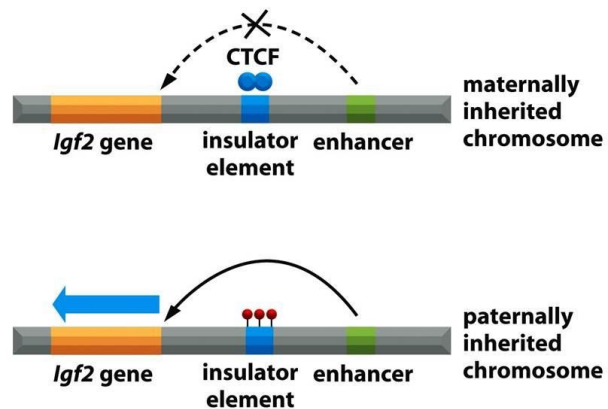
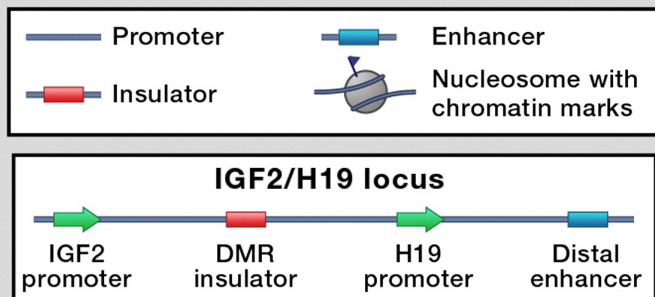
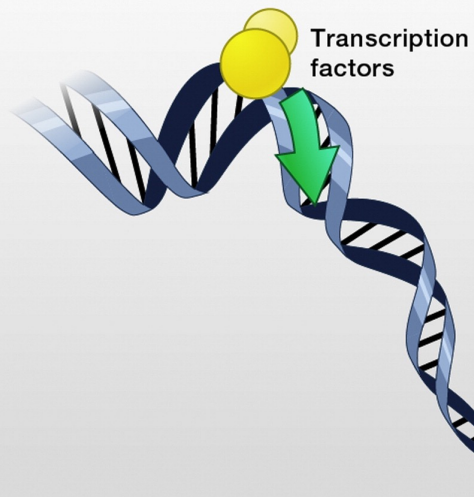
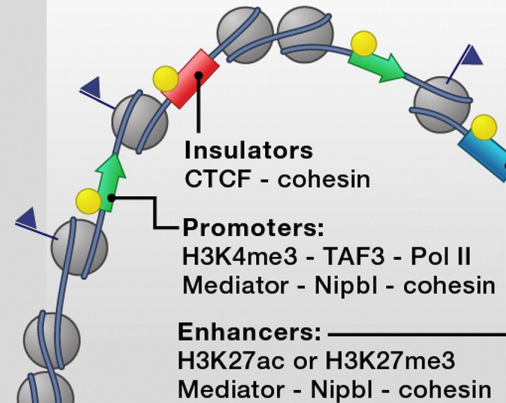


Figure 7-83 Molecular Biology of the Cell 5/e (© Garland Science 2008)

Transcription factors and DNA



Chromatin marks, readers and writers

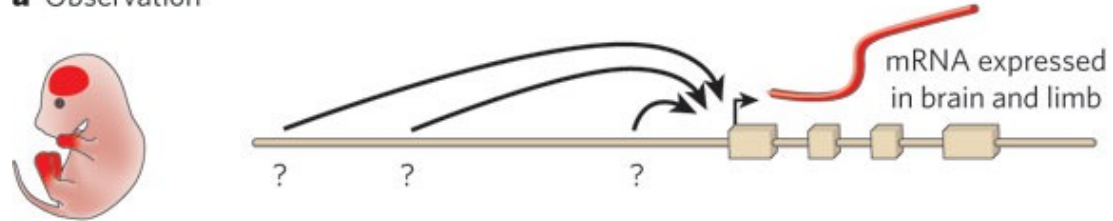


Paternal allele
DMR methylated
CTCF not bound
Insulator inactive
IGF2 transcribed

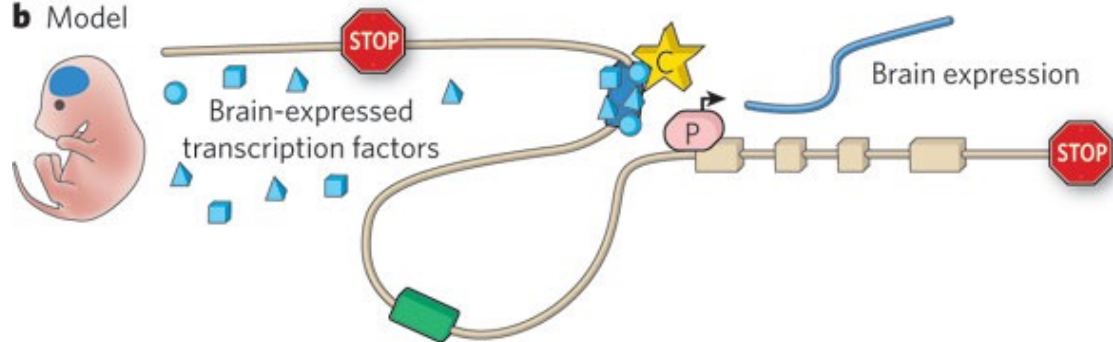
Maternal allele
DMR unmethylated
CTCF bound
Insulator active
IGF2 silent

Higher order chromatin organization

a Observation



b Model



c Model

